Deutsche Börse Group SUMMIT

Manual

Software Engineering/Development Lifecycle Framework

Internal Use Only

Internal

History

Version	Release Date	Description	Created / Amended by	Reviews and Sign-Offs
0.1	03-Dec-15	1st common draft CBF-CBL SUMMIT Document	Quality Management	
0.4	18-Dec-15	Various SDLC related additions	W. Mühlpfordt	
0.5	22-Dec-15	Final updates from Security Modules integrated	J.F. Gaudillat	
0.6	06-Jan-16	New AIP structure	J.F. Gaudillat	
0.7	15-Jan-16	2nd review CBF-CBL harmonization	J.F. Gaudillat	
0.8	22-Feb-16	3rd review CBF-CBL harmonization	J.F. Gaudillat	See Alfresco
0.9	03-Mar-16	Version for review & SO	J.F. Gaudillat	See Alfresco
0.10	12-Apr-16	Published version for Clearstream CBF-CBL	J.F. Gaudillat	
1.0	01-Jun-16	First DBAG SUMMIT Manual with Cash & Derivatives integration	J.F. Gaudillat	
1.3	21-Mai-19	Remove TO references and re-integrate security modules into standard modules	A. Schamber	
1.3	21.July-19	Included Review Comments	A. Schamber	
1.4	01-Oct.-19	Update History and TOC - Releasing V. 1.4	A. Schamber	
1.5	24-Jun-21	Prepare Version 1.5	A. Schamber	
2.0	19-Nov-21	Creation of combined manual	M. Rosenberg	See Alfresco
2.1	22-Nov-22	Removed reference to “DBAG IT Quality Guideline”; Replaced UML based “module views” with BPMN models and RACI matrices	M. Rosenberg	See Alfresco
2.1	14-Dec-22	Incorporated review feedback	M. Rosenberg	See Alfresco
2.1	16-Feb-23	Feedback from CBL and Risk IT added; final version	M. Rosenberg	See Alfresco

SUMMIT maintenance

Changes to the SUMMIT manual are co-ordinated among the IT functions and legal entities in accordance with the needs of the organisation and input from the parties involved.

Lifecycle Management

SUMMIT will be reviewed on an annual basis.

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1 Introduction

1.1 What is SUMMIT?

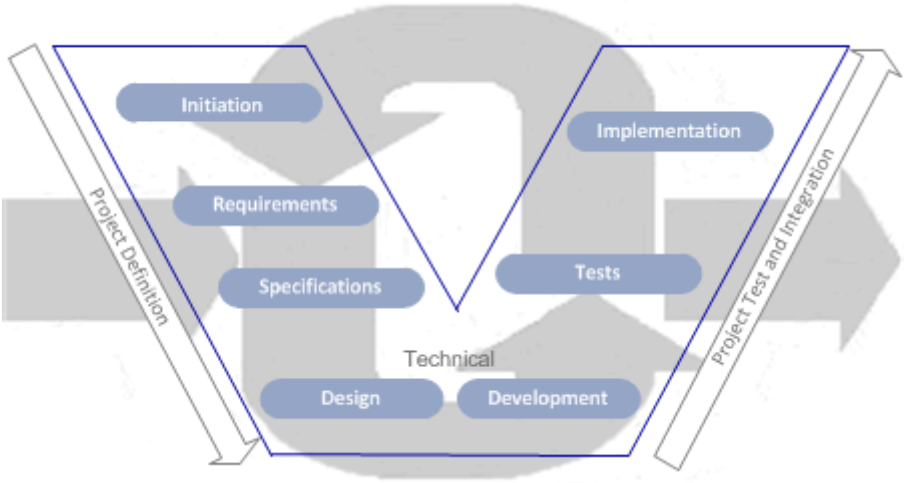
SUMMIT is a software engineering/development lifecycle framework. It also provides guidance on setting up and running IT Projects. It is not a standalone project management process by itself like PRINCE2.

Not all engineering/development activities are done in the context of a project. This doesn't preclude the use of SUMMIT's best practices for those engineering/development activities whether they are maintenance, changes or new development. The usage of SUMMIT for software engineering/development does not transform that development effort into an IT Project. For clarity and to reduce wordiness a reference to a “project” throughout this document also refers to a non-project based development activity.

This customised version of SUMMIT may be selected as a software engineering/development framework for use within IT and business areas.

SUMMIT describes the software engineering process by providing information on:

- potential SUMMIT modules to apply
- potential work products to deliver
- suggested roles and responsibilities of the parties involved



This non-exhaustive list highlights internal written rules, in line with their legal entity requirements, to which the project manager must adhere.

- Company policies including the Material Change Process Policy
- Information Security Framework
- Project Management checklist(s) if the development activity is also a project

2 SUMMIT Overview

2.1 Objectives

The objective of SUMMIT is to enable teams to develop systems in the end user's environment reaching the following goals:

- To ensure that systems meet the business objectives and requirements of the organisation.
- To develop systems that are sufficiently maintainable to evolve with the users, so that they continue to support changing business objectives in the future.
- To ensure systems are as easy as possible to use.
- To deliver systems when the customers need them.
- To deliver systems within the budget constraints.
- To ensure systems are as inexpensive to operate as is practical.

2.1.1 Key Considerations

Several of an organisation's goals may conflict with each other, such as time versus budget. The SUMMIT framework, therefore, provides approaches for:

- Determining the business objectives and requirements that support those goals.
- Evaluating alternative approaches, considering time and cost constraints.
- Developing an approach that meets those business objectives.
- Managing change as the team builds the system.
- Implementing the systems successfully and launching it into the business and technical environment.

The framework defines:

- A structure that guides the team and involves the main stakeholders in the decision-making process at key points.
- Work Products for each phase that address the needs of a well-run project.
- Roles for the team members that define the skill set required for planning and enabling them and users to understand how each performs successfully.

SUMMIT has been designed to address the entire engineering/development life cycle for a broad range of projects from small to large. It then guides the integration of deesigned systems into the user's business environment and fully implements these systems to achieve the desired result. SUMMIT also refers to non-IT activities, ensuring that no business delivery tasks are overlooked and that a successful launch is achieved.

To satisfy these requirements SUMMIT has been developed to be:

- Business Driven - Ensuring that all projects initiated are business priorities for the organization.
- Comprehensive - Both IT and business activities are covered by the framework.
- Action oriented - There is a bias to action with recommendations for activities that are needed and should be initiated immediately.
- Adaptive - The framework can be tailored to the requirements of the project/development activities.

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- Flexible - Techniques and tools can be used that are already used within the organization, while certain core techniques are defined that are key to producing certain deliverables.
- Documented - All modules are described with supporting diagrams giving a clear visual presentation of the framework and its structure.
- Usable by multi-disciplinary teams - A project requires the close collaboration of team members from several different functions in the organisation, each bringing their own skills, knowledge, experience and expertise.
- Strongly linked to delivery and implementation - Clear, comprehensive links from idea creation through to delivery is provided.

Using the framework requires judgement to ensure that the work products produced are appropriate to the environment in which it is being used. It permits the leveraging of skills and experience by embodying practical guidelines and key points for consideration. In keeping with any other project , it does not enable people who do not have the appropriate experience and skills to conduct a project with any guarantee of success.

SUMMIT achieves additional objectives, which are outlined below:

- SUMMIT encourages the use of the most effective and appropriate proven techniques, while providing opportunities to substitute an organisation's preferred technique, if applicable.
- SUMMIT supports the whole project/development life cycle on a wide range of projects/development activities, differentiated by scope, size, and technology and delivery method(s). It also supports individual components in a larger project in situations where a group does not have overall responsibility. It has been designed to avoid unnecessary increases in the time or cost of projects through unrealistic levels of analysis, documentation or project management activities.

2.2 Scope

SUMMIT promotes the active and informed participation of users and their management in all stages of a project and is suitable for distributed projects as well.

SUMMIT is designed to cover all project stages from the definition of business delivery requirements to the launch of the new/enhanced product or service. It covers all the business delivery (marketing, compliance, staff & customer readiness and pricing) as well as the technology, user, managerial and change elements involved in a project.

SUMMIT depends on trained managers and trained professional staff. It has many distinctive characteristics. The following key of characteristics illustrates the range of coverage:

- The definition of business requirements and constraints.
- Analysis and design techniques, with guidelines on identifying the underlying conceptual model from the analysis of management objectives and current systems.
- The construction of new systems involving disciplined progression through logical and physical descriptions of data and processes.
- Quality assurance and quality control, including risk management.
- Design for auditability, security and recoverability of system and data.

- Project management, covering project planning and reporting.
- Project documentation, including key work products / deliverables for users and working papers for the project team.
- User support, through education and training, and securing user involvement and commitment, and the use of work products, which are understandable to users.
- Product investigation, covering the evaluation and installation of application software, system software and hardware.
- Delivery support, including involvement of delivery management, service level agreements and development of delivery procedures and training.
- Infrastructure support and architecture.
- Testing.

2.3 Structure

2.3.1 Phases and Modules

SUMMIT is broken into phases, modules and work products. Phases and modules are logical units of the application development project. Phases are split in modules and modules are split into work products. The main criteria to combine different work-products within on module and multiple modules into a phase are: logical content, common ownership, common reviewer and the chronological sequence.

SUMMIT consists of a series of modules used to construct a project work plan. A SUMMIT module is a cohesive group of tasks and activities that are performed to produce a specific set of project work products. Modules can overlap two or more SUMMIT phases. During initiation the project representatives have to select the approach to be adopted and select the relevant modules from the appropriate route map.

The basic sequence of phases is Initiation, Requirement Analysis, System Definition, Design, Build and Test, and Service Implementation.

Although overlapping work can contribute to a quicker process in most cases, a phase exists for the following reasons:

- It establishes a set of formal work products that should be produced by the project/development team, before formally proceeding to the next phase. This enables the manager to divide a large project into a series of measurable milestones.
- The end of each SUMMIT phase requires a formal decision by the "Project Owner" to accept the conclusions that have been reached at this point and to decide to commit the resources necessary to proceed with the project.
- The major SUMMIT phase work products represent a solid base that can be used to define the scope and function of the system for subsequent phases. This provides a basis for change control that keeps the project focused and moving towards a successful conclusion.

The SUMMIT phase structure implies the basic sequence for the product development life cycle. Six standard phases have been defined for SUMMIT and are described below. For some assignments, it will be appropriate to define a phase structure combining or dividing the standard

phases. The SUMMIT phase structure for a particular assignment match the nature of that assignment and the preferences of the enterprise.

Activities within a SUMMIT module are not always performed sequentially; work can/shall begin as soon as required activity inputs are available irrespective of a formal signoff.

Initiation

This is the SUMMIT phase where the overall planning and organisation are set-up as a starting point for the project.

Requirements Analysis

This SUMMIT phase summarises all tasks associated with defining, reconciling (internally and externally) and analysing requirements (both functional and non-functional). It also aims to determine the project scope, tasks and mission. This also includes assessing the impact on the existing systems and identifying and evaluating options.

Solution Definition

The functional and non-functional requirements and/or the market and business needs, together with security requirements are portioned, detailed and comprehensively specified within the solution definition phase. This also includes estimating the cost for the subsequent phases.

Design

The design phase of the project consists of the tasks necessary to describe how the proposed solution is to be built and/or how work is to be done. In one or more steps the software coding is prepared by detailing and refining the specified requirements.

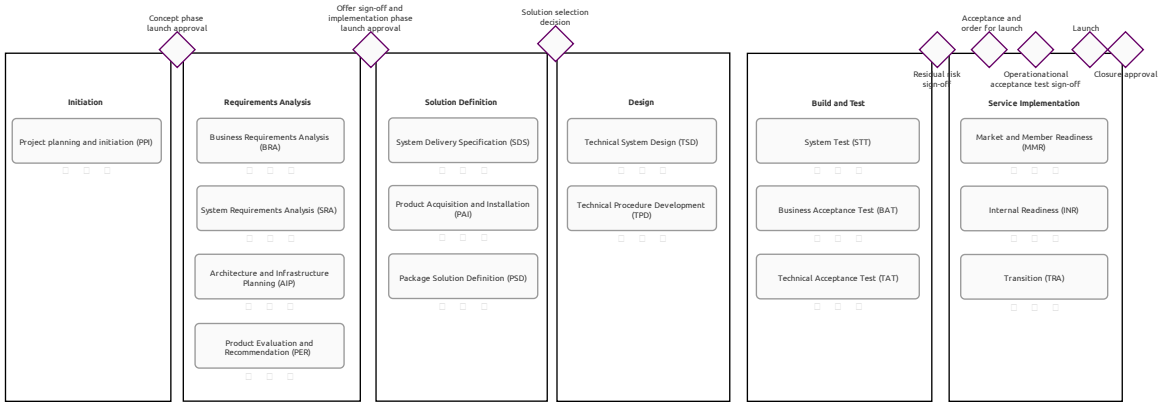
Build and Test

The SUMMIT build and test phase consists of those modules which provide the tasks necessary to construct and test the final systems. In this phase the specific technical components are programmed according to the documented output of the design phase. Furthermore, the planning, organisation, preparation and execution of the system test and the acceptance test (Business Acceptance Test and Technical Acceptance Test) is covered in this phase.

Service Implementation

The SUMMIT service implementation phase contains the internal migration and integration of the developed solution into its future technical and organisational environment. Additionally, it describes the process of launching the solution externally and to the customers.

The follow diagram summarises the general placement of the SUMMIT modules within the respective phases. The option for modules to overlap phases is not depicted but is permissible.



2.3.2 Work Products

SUMMIT work products are the smallest elements to determine the entire procedure and chronology of activities so that projects have a consistent, formalised and comparable structure. This structure can be adjusted to the characteristics of the project by tailoring the route and specifying the work product selection. The SUMMIT method contains building blocks (phases, modules and work products) that can be combined, reused and tailored to specific needs.

The work product structure establishes the ownership model by providing an explicit assignment of an owner to a work product and the sign off procedure for each work product. The work product matrix summarises the selection of work products. Preparation and execution of an activity is documented and is thus transparent for management and other stakeholders.

SUMMIT modules cluster logically related work products together due to their content and purpose, or due to organisational reasons, such as having the same author or reviewer. But according to the chronological overlapping of modules, work products often have to be started and generated when foregoing modules are still running.

Four readiness areas have also been identified:

- Project Readiness: all those work products related to the administration, planning and organization of the processes.
- Product Readiness: all those work products directly related with the description, specification, design, build and test of the new software, software release or product.
- Market and Member Readiness: all those work products related to the planning, organization, control, tracking and executing the readiness of external stakeholders. Market and Member Readiness is a module in its own right.
- Internal Readiness: all those work products related to the planning, organization, controlling, tracking and execution of the readiness of internal stakeholders. Internal Readiness is a module in its own right.

2.4 SUMMIT Routes

When planning a project using SUMMIT, one of the first steps is to classify the project by a set of criteria, like

- degree to which requirements can be defined

- size and complexity
- impact, …

Using this information, SUMMIT suggests a Base Route.

Base Routes are intended as a starting point for detailed planning. Planners should tailor the route once it has been chosen. The route, therefore, is not the plan. Its value is in saving the planner from starting on a blank page.

Please refer to section 3.1 for further details about routes.

2.5 Involvement of PM, PO and QM

Project managers are stakeholders on all modules.

It is recommended that project managers liaise with central functions such as Legal, Risk Management, Compliance, Information Security or the data protection officer at the beginning of the project, and assess with them their further involvement when appropriate.

Lifecycle Phase	Project Manager	Project Office/project support function (projects only)	Quality Management (optional)
Project initiation	Select appropriate route. Determine which deliverables to produce.	Agree PO support. Start project setup.	Provide support to use SUMMIT framework.
Project executing/controlling (ongoing during subsequent phases)	Monitor performance. Refine and update the plan and WP matrix (reviewers, approvers and deadlines). Ensure that the planned work products are created, reviewed and approved by the planned deadline. Track changes of scope. Monitor risks, issues, dependencies. Control costs. Communicate information status and report status. Manage team.	Provide support to Project Manager as needed. Support to refine and update the project plan. Monitor the work products creation, review, sign-off and safe-store against the WP matrix. Highlight delays in the work product completion to the Project Manager. Support tracking of scope, risks, issues, dependencies. Support communication and reporting.	Provide support to use SUMMIT framework and disseminate guidelines. Participate in Internal Quality Review (IQR) of the project on Management request to provide recommendations in compliance with IT processes and to collect recommendations to improve the IT Processes.

Lifecycle Phase	Project Manager	Project Office/project support function (projects only)	Quality Management (optional)
Project closing (service implementation phase and transition module)	Consolidate and store all work products and reporting. Describe lessons learnt.	Support the consolidation of work products and reporting. Participate in the lessons learnt description.	Capture Lessons Learnt.

2.6 Involvement of Solution Architect and Legal Entity Enterprise Architecture Function

Every project which has an impact on IT should nominate a Solution Architect (SA) or an IT Architect (ITA). The exact names of this role may vary, but the role includes the following aspects:

- Responsible for implementing the application, technology and data architecture enterprise vision through existing guidelines, policies and standards
- Minimizes architecture and technology risks by early alignment with IT Strategy and designing the system in line with the guidelines.
- Works together with the Business Architects, Business CIO, LE Enterprise Architecture function (LEEA), infrastructure team, experts and project managers.
- Supports the efficient fulfilment of business requirements, while taking functional and non-functional requirements into consideration.
- Has a focus on domain specific functional skills.
- Is considered an expert in system design and data modelling
- Supports issue escalation process if needed with preparation of different solutions
- The solution architecture covers all architecture domains: business, application, data, technology and security.

Lifecycle Phase	Solution Architect (SA)/IT Architect (ITA)	LE Enterprise Architecture Function (LEEA)
Project initiation	Determine which architecture deliverables to produce Understand architecture policies and guidelines Access application inventory to understand Group IT and to assess impact	Agree degree of LEAA support Review list of architecture deliverables Review impact assessment

Lifecycle Phase	Solution Architect (SA)/IT Architect (ITA)	LE Enterprise Architecture Function (LEEA)
Project executing/control (ongoing during subsequent phases)	Align solution technology stack with IT roadmap, reference architecture (SOA) and enterprise data models Foster re-use of common components and architecture building blocks Ensure that planned architecture work products are created Trigger review and approval processes for architecture deliverables Log architecture issues and risk and escalate to LEEA Participate in architecture community Ensure usage of centralized Development Architecture (DevOps) Align with infrastructure service for including operational requirements	Review and approve architecture deliverables Support and advise project in case of architecture decisions Solve escalations, Communicate updates to policies and guidelines Manage architecture community Train solution architects
Project closing	Update information in Application Repository List all technical debt accrued	

3 Process and Methodology Terms

Application of the terminology

SUMMIT terminology must be used across all projects/development activities. If SUMMIT has a word for a phase or work product, no other names should be used. This eases common understanding in cross-project communication and reporting in all locations.

If a project is outsourced to a service provider that uses its own methodology, a mapping of terminology and deliverables will suffice if the use of SUMMIT by that company would not be cost-effective.

Planning

Use of SUMMIT "route mapping" as basis for planning

At the start of a project or when planning a new phase, planners will use SUMMIT to select the 'base route' best suited to their project and to determine which modules and which work-products

will be produced. It is the project manager who must decide upon project start, which route to follow and how to customise the work products as required.

If work products are created, the names should be used according to the terminology provided in order to achieve a common language over all locations. By providing pre-defined routes for typical project, the Project Manager is presented with a pre-selection of work products, which should be customized as needed.

A set of metrics is also proposed. It is recommended to collect the various metrics for each project to enable a measurable continuous improvement process. The reporting of metrics will be introduced to measure process set-up and progress, respectively in order to assess the capability of the processes performed.

3.1 SUMMIT Base Route Selection

Base Routes are intended as a starting point for detailed project planning. Planners are expected to tailor the route once it has been chosen. This involves replicating, deleting, adding, and changing activities. The base route, therefore, is not the plan. Its value is in saving the planner from the arduous task of defining the work on a blank page.

Each route uses activities from the same superset. In some routes, appropriate for low risk kinds of projects activities are omitted from the full Work Breakdown Structure (WBS). Whereas higher risk project may include virtually all of the activities. Risk categories are among the criteria used in base route selection.

Technology provides another set of criteria. For example, one would not select the client/server route if the system to be delivered is to be centralized on a mainframe computer.

Project objectives and constraints provide another set of criteria. If time is severely limited, for example, one would look for a route with few "go – no go" decision points since these significantly extend a project's duration.

The task of route selection can be daunting when one realizes that the criteria are not mutually exclusive. Suppose you have severe time constraints but significant risk. Suppose you are forced to deliver a distributed Client/Server system but must, at the same time, conform to a strictly defined centralized data architecture based on Information Engineering. The decision scheme included here is meant to help a planner through the process of selecting a base route.

Again, the base route is not the plan. Its value is in saving the planner from the arduous task of defining the work on a blank slate.

Each base route uses activities from the same superset, the SUMMIT framework:

- In some routes, appropriate for low risks kinds of projects/development activities, significant numbers of activities are omitted from the full Work Breakdown Structure.
- The technology to be applied provides another set of criteria.
- Project objectives and constraints provide another set of criteria. If time is severely limited, for example, one would look for a Route with few "go – no go" decision points since these significantly extend a project's duration.

Independent of the route chosen for a project, the following points should be observed:

- When tailoring the project routes, please consider all Group Policies.

- The project manager is required to get formal approval of the chosen SUMMIT route and plan, in accordance with the project's governance, and agreements need to be recorded
- The Information Security Framework must be considered regardless of route and module chosen.

Degree to which requirements can be defined		Definable								Operational system			Not readily definable
Size and complexity		Simple and small		Large and complex						Large and/or complex	Small and simple		Large
Route	Module	SP	SC	LP	LC	NE	RE	IE	CS	AD	EN	MA	CI
Initiation	PPI												
Requirements analysis	BRA												
	SRA												
	AIP												
	PER												
Solution definition	SDS												
	PAI												
	PSD												
Design	TSD												
Build and test	TPD												
	STT												
	TAT												
	BAT												
Service implementation	MMR												
	INR												
	TRA												

Unlikely
Recommended
Highly recommended

3.2 Route details

3.2.1 Small Package (SP)Small Package (SP) Description

Route "SP" delivers a small, simple system, based upon a known application package, where requirements are already defined or readilsy definable.

The Small Package (SP) Route is used as the base route for projects where a known package is being installed for a single business function. If the package is to interface with other applications or data conversion is required, then these must be of limited scope and complexity. This route does not address the implementation of additional technology components.

It will have to be decided, in determining the implementation strategy, that the processes embodied in the package functionality will be adopted. The business processes, policies, and procedures will be changed to comply with these. The configuration of the package software will therefore be relatively straightforward requiring little or no detailed analysis of how the package functionality can be used.

3.2.1.1 Small Package (SP) Considerations

Phase	Module	Task
Initiation	PPI	Use PPI to address the key issues (e.g. scope, approach, ….) of a project/
Requirements Analysis	BRA	Module partly used; when deemed appropriate BRA, SRA and SDS could be merged in one document
	SRA	Manage SRA and SDS as one packet
	AIP	Module partially used in this route

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Phase	Module	Task
	PER	Module not used in this route
System Definition	SDS	System Delivery Specification becomes key deliverable.
	PAI	Use PAI to install products
	PSD	Use PSD to configure package.
Design	TSD	Manage TSD and PSD together if data conversion or interface.
Build and Test	TPD	Use TPD to develop data conversion or interface.
	STT	System Test is to be undertaken as separate part to Acceptance Test. Security testing is required on a regular/continuous basis and should always be considered.
	TAT	
	BAT	
Service Implementation	MMR	Normally undertaken in full unless entirely new system.
	INR	
	TRA	Module includes implementation strategy definition, preparation, execution and closedown phase of the project.
Unlikely		Recommended / Highly Recommended

3.2.2 Small Custom Built

3.2.2.1 Small Custom (SC) Description

Route "SC" delivers a small, simple, custom-built system, where requirements are defined or readily definable. This route is likely to enter into a fast track approach.

Route "SC" proceeds in three steps:

- Requirements Analysis and System Definition
  - System Requirements Analysis (SRA) and System Delivery Specification (SDS) are carried out as one stage. No Product Evaluation and Recommendation (PER) is done as Project Planning and Initiation (PPI) has determined that no package or product is suitable for this project. SRA verifies this decision and the choice of route. Product Acquisition and Installation (PAI) can be used, but only to install more of a known technology product.
  - The primary deliverable of this stage is the System Delivery Specification Document; no System Prospectus is produced.
- Design, Build and Test
  - Modules Technical System Design (TSD), Technical Procedure Development (TPD), and Internal Readiness (INR) are all performed. System Testing (STT) and Acceptance Testing (BAT, TAT) are performed but due to the small size of the system to be built the number of test cases and thus the conducted effort are low. Product and Regression Test can be combined.

- The primary deliverable of this stage is the Accepted System.

- Service Implementation
  - Parts of the Internal Readiness (INR) And optionally Market and Memer Readiness modules are performed.

3.2.2.2 Small Custom (SC) Considerations

Phase	Module	Task
Initiation	PPI	Use PPI to address the key issues (e.g. scope, approach, ….) of a project
Requirements Analysis	BRA	Module partly used; when deemed appropriate BRA, SRA and SDS could be merged in one document
	SRA	Manage SRA and SDS as one packet.
	AIP	Modules partially used in this route.
	PER	Module not used in this route.
System Definition	SDS	System Delivery Specification becomes key deliverable.
	PAI	Use PAI to install products.
	PSD	Module not used in this route.
Design	TSD	Manage TSD and PSD as one work packet.
Build and Test	TPD	Use TPD for program design, coding, and initial integration.
	STT	System Test to be undertaken as part of Acceptance Test (TAT to be performed when needed).
	TAT	
	BAT	
Service Implementation	MMR	Normally undertaken in full unless entirely new system.
	INR	
	TRA	Module includes implementation strategy definition, preparation, execution and closedown phase of the project.
Unlikely	Recommended	Highly Recommended

3.2.3 Large Package

3.2.3.1 Large Package (LP) Description

Route "LP" addresses the delivery of systems based on the use of one or more application products. Although this route addresses large package implementation projects/development activities, it does not address enterprise wide requirements or a program of linked implementations. Small, simple implementations particularly where there is no evaluation or selection will use the "SP" route.

The Large Package route is used as the base route for a wide variety of projects from a single package selection and implementation, to a multiple product selection (where more than one iteration of modules Product Evaluation and Recommendation (PER), Product Acquisition and Installation (PAI), and Package Solution Definition (PSD) is required). The stages below are suitable for all of these project types, however, the timing of certain modules within the stages (Product Evaluation and Recommendation, Product Acquisition and Installation, Package Solution Definition, Architecture and Infrastructure Planning (AIP), and Technical Procedure Development (TPD)) are flexible.

3.2.3.2 Large Package (LP) Considerations

Phase	Module	Task
Initiation	PPI	Use PPI to address the key issues (e.g. scope, approach, ….) of a project.
Requirements Analysis	BRA	Analyse and document business delivery requirements.
	SRA	Full use of SRA.
	AIP	Full use of AIP.
	PER	Use PER for application product.
System Definition	SDS	Use SDS for custom system definition.
	PAI	Use PAI for technology products.
	PSD	Use PSD for package solutions.
Design	TSD	Use TSD to design custom work.
Build and Test	TPD	Use TPD to develop data conversion and interfaces.
	STT	Run separate System Test and Acceptance Test (TAT to be performed when needed).
	TAT	
	BAT	
Service Implementation	MMR	Normally undertaken in full unless entirely new system.
	INR	
	TRA	Module includes implementation strategy definition, preparation, execution and closedown phase of the project.
Unlikely	Recommended	Highly Recommended

3.2.4 Large Custom Built

3.2.4.1 Large Custom (LC) Description

Route "LC" delivers a large and/or complex, custom-built system where requirements are defined or readily definable. This route covers almost all work products, except those related to package

software and the AIP module. It is the typical route choice in case of big release projects on an existing infrastructure.

Route "LC" proceeds in five steps:

- Requirements Analysis
  - The full System Requirements Analysis (SRA) module is performed and results in a System Prospectus. Product Evaluation and Recommendation (PER) can be used, but only to evaluate additional items of a known technology product. AIP module can be performed if necessary.
- System Definition
  - Module System Delivery Specification (SDS) is undertaken. Product Acquisition and Installation (PAI) can be used, but only to install more of a known technology product.
- Design
  - The Technical System Design (TSD) module is performed in its entirety.
- Build and Test
  - Modules Technical Procedure Development (TPD), System Testing (STT) and Acceptance Testing (BAT, TAT) are all performed and yield an Accepted System.
- Service Implementation
  - Parts of the Internal Readiness (INR) module and the Market and Member readiness (MMR) module are performed.

3.2.4.2 Large Custom (LC) Considerations

Phase	Module	Task
Initiation	PPI	Use PPI to address the key issues (e.g. scope, approach, ….) of a project.
Requirements Analysis	BRA	Analyse and document business delivery requirements.
	SRA	Full use of SRA.
	AIP	Full use of AIP.
	PER	Use PER for technology product.
System Definition	SDS	Full use of SDS.
	PAI	Use PAI for technology products.
	PSD	Use to define package configuration and specify any enhancements. Complete configuration in Build (Customize) and Test.
Design	TSD	Full use of TSD.
Build and Test	TPD	Full use of TPD.

Internal

	STT	Run separate System Test and Acceptance Test.
	TAT	
	BAT	
Service Implementation	MMR	Normally undertaken in full unless entirely new system.
	INR	
	TRA	Module includes implementation strategy definition, preparation, execution and closedown phase of the project.
Unlikely	Recommended	Highly Recommended

3.2.5 New Technical Environment

3.2.5.1 New Technical Environment (NE) Description

Route "NE" encompasses those activities required to identify, evaluate, acquire, and install one or more substantial components of a technical environment. This route is designed to be used as a basis for stand-alone projects/development activies.

If a big custom system is to be built on a greenfield approach, a combination of both routes LC and NE would be appropriate choice, either in combination or in a sequential approach.

Route "NE" proceeds in three steps:

- Environment Definition
  - A partial System Requirements Analysis (SRA) module is performed to set scope and determine whether or not a prototype of application requirements will be defined and developed to enable the testing of critical assumptions made about the target technical environment and infrastructure.
  - Full use of the Architecture and Infrastructure Planning (AIP) module is made to define the target environment, whether it be a production environment, test environment, development environment or combination.
  - A limited System Delivery Specification (SDS) module is optionally performed to build any functional prototypes identified in SRA.
  - Full use of the Product Evaluation and Recommendation (PER) module is made to procure the required technologies identified in AIP.
- Build and Test
  - A full Product Acquisition and Installation (PAI) Module is performed resulting in the installation and configuration of the required technologies in the target environment.
  - A full System Testing (STT) and Acceptance Testing (BAT, TAT) modules are performed to ensure integration with other existing elements of the enterprise-wide operating environment, yielding the Production Environment. It will also ensure that the existing functionality is correctly carried over the new technological basis, respectively is correctly implemented in care of new functionalities to be included.

- Service Implementation
  - Parts of the Internal Readiness (INR) module is performed yielding a reviewed production system.

3.2.5.2 New Technical Environment (NE) Considerations

Phase	Module	Task
Initiation	PPI	Use PPI to address the key issues (e.g. scope, approach, ….) of a project.
Requirements Analysis	BRA	Partially used when needed.
	SRA	Some use of SRA.
	AIP	Use AIP/PER to assess and establish technology infrastructure.
	PER	Use AIP/PER to assess and establish technology infrastructure.
System Definition	SDS	Some use of SDS.
	PAI	Use PAI to acquire and install technology products.
	PSD	Modules not used in this route.
Design	TSD	Modules not used in this route.
Build and Test	TPD	Modules not used in this route.
	STT	Use for final test.
	TAT	
	BAT	
Service Implementation	MMR	Use for review
	INR	
	TRA	Module includes implementation strategy definition, preparation, execution and closedown phase of the project
Unlikely	Recommended	Highly Recommended

3.2.6 Re-engineering

3.2.6.1 Re-engineering (RE) Description

Route "RE" provides a framework for capturing and transferring existing programs, data, screens, and reports into a CASE (computer-aided software engineering) tool environment. There they are accessible to several approaches including forward engineering.

Route "RE" proceeds in three cycles, the last of which comprises the traditional five stages:

- Assessment

- This cycle takes inventory of the existing environment, then analyses and defines target objectives. The principal deliverable is an Inventory and Analysis.

- Positioning
  - This cycle achieves restructuring, loading into CASE, and re-documenting using parts of TPD, SRA and SDS to produce the desired models.
- Re-engineering
  - Five steps are performed to achieve conversion, migration and/or enhancement:
  - Analysis: Tasks from SRA.
  - System Definition: Module SDS.
  - Design: Module TSD.
  - Build and Test: Modules TPD, STT, BAT, TAT.
  - Service Implementation: Module INR.

3.2.6.2 Re-engineering (RE) Considerations

Phase	Module	Task
Initiation	PPI	Use PPI to address the key issues (e.g. scope, approach, ….) of a project
Requirements Analysis	BRA	Create inventory of the existing environment, then analyse and define target objectives
	SRA	Use tasks from SRA/SDS/TSD/TPD to assess and position.
	AIP	Modules not used in this route.
	PER	Modules not used in this route.
System Definition	SDS	Use tasks from SRA/SDS/TSD/TPD to assess and position.
	PAI	Modules not used in this route.
	PSD	Modules not used in this route.
Design	TSD	Undertake in full to deliver.
Build and Test	TPD	Undertake in full to deliver.
	STT	Undertake in full to deliver.
	TAT	Undertake in full to deliver.
	BAT	Undertake in full to deliver.
Service Implementation	MMR	Used to organize, control and execute service implementation
	INR	Used to organize, control and execute service implementation
	TRA	Module includes implementation strategy definition, preparation, execution and closedown phase of the project.

Phase	Module	Task
Unlikely	Recommended	Highly Recommended

3.2.7 Information Engineering (IE)

3.2.7.1 Information Engineering (IE) Description

Route Base Information Engineering covers delivering systems under Information Engineering (IE).

IE is a well-known system planning and development approach, characterized by enterprise models and logical systems architectures, and making extensive use throughout of IE tool sets.

3.2.7.2 Information Engineering (IE) Considerations

Phase	Module	Task
Initiation	PPI	Use PPI to address the key issues (e.g. scope, approach, ….) of a project.
Requirements Analysis	BRA	Use BRA /first WP in SDS / PER to perform IE stage.
	SRA	Use SRA / first WP in SDS / PER to perform IE stage.
	AIP	Use SRA / first WP in SDS / PER to perform IE stage.
	PER	Use SRA / first WP in SDS / PER to perform IE stage.
System Definition	SDS	Use SRA / first WP in SDS / PER to perform IE stage.
	PAI	Undertaken in full to implement IE stages.
	PSD	Module not used in this route.
Design	TSD	Undertaken in full to implement IE stages (TAT to be performed when needed).
Build and Test	TPD	Undertaken in full to implement IE stages (TAT to be performed when needed).
	STT	Undertaken in full to implement IE stages (TAT to be performed when needed).
	TAT	Undertaken in full to implement IE stages (TAT to be performed when needed).
	BAT	Undertaken in full to implement IE stages (TAT to be performed when needed).
Service Implementation	MMR	Used to organize, control and execute service implementation
	INR	Used to organize, control and execute service implementation
	TRA	Module includes implementation strategy definition, preparation, execution and closedown phase of the project/.

Phase	Module	Task
Unlikely	Recommended	Highly Recommended

3.2.8 Client/Server

3.2.8.1 Client/Server (CS) Description

Route "CS" provides a project framework for the delivery of information systems characterized by a distribution of functionality over more than one environment. Typically, some functionality resides on the personal computers of end-users ("clients") and some on larger computers which provide services such as data storage and access to multi-user applications ("servers"). A mainframe environment ("host") is often involved as the ultimate source and/or destination of data.

Route "CS" provides a comprehensive linkage of modules and tasks to deliver systems for enterprise-wide applications in the target C/S environment.

The CS Base Route is organized into four steps after Project Planning and Initiation: (1) Architecture and Analysis, (2) Specification, (3) Design, (4) Build and Test, and (5) Service Implementation.

(1) Architecture and Analysis

- System Requirements Analysis (SRA) (entirely).
- Architecture and Infrastructure Planning (AIP) (entirely).
- If prototyping is to be employed as a means of understanding user requirements, a set of activities from TPD and SDS can be invoked. Optionally:
- Product Evaluation and Recommendation (PER) may be invoked to select and/or identify vendor products.
- Product Acquisition and Installation (PAI) may be invoked to negotiate vendor contracts and install either technology or application products.

(2) Specification

This stage borrows activities from several modules and reorganizes them to produce complete, detailed test plans and technical specifications based upon the validation accomplished by small-scale, iterative development of application prototypes. The activities have been extracted from:

- System Delivery Specification (SDS).
- Architecture and Infrastructure Planning (AIP).
- Technical System Design (TSD).
- Technical Procedure Development (TPD).

If prototyping is to be employed as a means for design and preliminary development, a set of activities from TPD and SDS can be invoked. Optionally, as in the previous stage:

- Product Evaluation and Recommendation (PER) may be invoked to select and/or identify vendor products.

- Product Acquisition and Installation (PAI) may be invoked to negotiate vendor contracts and install either technology or application products.

(3) Design

This stage borrows activities from a number of modules and reorganizes them to produce complete, detailed test plans and technical specifications based upon the validation accomplished by small-scale, iterative development of application prototypes. The activities have been extracted from:

- System Delivery Specification (SDS).
- Architecture and Infrastructure Planning (AIP).
- Technical System Design (TSD).
- Technical Procedure Development (TPD).

If prototyping is to be employed as a means for design and preliminary development, a set of activities from TPD and SDS can be invoked. Optionally, as in the previous stage:

- Product Evaluation and Recommendation (PER) may be invoked to select and/or identify vendor products.
- Product Acquisition and Installation (PAI) may be invoked to negotiate vendor contracts and install either technology or application products.

(4) Build and Test

This stage uses tasks and activities from two modules in their entirety:

- Technical Procedure Development (TPD).
- System and Acceptance Test (STT, BAT and TAT).
- Prepare System Tests, Create System Testing Environment and Perform System Tests.

(5) Service Implementation

This stage uses the Internal Readiness (INR) and the Member & Market Readiness (MMR) module in its entirety. Because the system may have to be installed in more than one end-user environment, it may be necessary to plan separate iterations of the transition part of INR module for each environment. Further, because identical applications can perform differently in the different environments, three tasks from System and Acceptance Testing may be executed in parallel with each iteration of the transition part of INR module.

3.2.8.2 Client/Server (CS) Considerations

Phase	Module	Task
Initiation	PPI	Use PPI to address the key issues (e.g. scope, approach, ….) of a project.
Requirements Analysis	BRA	Analyse business requirements for enterprise-wide applications in the target C/S Environment.
	SRA	Full use of SRA, invoke selected TPD/SDS tasks to prototype.

Phase	Module	Task
	AIP	Full use of AIP.
	PER	Use PER to select and/or identify vendor products.
System Definition	SDS	Extract SDS/AIP activities to design.
	PAI	Use PAI to install products.
	PSD	Module not used in this route.
Design	TSD	Extract TSD/TPD activities to design.
Build and Test	TPD	Undertake TPD/SAT/STT in full to integrate.
	STT	Undertake STT, TAT, BAT in full to integrate (TAT to be performed if needed).
	TAT	
	BAT	
Service Implementation	MMR	Undertake in full.
	INR	
	TRA	Module includes implementation strategy definition, preparation, execution and closedown phase of the project.
Unlikely	Recommended	Highly Recommended

3.2.9 Adaptive Development

3.2.9.1 Adaptive Development (AD) Description

Route "AD" delivers a system based on a substantial adaptation of an existing system where the requirements and target system are defined or readily definable. It typically represents a medium to release project, implementing minor enhancements to an existing large application.

Route "AD" proceeds in four steps:

- Requirements Specification
  - The System Requirements Analysis (SRA) and The System Delivery Specification (SDS) modules are undertaken as a combined work packet which results in changes to the existing systems System Delivery Specification.
- Design
  - The Technical Systems Design (TSD) module is performed resulting in updated Technical System Design documentation.
- Build and Test
  - The Technical Procedure Development, System Test (STT) and Acceptance Testing (BAT, TAT) modules are performed.
- Service Implementation

- The Internal Readiness Module (INR) and optionally the Market and Member Readiness module are performed.

3.2.9.2 Adaptive Development (AD) Considerations

Phase	Module	Task
Initiation	PPI	Use PPI to address the key issues (e.g. scope, approach, ….) of a project/development activities.
Requirements Analysis	BRA	Module partly used; when deemed appropriate BRA, SRA and SDS could be meged in one document.
	SRA	Limited use of SRA.
	AIP	Use AIP if there are significant architectural changes
	PER	Module not used in this route.
System Definition	SDS	System delivery specification is key deliverable.
	PAI	Module not used in this route
	PSD	Module not used in this route
Design	TSD	Use TSD/TPD to make changes.
Build and Test	TPD	
	STT	Partially performed.
	TAT	
	BAT	
Service Implementation	MMR	Undertake in full.
	INR	
	TRA	Module includes implementation strategy definition, preparation, execution and closedown phase of the project.
Unlikely	Recommended	Highly Recommended

3.2.10 Enhancement

3.2.10.1 Enhancement (EN) Description

The Enhancement (EN) base route is applicable when small changes to an existing information system are defined or readily definable. It typically represents a small release project, implementing little enhancements/updates to an existing application.

Characteristics of this route are:

- No Systems Prospectus is produced.
- Modules PER and PAI are not performed.

- This route map must be used in stand-alone mode; it should not be combined with any other route map.
- Minimal changes to conceptual process and data models required.
- Minimal documentation changes required.
- Minimal changes to the application function.
- Minimal changes to database or data set structures.
- Minimal technical system configuration changes.
- Minimal visibility/risk.

3.2.10.2 Enhancement (EN) Considerations

Phase	Module	Task
Initiation	PPI	Use PPI to address the key issues (e.g. scope, approach, ….) of a project.
Requirements Analysis	BRA	Module partly used; when deemed appropriate BRA, SRA and SDS could be merged in one document.
	SRA	Use some of SRA.
	AIP	Module not used in this route.
	PER	Module not used in this route.
System Definition	SDS	Use SDS to define change specifications.
	PAI	Module not used in this route.
	PSD	Module not used in this route.
Design	TSD	Use to make changes.
Build and Test	TPD	Use to make changes.
	STT	Perform reasonable tests ot ensure integration with other existing elements of the enterprise-wide operating environment.
	TAT	Acceptance Test only (TAT to be performed when needed).
	BAT	
Service Implementation	MMR	Use to close.
	INR	
	TRA	Module includes implementation strategy definition, preparation, execution and closedown phase of the project.
Unlikely	Recommended	Highly Recommended

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3.2.11 Maintenance

3.2.11.1 Maintenance (MA) Description

Route "MA" covers the work undertaken to keep a production system operating according to its currently agreed requirements.

If the work to be undertaken is more in the nature of an enhancement (there will be an alteration to support new or modified business operations or policies) the "EN" route may be more appropriate.

Route "MA" is conducted in one stage; no approvals are required in order to proceed from one set of activities to another. The route is most suitable for the following categories of work:

- Adaptive - Minor modification of data value (not data structure) to reflect changes in Business environment.
- Corrective - Removal of defects.

There are other categories of maintenance which are often less urgent in character. In such instances consider performing the work as part of a scheduled release. Another route, such as "AD", may be more appropriate if it is decided that the maintenance can wait and be dealt with along with several other changes. These discretionary categories of maintenance include:

- Perfective - Modification, restructuring or documentation of system components to reduce future maintenance.
- Tuning - Minor modification of system architecture to improve performance (but not system function).
- Upgrade - Minor modification of a system to reflect changes in technical operating environment.

3.2.11.2 Maintenance (MA) Considerations

Phase	Module	Task
Initiation	PPI	Use PPI to address the key issues (e.g. scope, approach, ….) of a project.
Requirements Analysis	BRA	Use to capture business requirements if business driven maintenance.
	SRA	Module not used in this route.
	AIP	
	PER	
System Definition	SDS	Module not used in this route.
	PAI	
	PSD	
Design	TSD	Use to make corrections.
Build and Test	TPD	Use to make corrections.

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	STT	Module partially used.
	TAT	Business Acceptance Test only (TAT to be performed when needed).
	BAT	
Service Implementation	MMR	Use to close.
	INR	
	TRA	Module includes implementation strategy definition, preparation, execution and closedown phase of the project.
Unlikely	Recommended	Highly Recommended

3.2.12 Controlled Iteration

3.2.12.1 Controlled Iteration (CI) Description

Route "CI" delivers a system characterized by vaguely known, rapidly evolving requirements and rapidly changing design and solution techniques.

Controlled Iteration operates by setting up an executable model of user requirements, based upon and supported by appropriate conceptual data and process models. This executable model is progressively expanded and refined through a series of passes (iterations) until it is shown to meet key user requirements as defined in the Project Charte.

Route "CI" proceeds in four stages; each, however, is used differently in different types of iterations:

- Requirements Analysis and System Definition
  - System Requirements Analysis (SRA) and System Delivery Specification (SDS) are carried out.
- Design
  - Module Technical System Design (TSD) is performed.
- Build and Test
  - Modules Technical Procedure Development (TPD) and a portion of System Testing (STT) and Acceptance Testing (BAT, TAT) are performed, resulting in a Reviewed Prototype.
- Service Implementation
  - Some tasks of the Internal Readiness (INR) module and optionally the Market and Member Readiness (MMR) module are carried out.

3.2.12.2 Controlled Iteration (CI) Considerations

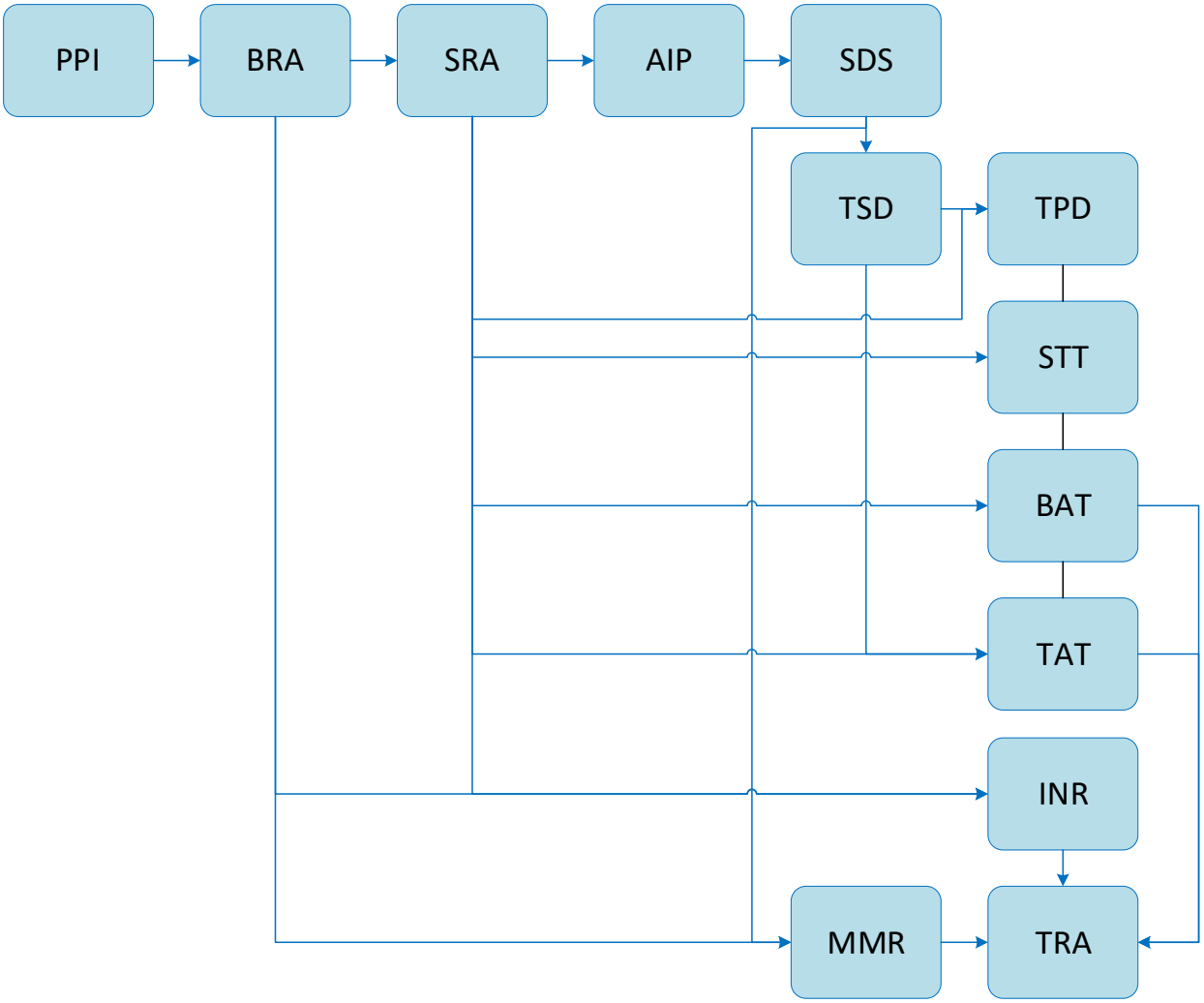
Phase	Module	Task
Initiation	PPI	Use PPI to address the key issues (e.g. scope, approach, ….) of a project.
	BRA	Confirm iterative approach in BRA.

Requirements Analysis	SRA	Confirm iterative approach in SRA.
	AIP	Use once to determine development environment.
	PER	Use once to determine development environment.
System Definition	SDS	Undertake rapid iterations through SDS/TSD/TPD using high productivity tools and omitting activities; reviewing go/no go decision.
	PAI	
	PSD	Module not used in this route.
Design	TSD	Undertake rapid iterations through SDS/TSD/TPD.
Build and Test	TPD	Undertake rapid iterations through SDS/TSD/TPD.
	STT	Run separate System and Acceptance Tests (TAT to be performed if needed).
	TAT	
	BAT	
Service Implementation	MMR	Undertake on final pass only.
	INR	
	TRA	Module includes implementation strategy definition, preparation, execution and closedown phase of the project.
Unlikely	Recommended	Highly Recommended

3.3 Data Flow Diagram

The following diagram depicts the primary data flows between modules. This is not limiting nor exhaustive and data may flow amongst all modules.

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4 Module descriptions

The following sections provide more details about the different modules and include an overview in the form of process model, a RACI matrix, descriptions of work products, key considerations and other elements. Note: the Business Operations and Corporate Functions are only involved when appropriate.

The roles appearing in the RACI matrices are listed here. The role names are indicative and may not correspond to role names used by a particular project or department.

Role list

Role name
Business Operations
Business Operations & Corporate Functions

Role name
Change Management
Change Management (for Core Product Projects)
Change Management (for IFS, GSF and Corp. Center projects)
Corporate Functions
Customer Service
External Suppliers
Group Marketing
Information Owner
Information Technology
IT Development
IT Development & IT Implementation Collaboration Group
IT Functional Analysts
IT Implementation
IT Infrastructure
IT Operations
IT Test
IT Transition Team
Legal
Legal and purchasing
Marketing
Network Management
Product Management
Product Management (for Core Product Projects)
Product Management (for IFS, GSF and Corp. Center projects)
Project management
Quality Management, Controls & Change Management
Relationship Management

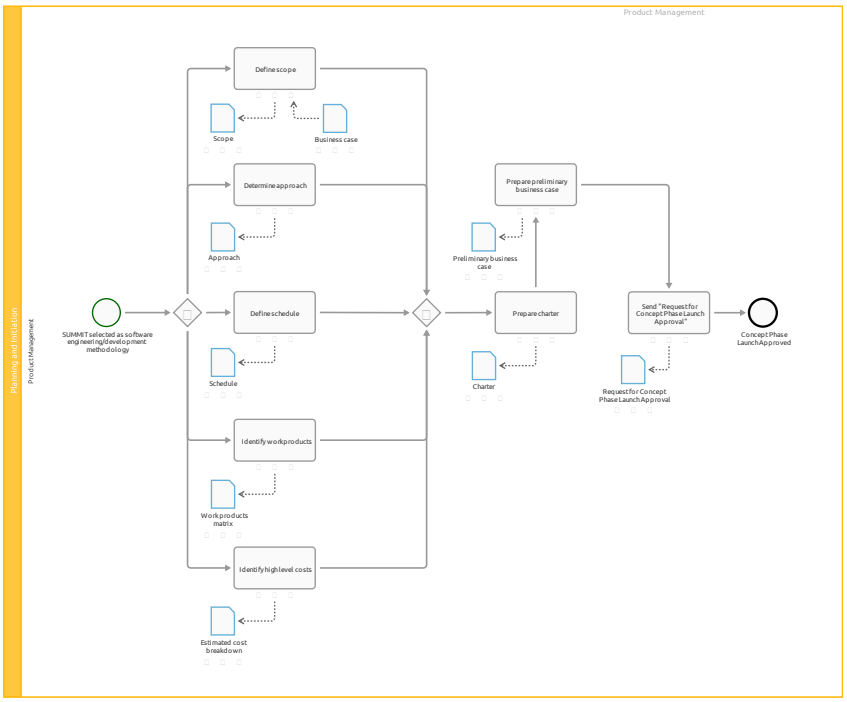
Description of RACI+S

Terminology	Roles and Responsibilities
Accountable	This is the class of people that are accountable to oversee that the work gets done. This usually means the immediate manager overseeing the work.
Consulted	These may be subject matter exerts who need to be consulted at the time of an exception. There is a possibility that am unanticipated scenario arises in a process. These are the people who will do the thinking and suggest any deviations from the Standard Operating Procedure (SOP).
Informed	This is the class of people who have some interest in the performance of a given task. This may be a manager trying to control the execution of the task at hand. Also, this could be an input signal to the other process.
Responsible	This is the class of people who are ultimately responsible for getting the work done. This may refer to the individual workers that perform the given task or it could refer to a system in case the task is automated.
Sign-off	Repsonsible role for providing sign-off on important decisions and select work products.

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4.1 Project Planning and Initiation (PPI)

4.1.1 Module View



1

PPI RACI+S	Product Management	Business Operations & Corporate Functions	Information Technology
Define schedule	A, R	C	C
Identify high level costs	A, R	C	C
Define scope	A, R	C	C
Determine approach	A, R	C	C
Identify workproducts	A, R	C	C
Prepare charter	A, R	C, S	
Prepare preliminary business case	A, R	C, S	
Send "Request for Concept Phase Launch Approval"	A, R		

[1] Due to a rendering bug the gateway type symbols are not displayed. This model contains only parallel gateways.

4.1.2 Module Description

The PPI module is the required starting point, except for non-project based “maintenance”, for all project/development activities, whether identified as an idea or a request from a business user or as the result of a strategic plan. The Project Planning and Initiation (PPI) module addresses the key issues before the project can formally begin, it also provides the input for the request to proceed to the concept phase at the Steering Committee if a steering committee has been established..

Objectives include: definition of targets, scope, approach to be adopted and mobilization, including detailed planning of the initial phase, preparation of the project team, work product matrix and a breakdown of costs.

PPI requires liaison between the project managers, project owner and key business and operational stakeholders who must gather and augment any existing project documentation, for presentation to the appropriate levels of management for final approval and authorization to commit resources. They will be responsible to guide the project through the approval process, dealing with all objections and delays. The PPI module determines the feasibility and alters the project accordingly to manage the risk.

While the BRA module focuses on all product-related considerations (i.e. working out a concept for the product/service to be developed), the PPI module sets up the actual project, embedding it in the appropriate context and identifying all stakeholders, interfaces, dependencies and constraints the project will encounter during its life cycle.

4.1.3 Key Considerations

The considerations undertaken in this module will ensure that resources are employed productively and the project is mobilized on a sound basis. It is important, however, to match the level of detail, and hence effort, to the size and complexity of the assignment, and the needs and standards of the organization. The level of effort required by this module could vary considerably depending upon the complexity, risk, criticality and scale of the project.

For a successful project, genuine commitment must be established at the outset of the project, particularly among the affected Senior Managers. This commitment must be maintained throughout the project’s lifecycle by: continuous involvement of stakeholders, managing expectations, addressing conflict as it arises, listening to and dealing with concerns raised and delivering benefits early and regularly.

A clear statement of goals, objectives, and anticipated benefits will determine:

- Project scope
- Resources required (skill sets and availability)
- Business events that require study
- Appropriate or inappropriate timeframe (market deadlines or competitive threat)
- Potential need for external Service Providers
- External systems or products that require evaluation
- Other systems and planned projects that will have an impact upon the success of the project
- Stakeholders list

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- Information security needs and risks
- High-level impacts on future organization

4.1.4 Activities and Work Products

4.1.4.1 Define Project Scope

- Project Scope
  - The purpose of the Project scope is to understand and document the factors which govern, limit and define the work necessary to satisfy the request.
  - Objectives of the project must be defined to ensure that the project manager and any assigned members of a PPI team have a clear understanding of the goals and rational of the requesting organization. The objectives will guide all subsequent PPI efforts.
- The outline of Project Scope covers:
  - Business objectives
- The Business Owner presents clear business objectives for the project with a comprehensible business justification in the form of a Business Case.
  - Project dependencies
- Identification of any related or concurrent projects or events that could impact the project.
  - Preliminary project context
- The Business Owner briefly presents the company strategy and the positioning of the project within this strategy. The Business Owner describes the links with other projects and its priority in comparison to other initiatives.
  - Organization chart
- The Organization Chart includes all principal staff assigned to the project. This will be refined as the project proceeds through the project approval lifecycle.
  - List of Stakeholders
- The list of areas impacted by the project will help identify the key stakeholders and the composition of the project Committee.
  - Work products mandated by Information Security Framework (Intranet -> Chief Risk Officer -> Information Security Framework & Governance -> IS Framework and Intranet -> CIO/COO -> Group Security -> Information Security Guidelines).

4.1.4.2 Determine Project Approach

Project Approach

- The purpose of this work product is to set the direction for the project . This involves consideration of alternative approaches to satisfy the requirements. Although the work is necessarily at a high level, the project plan requires an idea as to how the project will be structured. The Project Approach must identify the factors that will be used to determine the recommendation of a preliminary "build," "buy," or "reuse/re-engineer" approach.

- The impact and risk of the proposed project approaches are assessed and the related benefits and disadvantages documented to enable management to make an informed decision on progressing further the project. The selection and tailoring of the appropriate SUMMIT route map(s) is done at this stage. All planning assumptions are fully documented prior to approval of the project approach.
- The outline of Project Approach covers:
  - Constraints and assumptions
  - Provisional delivery strategies
  - Impact and risk assessment

4.1.4.3 Define Project Schedule

- Project Schedule

The Project Plan will provide the basis for effective project control during the first phase and therefore has to document project factors in a manner sufficient to establish realistic expectations and will facilitate the commitment of resources from senior levels of management.

- Master Plan

The Master Plan is the high-level plan to be used as a reference for managing the project. It contains the overall planning, monitoring and implementation of a project led by the Project Manager on behalf of the process owner.

This contains a first timeline of the phases, the most important Work Products, the start and completion dates and the assignment of responsibilities.

Wherever possible, key milestones for design, specification and implementation should be listed. However, it must be understood that any milestone denoted in this stage is a preliminary basis for the assessments and analyses to follow in the later stages of the project.

- Detailed Plan for Concept Phase

According to the principle of rolling wave planning, in addition to the master plan a detailed project plan must be developed for the Concept Phase.

The "Concept Phase Plan" describes the required resources and budget for documentation, review and sign-off of the Business Requirements document forming part of the SUMMIT module BRA.

Dependent on circumstances with respect to complexity of effort involved, resource availability and/or time to market, project managers may elect to combine the Concept and Specification phases. Normally this means that the team combines the Business Requirements effort and System Requirements effort within the SRA documentation.

4.1.4.4 Identify Work Products

- Work Products Matrix

A first version of the WP Matrix is defined reflecting the project approach and the main work products to be delivered with planned sign-off dates against each project and their relevant stakeholders (reviewers and signatories).

From regulatory and security perspective projects need to produce/update a set of mandatory work products. Even if the chosen route does not contain the respective module. These work products are:

- System Documentation
- Business/User Guide
- End user Guide
- Business Operation's Guide
- Handbook for Technical Operators
- Installation Guide
- Technical Operations Guide
- Development Guide (optional)
- Software Development Process
- Tool Chain Overview
- Software Component Overview
- Documentation mandated by the Information Security Framework

4.1.4.5 Identify High Level Costs

- Breakdown of the Estimated Costs
- Breakdown of the Estimated Costs for the phase to the level that subsequent reporting will be performed.

Deliverables:

- Project Charter

The Project Charter defines the high-level outline of the project with sufficient detail to clearly identify what the project delivers and the relevant background material. The Project Charter will be used as the basis for the decision to progress the project to the Concept Phase. It will be a consolidation and augmentation of any previous work done to define the project.

The Project Charter includes the following elements:

- Definition of the Scope (inclusions, exclusions and constraints)
- Business Case (financial and business objectives with justification)
- Organization (key project resources)
- List of Stakeholders
- Approach
- Constraints and assumptions
- Delivery strategy
- Risk and Impact Assessment (Self-Assessment)
- Work Schedule

- High level Master Plan
- Detailed plan for Concept Phase
- Work Product Matrix
- Breakdown of Estimated Costs

The level of detail included in the Project Charter should be in accordance with the level of complexity of the project.

The Project Charter will be the baseline document against which the progress, change management issues and viability of the project can be assessed.

- Preliminary Business Case

The Preliminary Business Case details the business justification for the project and must be created in compliance with formal standards of Deutsche Börse Group. The assumptions used in the BC must be explicit and transparent. Quantitative and qualitative benefits are only accepted if a method to trace and capture them over time after delivery of the product change is specified.

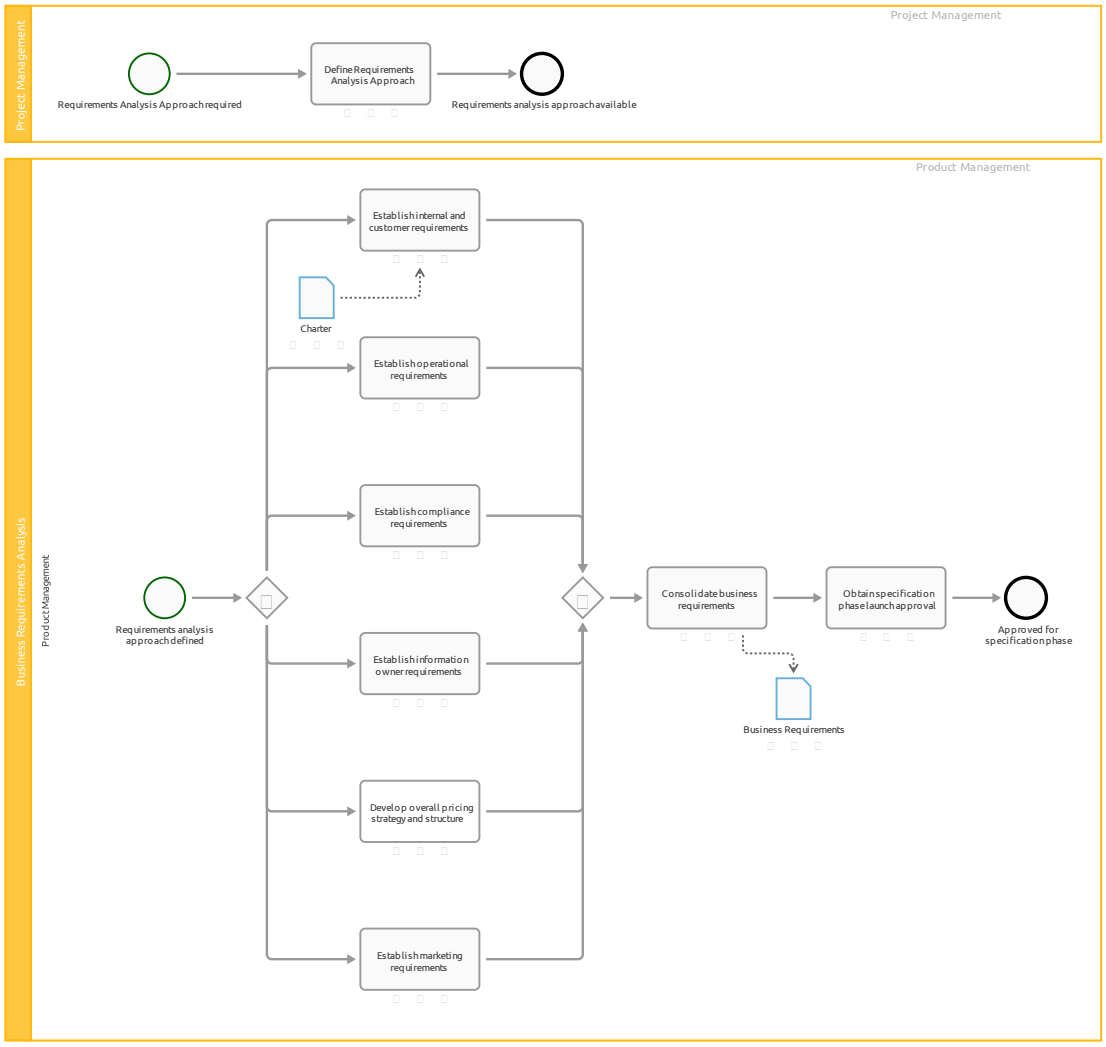
Inputs are the project scope and the high-level costs.

The Business Case will be refined as the project proceeds throughout the project approval lifecycle, particularly when detailed development and testing costs become available.

Internal

4.2 Business Requirements Analysis (BRA)

4.2.1 Module View



[2]

[2] Due to a rendering bug the gateway type symbols are not displayed. This model contains only parallel gateways.

BRA RACI+S	Product Management	Business Operations & Corporate Functions	Information Owner	Marketing	Information Technology	External Suppliers	Project management
Define requirements analysis approoach					C		A, R
Establish internal and customer requirements	A, R	C			C		
Establish operational requirements	A,R	C			C		
Establish compliance requirements	A, R	C			C		
Establish information owner requirements	A, R	C	C		C		
Develop overall pricing strategy and structure	A, R	C		R	C		
Establish marketing requirements	A, R	C		R	C		
Consolidate business requirements	A, R, S	C, S	C, S	C	C	C	
Obtain specicifation phase launch approval	A, R						

4.2.2 Module Description

The Business Requirements Analysis (BRA) module analyses and documents the Business delivery requirements (i.e. functional and non-functional requirements) to be addressed in order to successfully deliver projects/development activities.

This module builds upon the high-level business delivery approach developed during the Project Planning and Initiation (PPI) phase.

Once an in-depth understanding of the requirements is reached, the detailed Business delivery approach can be defined.

This module ensures that the correct level of emphasis is given to up-front preparation work. This is vital before a project moves into the development / implementation phase. The project must start with clear requirements (Business delivery as well as system).

4.2.3 Key Considerations

The project manager has overall responsibility for all Business delivery tasks, and for ensuring that they are performed within the necessary timeframes.

The Project Manager checks that all available information has been gathered from the whole organization, particularly from Customer Relations, Corporate Strategy and Marketing and validate if the scope, objectives and concept have been investigated to a level of detail appropriate for this size of project. The project manager will also ensure that recommendations are well supported by clearly presented arguments and that all inputs from internal and external parties have been provided.

Where deemed appropriate by all parties, the BRA and SRA can be performed in parallel and the respective results may be merged into one work product. This approach may have a positive impact on the timeline since both business requirement and system requirement can be presented in a concise and comprehensive way. Activities and Work Products

4.2.4 Activities and Work Products

4.2.4.1 Establish Internal & Customer Requirements

The focus of this activity should be on establishing the requirements rather than on a definition of a system.

The project manager is responsible for identifying and involving all relevant areas, i.e. Training, HR, Customer Services, and Sales. The relevant experts are responsible for providing the appropriate support.

Communication requirements in this module relate to internal communications. It does not cover marketing communications to prospective customers for the new product, or communications to operational staff regarding a new product.

4.2.4.2 Establish Compliance Requirements

This task addresses key legal aspects for the project manager's consideration.

An investigation should be undertaken to review and investigate requirements for compliance matters and concerns, including preparation and negotiation of new legal documentation, and regulatory requirements, as well as requirements coming from outside.

This task is the responsibility of the Project Manager.

The relevant areas (i.e. Legal, Finance, Regulatory Affairs, Credit, Treasury, and Audit) will provide the necessary support to complete this task.

The project manager is responsible for ensuring that relevant areas have been consulted. Where appropriate these departments are responsible for assigning the suitable specialists to identify requirements and co-ordinate with the project team.

The project manager will identify if significant barriers or constraints exist that will affect the company's ability to take advantage of these opportunities and that require resolution in order for the project to proceed.

4.2.4.3 Establish Information Owner Requirements

This task addresses key aspects of handling information owners' requirements for project manager's consideration.

An investigation should be undertaken to identify the relevant information owner(s) and subsequently identify, investigate and review their potential requirements and concerns regarding internal and external security requirements in the light of new functionality and data processing included in the project scope.

This task is the responsibility of the project manager.

The relevant Information Owner's area will provide the necessary support to complete this task.

The project manager is responsible for ensuring that relevant areas have been consulted. Where appropriate these departments are responsible for assigning the suitable specialists to identify requirements and co-ordinating with the project/development team.

The project manager will identify if significant barriers or constraints exist that will affect the company's ability to take advantage of these opportunities and that require resolution in order for the project to proceed.

4.2.4.4 Develop Overall Pricing Strategy & Structure

The purpose of this task is to develop a pricing strategy and structure for the new/enhanced product or service.

Marketing is responsible for development of a high-level pricing strategy consistent with the product strategy, for development of scenarios for different pricing structures and for the pricing decision.

4.2.4.5 Establish Marketing Requirements

The purpose of this task is to identify all Marketing Requirements for the new/enhanced product. This task builds upon the high-level Business delivery approach developed during the Project Planning and Initiation (PPI).

This task is under the responsibility of the project manager.

Marketing will provide the necessary support to complete this task.

It is likely to require liaison, consultation and co-ordination with customers, Product Development, Sales, Customer Services, Corporate Strategy, Corporate Communications and Public Affairs.

Marketing should be involved in the project team from the very beginning of the product development process and has a critical role to play. Marketing supports the project manager in the development and implementation of marketing plans, and commercialization and roll-out of the product. Marketing is responsible for competitive analysis for product placement / positioning.

Marketing should identify corporate policies affected by this product in terms of branding and communications. They should identify if the organization has already established relationships with suitable external parties.

4.2.4.6 Consolidate Business Requirements Deliverable

Deliverable: Business Requirements

The purpose is to understand the needs of the Business and to identify the strengths and weaknesses of any existing processes, procedures and tasks (automated or manual) that support the user within the scope of the proposed functions.

Following aspects will be considered:

- Business Case
- Business Process Model
- Business Function description
- Input and Output of the Business Process
- Traceability matrix (mapping business needs described in the product description and Business Requirements).

Non-functional requirements and Constraints, e.g.:

- Confidentiality, integrity, availability, authenticity, non-repudiation (aka Information Classification).
- Information security requirements derived from the Information Security Framework as applicable
- Service Levels Requirements (i.e. service quality and business hours)
- Timeline
- Functional and Technical Operability
- Budget
- Strategy (e.g. Group Strategy, IT Strategy)

4.2.5 Quality Keynotes

The Business Requirements address the end-to-end process. Business Requirements contain detailed quantifiable and qualifiable requirements.

Objective criteria are to be established for the acceptance of the requirements:

- Clearly and properly stated
- Complete
- Consistent with each other
- Uniquely identified

Use the recommended BRA template and describe:

- The Detailed Business Project Scope:
  - List business processes in scope.
  - List all assumptions.
  - Provide a high level end-to-end business model.
  - Confirm the end-users that will participate in the BRA.
  - Model the business processes and project context.

The work deliverables within the module are well-defined, i.e. a phase work breakdown structure has been defined. The cost, elapse time and expense estimate are respected.

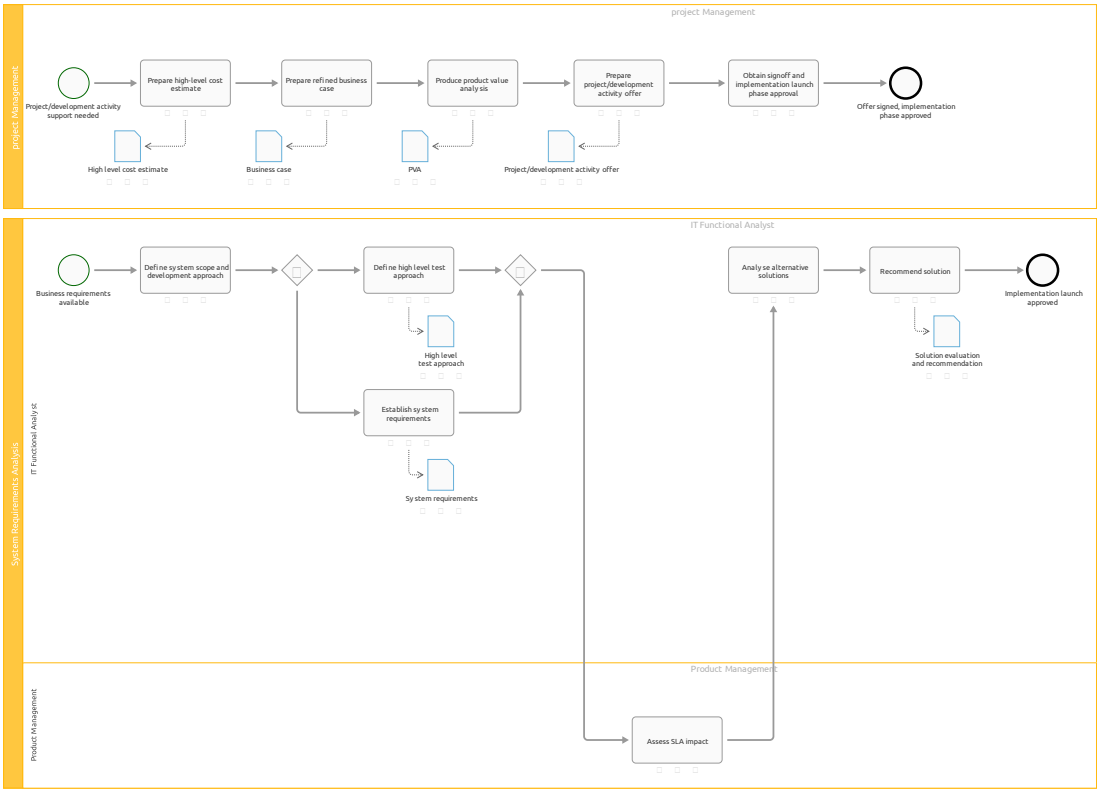
- Detail the User Needs:
  - Gather information about current work flows, procedures and systems used to perform the work (if relevant).
  - Document current business processes and systems used.
  - Refine the new system's business objectives and constraints.
  - Build a high-level current business model (if relevant).
  - Evaluate the current processes/system situation.
  - Provide the rationale for change.

- Document the impacts the new systems might have on other parties other than the project initiator.

- Detail End-to-End Business Model and Requirements:
  - Avoid computerization or re-implementation of the current processes.
  - Use Business data and modelling techniques.
  - Do not distinguish between manual and automated components.
  - Do not dictate how automation or system might be achieved.
- Cover all aspects of the new system to be defined.
  - Marketing Requirements
  - Compliance Requirements
  - Staff and Customer Readiness Requirements
- Overall Pricing Strategy and Structure

4.3 System Requirements Analysis (SRA)

4.3.1 Module View



[3]

[3] Due to a rendering bug the gateway type symbols are not displayed. This model contains only parallel gateways.

SRA RACI+S	Product Management	Business Operations & Corporate Functions	IT Development	IT Functional Analysts	IT Test	External Suppliers	Project management
Define system scope and development approach	C	C	R	A, R			
Define high level test approach	R, S	C, S	R, S	A, R, S	R, S		
Establish system requirements	C, S	C, S	R, S	A, R, S		C	
Assess SLA impact	A, R	C	R	R			
Analyse alternative solutions			R	A, R		R	
Recommend solution			R	A, R		R	
Prepare high-level cost estimate							A, R
Prepare refined bsuiness case							A, R
Produce product value analysis							A, R
Prepare project/development activity offer							A, R
Obtain signoff and implementation phase launch approval							A, R

4.3.2 Module Description

The System Requirements Analysis (SRA) module analyses and refines business requirements (BR). Considering the business requirements, IT-Strategy and Guidelines the SRA defines a comprehensive and concise set of System Requirements.

Once an adequate understanding of the requirements is reached, alternative delivery approaches can be identified and evaluated. Alternatives may include "build or buy" or differing extents to which processes and data should be automated. The impact of alternative solutions, in terms of effort, cost and risks incurred are assessed in order to come up with a profound system recommendation.

Project Manager and Solution Architect establish an initial version of the high-level blueprint, serving architectural discussion and to support the system recommendation. The blueprint covers business, application, data and technology to an extent necessary for the recommendation. Therefore, the blueprint utilizes a number of architecture viewpoints highlighting the different aspects of the system alternatives e.g.

System Independent Viewpoints

- Business Process Flow (the sequence of functions required to produce outcome e.g. using a BPL)
- Functional Model (a logical grouping of business functions)
- Information model and logical data model (a description of input, output and intermediate business objects using the business domain’s language)
- Information Flow, Roles and organizations (involved organization and roles executed by those organizations)
- Business Service Model (Logical grouping of the interfaces along functions and their respective security context)

Viewpoints for design alternatives

- Application Interaction Model (impacted applications, services and their dependencies in terms of functions)
- Data Flow Diagrams (logical data flow between the involved applications/application service)
- Technology (proposed technology stack for the envisaged system)

4.3.3 Key Considerations

Analysts should focus on the future business functions and process and avoid computerizing or re-implementing the current system, which may have developed over a long period and include facilities intended to fulfil requirements that no longer exist.

It is recommended to lead this module in a collaborative mode, setting up a joint team involving specialists from the Business and IT areas.

The information security risks must be assessed based on the information security framework.

The System Requirements present a high-level end-to-end view of the areas and systems impacted by the proposed system or alternatives (e.g. main systems, interfaces, offshore systems, local applications, etc).

They are derived from the functional and non-functional Business Requirements and Constraints.

- Functional System Requirements are based on a detailed analysis of the business functions to be performed and the level of automation required. Hence a description of both the required functions and the scope of the implementation (System Requirements) must be provided.
- Architecture requirements identify a set of quality criteria to be met by the system in addition to the functions to be performed. These include but are not limited to information security, performance, availability, usability, maintainability, operability, etc. Non-functional requirements are the drivers of the architecture.
- Constraints describe the limitation or external factors the system **must** fulfil. Constraints may be derived from business needs, budget or market needs and are also defined by company or IT strategy, thus impacting both project and/or proposed system. While requirements can be prioritized or descoped, constraints cannot be descoped but have to be fulfilled by the envisaged system. Constraints are often non-functional.e.g. recovery time objective of 2 hours enforced by regulation, SWIFT message according ISO20022, a certain technology stack as required by the IT-Strategy, or an external defined production date set by national or international law.

Requirements and constraints have to be reviewed with all parties. The implementation scope must be mutually agreed on with all involved parties in order to prevent misunderstandings. They are input to the project plan for the subsequent phases.

Descoped system requirements must be communicated explicitly to sponsors and stakeholders. Business Operations or Customers will have to compensate the lack of automation.

It is important to capture requirement changes and rationale during the process as well as to maintain bi-directional traceability among business requirements and system requirement.

While business requirements describe the needs of the business with respect to function and quality the system requirements form the basis of the System Specification. Regulators require System Requirements must be traced back to business requirements or constraints.

If a system based on the use of third party application software products is deemed feasible the Product Evaluation and Recommendation (PER) module may be initiated to identify and evaluate available products.

In order to allow for tangible bottom-up estimating and productivity measurement, Fast Function Point Analysis (FFPA) may be applied. This effort will be performed by Functional Analysts, and monitored and supported by the stakeholders.

4.3.4 Activities and Work Products

4.3.4.1 Define System Scope and Potential Development Approach

The purpose is to detail the functional scope and to identify and document Business processes impacted:

- Overall System Scope:
  - Key principles, assumptions, constraints and dependencies.
  - High-level view of the Business processes.
  - High-level system blueprint, outlining key architecture aspects.
- Detailed scope definition:
  - Processes and systems impacted.
  - Business inputs, outputs and interfaces to other Business functions.
  - Business events / transactions the system must process.
  - Processes and systems out of scope.
  - Information Security specifics based on Information Security Framework.

4.3.4.2 Define High Level Test Approach

- High Level Test Approach
  - Identify high level test plans
  - Identify test activities (STT, TAT, BAT, Regression, external, ….)
  - Identify high level resources

4.3.4.3 Establish System Requirements

- System Requirements
  - System Requirements consolidate the definition of the Business Requirements (functional and non-functional) within the scope of the proposed system and detail the end-to-end design to meet the Business Requirements considering all design constraints.

- An exercise is carried out in order to assess the feasibility of the Business idea in general and to obtain an indication of the possible timeline. The cost associated with implementing and operating the new product is also assessed at a high level.
- Traceability matrix (mapping Business Requirements and System Requirements).
- Conceptual process model and data model.
- The System Requirements are related to testable scenarios.

4.3.4.4 Assess SLA Impact

The project managers will identify existing internal SLA that will be impacted by the new system and will identify possible impacts (e.g. response time, volumes, timeline...).

4.3.4.5 Analyse Alternatives

The purpose is to identify alternatives capable of meeting the requirements as defined and to involve the users in the evaluation. In many application areas the use of software products or "package" solutions available from third parties is an alternative that should be considered. Other alternatives are likely to represent differing degrees of automation and different environments.

The work product describes why the selection is the most appropriate solution to the requirements at a sufficient level to support a commitment to the next stage of system delivery. Strategies for delivering and installing the new system are completed.

Content consists of:

- Business and technical constraints
- Project delivery constraints
- Product description
- Alternatives
- Evaluated alternatives and recommendation
- Rationale for the selection
- Overview
- Constraints
- Delivery and transitions strategies

4.3.4.6 Recommend Option

Evaluation and Recommendation

Deliverables:

- System Requirements
- High Level Test Approach
- Documentation required by the Information Security Framework
- Impact Check

4.3.5 Quality Keynotes

The System Requirements contain detailed quantifiable and qualifiable as well as traceable requirements.

Analysts should respond to intermediate findings and confirm them. They should identify proposals to adjust the scope of SRA where necessary.

An appropriate range of design alternatives should be selected for evaluation (when third party application software products are considered).

All deliverables should be agreed adequately with users. User understanding and commitment should be obtained.

Sufficient attention should be paid to reviewing the expected costs, benefits and risks of the system.

A thorough re-planning exercise should be undertaken. The route map should be reviewed.

Plans for phasing delivery and/or transition must be coordinated with other projects/development activities. Practical conversion issues must be addressed.

The owners of the different tasks are responsible and accountable for the rationale used, decision taking and completeness of the documents to be produced. The documents are deemed by all parties (Users, Product Management and IT) to be clear, unambiguous and well defined enough to be handed over for the next deliverable work.

Clarifications and decisions on scope and user needs/requirements as well as alignment with third parties for common requirements must be agreed adequately and promptly in order to gain understanding and commitment from all parties involved.

The owner of the process also ensures that:

- The work plans and schedules are realistic, well-defined and adhered to. Schedules/plans are established by activity and team members.
- All activities necessary for the accomplishment of the deliverables are assigned to the team members that have the appropriate qualifications to deliver recommended work, ensuring the right skills set and techniques are used.

End-users, business, technical and application experts have been appropriately used, i.e. roles and responsibilities within the deliverable have been determined and a team structure is in place.

Use the recommended SRA template and describe:

- Refined IT Project Scope, by:
  - Reviewing consistency and completeness of E2E Business model.
  - Tracking changes in scope.
  - Confirming key principles, dependencies, assumptions and constraints.
  - Evaluating current processes and systems and confirming high-level model and assessment of the currents processes and systems.
  - Elaborating on the rationale for change.

- Providing a high-level view of the future Business processes.
- Detailing anything that is "out of scope".

- User Needs and E2E Business model, by:
  - Describing context and end-to-end conceptual models.
  - Defining user interfaces, dialogues, outputs and preliminary prototypes.
- Functional Fine Specifications, by:
  - Reviewing sub-process/system partitioning.
  - Specifying process or product usage.
  - Specifying external interfaces.
  - Describing automated and manual processes.

Consolidating functional requirements related to Business processes (access management, user interfaces, outputs required, and validation rules.

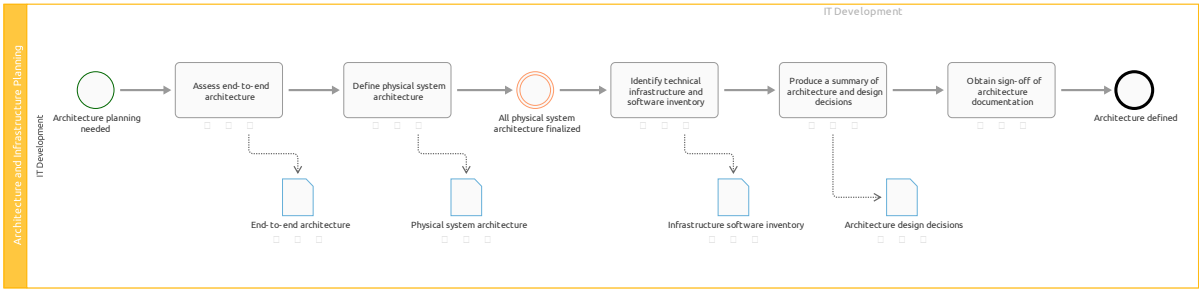
Providing non-functional requirements as per SRA template (Interoperability, cutover, data conversion, operational requirements, audit, data retention, archiving).

- Traceability between business requirements and system specifications.
- Cross system impacts (where relevant).

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4.4 Architecture and Infrastructure Planning (AIP)

4.4.1 Module View



AIP RACI+S	IT Development	IT Infrastructure	IT Implementation	IT Operations
Assess end-to-end architecture	A, R, S	C	C	C
Define physical system architecture	A, R, S	C	C	C
Identify technical infrastructure and software inventory	A, R, S	C	C	C
Produce a summary of architecture and design decisions	A, R	S	S	S
Provide workproducts for integration into offer	A, R			

4.4.2 Module Description

4.4.2.1 Overview

The AIP module determines and documents architecture from the end-to-end processes and cross systems aspects, defines the physical system architecture, interface specifications and identifies the technical infrastructure and technical inventory. The AIP defines the architecture and infrastructure for the system to be delivered in production, testing and development environments. The document describes the software and hardware requirements and deployment configuration required to integrate the proposed system into the existing production system landscape.

Therefore, the AIP is necessary if there are architectural modifications. Consequently, functional changes with no impact on the architecture do not require the use of the AIP module. In this case, impacts should be identified in the SDS.

The overall goal of the AIP is to specify the system. Sometimes, a conceptual system is outlined as the result of a conceptual phase before the AIP is produced.

In order to streamline the AIP process, tasks are divided and apportioned appropriately. The process starts with a global view looking at the End-to-End architecture (E2E), the goal of which is to identify new or impacted components and to distribute responsibilities for these components to a level where work can be performed concurrently and independently of each other.

The End-to-End architecture focuses on cross system aspects and finishes when all cross system impacts and all the system interfaces have been identified. Due to its cross-system nature the E2E architecture must use system independent languages and concepts.

Either sequentially or concurrently with the E2E architecture, the Physical System Architecture (PSA) is defined. The different physical system architectures detail the architecture of new systems and changes on existing systems. A PSA expands the processes described in the E2E and looks at subsystems of a given system. While the E2E architecture focuses on the overall structure of the system and its building blocks, the PSA focuses on internals of those building blocks.

The interface agreements finalize the details of the interfaces identified in the E2E architecture on a bilateral basis. At this stage, system interfaces resemble programming interfaces and the agreement specifies which data is passed to the interfaces. Note that the interface identification process is part of the E2E architecture and the AIP, the interface detail specification is part of the TSD phase.

Finally, the technical infrastructure and software inventory (TISI) is defined. The TISI phase is the final phase where global decisions impacting all systems are taken and documented. Typical decisions taken here are technology decisions e.g. middleware, operating systems, data bases etc. The corresponding deployment constraints are documented. This activity is started when all PSA and all agreements have been finalized. It concludes the AIP process.

4.4.2.2 Process Analysis

The proposed AIP process is based on the notion of a system and requires a formally agreed definition of what a system is. For the time being, we assume that a system can be decomposed into subsystems allowing a streamlined approach of the architecture.

The AIP requirement is driven because key decisions and choices must be traced back to requirements and when this includes requirements for other or future projects they must be made explicit. The requirements include functional, non-functional and technical requirements. The AIP can only be finalized once the SRA(s) is finalized.

- The AIP is architecture centric in the sense that the solution of any recurrent problems should be reused as much as possible. These solutions are materialized in architecture guidelines which guide the choice from the set of alternatives fulfilling the requirements. Any deviations from the guidelines should be made explicit and rationalized.
- The AIP is solution oriented. The requirements state the needs or conditions to be met and the architecture describes the solution to the requierements. The AIP should analyse how well the proposed approach will fulfil non-functional requirements such as performance characteristics
- The AIP is largely process focused. It focuses on processes and control flows rather than data. The analysis of the proposed system in terms of its ability, scalability, maintainability, etc. focuses on the process specific aspects. The interface agreement provides the link between the processes and the data aspects.
- The process is adaptable. The phases to be executed and which deliverables will be produced can vary depending on project scope, complexity, criticality etc.

It is recommended to have in an iterative approach to sign-off and approval of the AIP deliverables, and have weekly meetings with the stakeholders to keep them up-to-date with the key decisions which are being discussed.

Signed-off of the Architecture deliverables are maintained within projects/development activities.

The architecture must therefore be recovered cross- project and the part of the "current" architecture impacted by a project must be updated or adapted by the originating project .

4.4.2.3 Systems

Preparation of the AIP requires knowledge of which entities are considered to be systems.

Systems are run time entities ("processes") as opposed to applications which are more development entities ("pieces of code"). Systems usually have (but not always) the following characteristics:

- A system usually implements a subset of business processes which are largely independent of other business processes.
- A system is usually the source or the "master" of some data and has its own data base or data base schema.
- A system is usually a set of technical (Unix, Mainframe) processes running on the same run time technology or middleware and built out of a component developed in a single programming language. The noticeable exception is the case where front end (GUI) and back end are developed on two different technologies.

In absence of a comprehensive representation of the concept of a system, an explicit list of systems, such as the E2E architecture diagram can be produced.

4.4.3 Key considerations

It is important that the architect consult with the broader architecture community and review available architecture guidance before producing the architecture relevant work products.

4.4.4 Activities and Work Products

4.4.4.1 Assess End-to-End Architecture

End-to-End Architecture

- The E2E architecture focuses on E2E processes and cross systems aspects. It defines which pieces of the E2E processes will be implemented by which systems and identifies the system interfaces.
- The processes must be described in a business domain language independent of the way pieces of these processes are expressed, implemented or executed in the different systems.
- The description must be detailed enough to ensure that all cross system (cross technology), cross team and cross process aspects are covered and all system interfaces are identified. In order to ensure a comprehensive and concise description, the process model needs to reach an adequate level of detail, e.g. down to logical unit of work. However, a complete reverse engineering of all processes embedded in all systems during AIP/SRA is not feasible. To overcome this problem, the approach proposed is to model the main business scenario.
- The same problem appears with the breadth of the E2E processes description or when trying to answer the question "where does an E2E process start and where does it end". The answer is simple, it starts and ends with user interactions. However, for the sake of the architecture design the interaction will end at a systems boundary at the edge of the current project scope.
- For the sake of abstraction, the E2E processes should be less detailed and expanded than those found in the Physical System Architecture (PSA). The general rule is "as

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detailed as needed" to allow the different teams to be able to work independently of each other on the different systems.

- The E2E architecture provides an analysis of the ability of the system to fulfil the non-functional requirements (ities). A set of process properties has been selected and should be discussed for each system. These are:
  - Maintainability and Testability: ease of changing the solution and ease of testing it.
  - Performance: ability to meet the process execution time constraints and monitoring.
  - Scalability: ability to increase performance without changing the code.
  - Recovery and contingency: ability to recover from a process failure, monitoring and detection of such failure and ability to execute alternative processes.
  - Availability: ability to provide a given service over time.
  - Security
  - Data Base Archiving.

4.4.4.2 Define Physical System Architecture

Physical System Architecture

The first role of the Physical System Architecture is to detail and expand pieces of E2E processes allocated to a given system in the E2E architecture. It details the E2E architecture and its main specificity is that it is usually performed by a single team, the team responsible for the system at hand. Also, a single system at the E2E level is sometimes considered internally as two subsystems maintained by two smaller teams. In this case the role of the PSA is an "E2E" architecture for these subsystems. In any case, for the PSA, the processes should now be detailed at least down to the impacted logical unit of work.

- The specificity of the PSA is that it maps the functionality into deployable unit (e.g. process, service). This consists of the following:
  - Specifying how cross Logical Unit of Work (LUW) processes are implemented (i.e. UC4, point to point etc.)
  - Specifying the mapping between the physical processes (or light weight processes) and the LUW of the system. (ex. N-tier or single process architecture, pipeline architecture etc.).
  - Optionally identifying the main components and specifying how they interact with each other to implement the functionality of the different LUW provided by the system (e.g. component and interaction diagrams).
- Hardware is usually dedicated to the system in the production environment. The PSA will contain the sizing of new hardware when substantial changes are foreseen at this level. The Technical Infrastructure and Software Inventory (TISI) will indicate if new third-party software or tools must be installed.

4.4.4.3 Identify Technical Infrastructure and Software Inventory

Infrastructure Software Inventory

The Technical Infrastructure and Software Inventory (TISI) are the last global - cross system phases. This is where the choice of the versions for the main third-party products, middleware and databases, is specified. The compatibility between the different versions is checked and the deployment architecture is drawn. This consists of regrouping the different processes from the different systems into the UNIX domain or servers foreseen in production. The deployment constraints should also be identified so that they can be checked in the different testing environments.

4.4.4.4 Produce a Summary of Architecture and Design Decisions

Architecture and Design Decisions

The purpose of this deliverable is to describe in non-technical terms the decisions in terms of architecture or infrastructure made by IT (Application development, Architecture Team, ...). The document is a summary of the AIP work product for the Business teams and Business manager.

Deliverables:

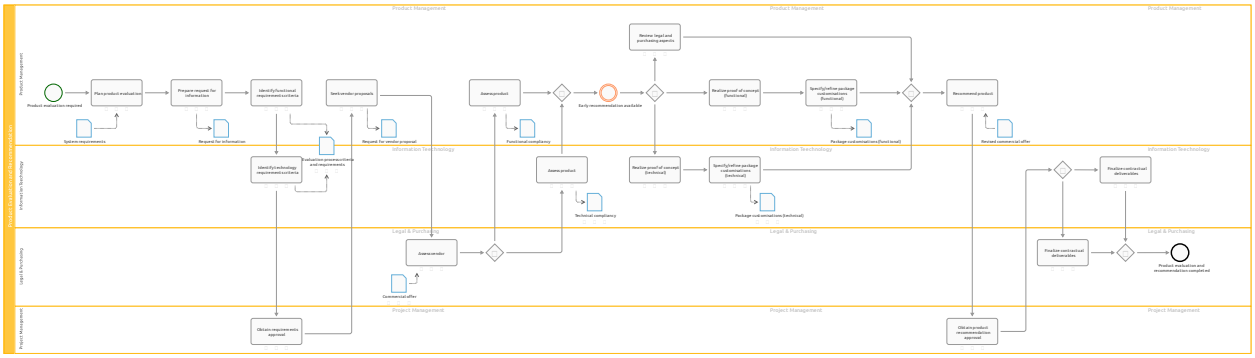
- End-to-End Architecture
- Physical System Architecture
- Infrastructure Software Inventory

4.4.4.5 Quality keynotes

Use of agreed architecture templates and sign-off on final architecture planning.

4.5 Product Evaluation and Recommendation (PER)

4.5.1 Module View



[4]

[4] Due to a rendering bug the gateway type symbols are not displayed. This model contains only parallel gateways.

PER RACI+S	Product Management	Business Operations & Corporate Functions	Information Technology	External Suppliers	Legal and purchasing	Project Management
Plan product evaluation	A, R	C	C			
Prepare request for information	A, R	C	C			
Identify functional requirements criteria	A, R	C	C			
Identify technology requirements criteria	C	C	A, R			
Obtain requirements approval						A, R
Seek vendor proposals	A, R	C	C	C		
Assess vendors	R	R	R	C	A, R	
Assess product	A, R	C	A, R	C		
Review legal and purchasing aspects	A, R				R	
Realize proof of concept (technical)	C	C	A, R	C		
Realize proof of concept (functional)	A, R	C	C	C		
Specify/refine package customisations (technical)	C		A, R	C		
Specify/refine package customisations (functional)	A, R		C	C		
Recommend product	A, R	C	R		C	
Obtain product recommendation approval						A, R
Finalize contractual deliverables	C		A, R		A, R	

4.5.2 Module Description

4.5.2.1 Overview

The purpose of the Product Evaluation and Recommendation (PER) module is to evaluate and recommend products and vendors that best meet the users' needs. The delivery of the proposed system uses a structured process to maximize the chance of success.

The recommended product(s) will form part of the overall recommendation in Business and System Requirements.

"Product" in this context refers to products which can be provided by a third-party supplier.

Several different categories of product are addressed by this module:

- **Application products** which provide all or part of the application system itself (normally software).
- **Technology products** which form part of the technical environment necessary to either operate or deliver the required system:
  - Operational products are those which are required to actually run the system in the production environment or data management.
  - Delivery products are those required during the delivery project itself, out-of-the box applications, applications outsourced to external parties, prototyping tools, and workbenches.

4.5.2.2 Prerequisites and Related Efforts

The necessary inputs for the product evaluation exercise are:

- Product descriptions: The product descriptions are brief descriptions of all the vendor products being considered for evaluation and are particularly useful for explaining the product, and the differences between them, to the user.
- Approved requirements definition (for an application product): The requirements definition consists of the business and functional requirements to be supported by the proposed product.
- Technology product requirements packet (for a technology product): The "packet" consists of a brief description of the required product, together with (if appropriate) the most comprehensive available definition of the proposed system. Note that in some cases involving delivery technology, this system definition is not necessary. It is necessary, however, to include in some form a definition of the likely volumes of transactions and data to be handled by the product and the number of users to be supported.

4.5.3 Key Considerations

The PER module is structured to address a single product. It is envisaged that the module may be executed several times, perhaps at different stages of the project, but always after a requirement has been identified in Business Requirements.

The PER module should be initiated after the Business Requirements or System Requirements including security requirements for application products are available. For technology products it may be alternatively initiated during the Architecture and Infrastructure Planning module.

In defining the product requirements, it is essential to consider the following:

- Functional requirements (partially extracted from the approved requirements definition from the Business Requirements, System Requirements including security requirements).
- Non-functional requirements: volumes of data and/or transactions to be handled by the relevant system component and the required performance and security features. For some technology products, it is critical that the number of environments needed both from the development and the operational viewpoint are considered (partially extracted from the System Requirements).
- Production requirements: requirements from production on how to fit the desired system into the existing landscape in a way that the existing service levels and security requirements are maintained, and that unnecessary cost can be avoided.
- Vendor evaluation (cost, schedules, services and vision).

The PER module should be led in accordance with the Corporate Purchasing Process.

4.5.4 Activities and Work Products

4.5.4.1 Plan Product Evaluation

Agree on the scope of the package (functional and non-functional requirements).

Agree on the evaluation process.

4.5.4.2 Prepare Request for Information

Request For Information

A request for information (RFI) is a standard business process of which the purpose is to collect written information about the capabilities of various suppliers. Normally it follows a format that can be used for comparative purposes. An RFI is primarily used to gather information to help decide on what steps to take next.

4.5.4.3 Identify Functional Requirements Criteria

4.5.4.4 Identify Technology Requirements Criteria

- Evaluation process criteria and requirements
- This work product is used as a starting point for the selection of the appropriate product. It contains the following:
  - The reason why vendors are being considered, referring to any internal documentation that supports the decision.
  - Business and / or product requirements (or technical requirements, if a technical product is to be selected) in scope.
  - Requirements regarding the vendor selection.
  - Definition of the evaluation process and criteria.
  - List of products which can meet the requirements in scope. The vendors providing these products will receive a request for a proposal.

4.5.4.5 Seek Vendor Proposals

Request for vendor Proposal

- This work product will be sent to the pre-selected vendors to request the necessary information to make a final choice.
- Describe the background of the request: business context, expected vendor involvement, deliverables to be provided by the vendor, schedule of events, reference terms and conditions.
- Provide a summary of the project outline, current system, core processes, system requirements and constraints.
- Describe the functional requirements applicable and the requirements regarding vendor involvement.
- Request the necessary product descriptions and references from the vendor.
- Present the applicable evaluation criteria and test scenarios.

4.5.4.6 Assess Vendor

Commercial Offer is input for the activity.

4.5.4.7 Assess Product

Ensure compliance with technical architecture and infrastructure.

Document what the gaps are (functional and technical).

Possible commitment to fill in the gaps (also reported in traceability matrix).

- Technical Compliancy
- Functional Compliancy
- Security Compliancy

4.5.4.8 Review Legal and Purchasing Aspects

- License Model
- License Agreement
- Statement of Work
- Service Level Agreement
- Maintenance Agreement
- Company viability
- Company skills
- Company references
- Company capacity to deliver
- Cost projection

4.5.4.9 Realize Proof of Concept

Proof of concept is a short and/or incomplete realization of a certain functionality to demonstrate its feasibility, or a demonstration in principle, whose purpose it is to verify that some concept or theory is probably capable of exploitation in a useful manner.

4.5.4.10 Specify/Refine Package Customization

- Package Customization (Technical Aspects)
- Package Customization (Functional Aspects)

4.5.4.11 Recommend Product

- Product and Vendor Assessment and Evaluation Report

4.5.4.12 Finalize Contractual Deliverables

- Contractual Deliverables Finalized

Finalize documents such as License Agreement, Maintenance Agreement, Statement of Work, Service Level Agreement, …

Deliverables:

Product Recommendation

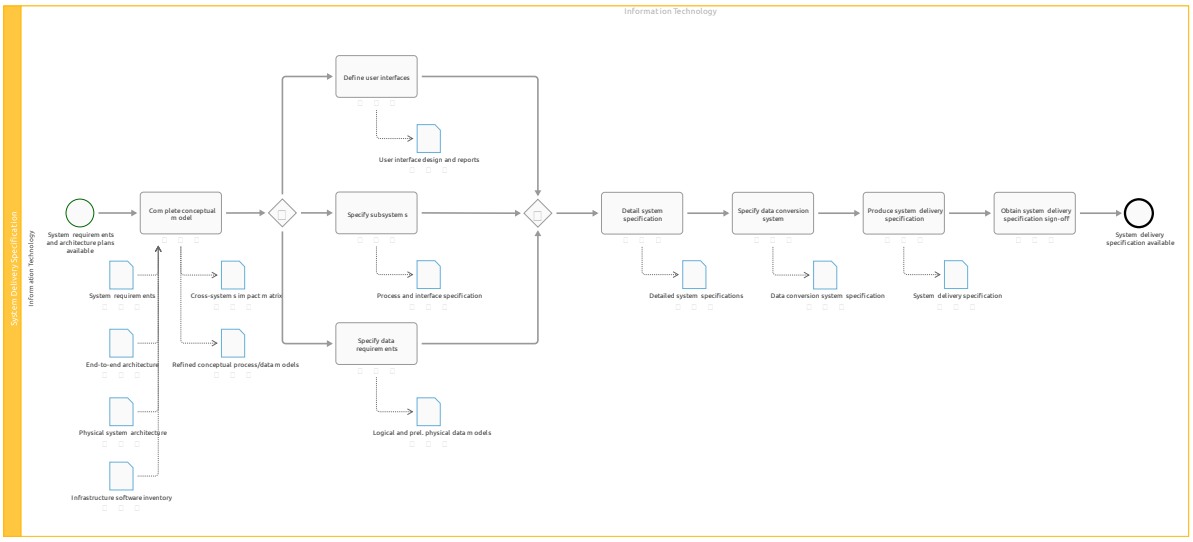
- Once the information from all vendors has been evaluated against the defined criteria, the purpose is to recommend a vendor and a product and to highlight the impact of this decision.
- Consolidate and compare the differences between the proposals.
- Recommend a vendor.
- Document the key milestones, costs and schedules agreed with the chosen vendor.
- Identify what impact the choice made has on the current project plans.
- The Product Recommendation Report is created and reviewed according to the evaluation process agreed within this module. The parties involved will depend on the nature of the product.
- Product and Vendor Assessment and Evaluation Report.
- There shall be 2 main sections in the document: one related to Functional Requirements and another one related to System Requirements.
- The purpose is to evaluate the responses received by each vendor according to the requirements and criteria identified in the "Request for Vendor Proposal" work product.
- Indicate how every requirement is met by the product (functional and technical), vendor services, costs and schedules.
- Draw a short list of products selected, including costs.
- In order to refine this list tasks 6 "Assess vendor" and 7 "Assess Product" can be performed in different iterations.

4.5.5 Quality Keynotes

- The PER module should be called at the right time: late enough for requirements to be defined and early enough to provide cost estimations for the business case and the business case update.
- Where technology products are to be selected, perhaps the selection needs to be carried out in a wider context than the current project.
- Products and vendors should be qualified objectively and realistically (and in accordance with constraints set by the user) during short-listing.
- The vendor relationships are to be handled satisfactorily, e.g.:
  - Have all vendors understood the requirements as described in the "Request for Vendor Proposal" work product?
  - Has a record been kept of all meetings with vendors and of the problems raised by them at each stage of the evaluation?
  - Are similar facilities extended to each vendor?
  - Has impartiality been maintained?
  - Has confidentiality been maintained?

4.6 System Delivery Specification (SDS)

4.6.1 Module View



[5]

SDS RACI+S	Product Management	Business Operations	Corporate Functions	Information Technology	External Suppliers
Complete conceptual model				A, R	
Define user interfaces				A, R	
Specify subsystems				A, R	
Speicy data requirements				A, R	
Detail system specification				A, R	
Specify data conversion system				A, R	
Produce system delivery specificiation				A, R	
Obtain system delivery specification sign-off	S	S	S	A, R, S	

4.6.2 Module Description

Overview

The System Delivery Specification (SDS) module expands upon the System Requirements documented in the SRA module to produce the definitive specification on which all subsequent functional design and build activity will be based. The emphasis of this module is on defining the system from a user perspective while still respecting the Business and security requirements. Technical limitation should be considered but should not restrict creative problem solving. Technical implementation aspects will be addressed at a later stage.

The SDS includes information on functions, user interfaces, manual processes, automated processes, data structures, security, control, back-up and contingency requirements, as well as relevant error handling procedures. It is further expanded into a full delivery specification with the

[5] Due to a rendering bug the gateway type symbols are not displayed. This model contains only parallel gateways.

addition of data conversion specifications, delivery, test and transition plans and Acceptance Test Strategy.

Prerequisites and Related Efforts

Prerequisites for this module are the Business Requirements, and System Requirements (including non-functional requirements) and the specifications of any products - particularly application products - to be employed in delivery of the system.

4.6.3 Key Considerations

The successful acceptance of the system depends on user commitment to providing a sufficient level of input. Any prototyping or other tools and techniques to be used during this module need to be installed and fully available before work commences.

The project managers are responsible for the consistency of all work products, whether focused on functionality or technology. The module is a collaborative effort involving specialists from relevant Business and IT areas.

IT testing is based on the SDS to ensure that the delivered system meets the requirements. Detailed functional testing is done (strong functional skills required) during the Product and System Test Module, before handling over the system for Acceptance Testing.

There should be a continuity of resources throughout the requirements, specification and testing stages. The resources involved in the requirements process should carefully review the SDS in order to quickly identify and rectify differences between Business intent and IT interpretation.

If evaluation and selection of tools or 3rd party applications are required, the Product Evaluation and Recommendation (PER) module must be used. The Product Acquisition and Installation (PAI) module provides guidelines for ensuring that these tools or applications are properly installed. If new techniques are to be used it is essential that sufficient time and effort is allocated for staff training and familiarization.

In order to allow for tangible bottom-up estimating and productivity measurement, Fast Function Point Analysis (FFPA) may be applied to functional requirements. This effort will be performed by Functional Analysts, and monitored and supported by the stakeholders.

4.6.4 Activities and Work Products

4.6.4.1 Complete Conceptual Model

- Cross-Systems Impact Matrix
  - The work product is to list all systems that are potentially impacted by the target system/application/product [e.g. main systems, interfaces, offshore systems, local applications etc.].
- Refined Conceptual Process/Data Models
  - The work product is to complete the conceptual models (both process and data models) started in the SRA and to ensure that these models are consistent (both internally and with each other). It is composed of and includes:
  - Data activities needed to support the project.
  - Business rules as reflected in the data relationships.

- Entity identifiers which distinguish one data entity occurrence from another.

- Update of the documentation of the data available for conversion.

4.6.4.2 Define User Interfaces

- User Interface Design and Reports
  - To specify the interfaces between system and user, in terms of screen formats, dialogues, report layouts, and input forms (if required).
- The emphasis is on the user requirement rather than technical definitions, although the limitations imposed by the technical environment should be recognized (where they are known):
  - Screen and report formats.
  - Layouts for input forms (if required).
  - Design standards with respect to error handling, menu structures, recovery and overall design philosophy.
  - The preparation of prototypes of these interfaces is assumed; these prototypes may range from rudimentary pencil-and-paper representations to interactive screen-based simulations. Depending on the prototyping tools used the prototypes themselves may also be valuable at a later stage, either as a basis for development or as part of the actual delivered system.

4.6.4.3 Specify Subsystems

- Process and Interface Specification
- This work product is to specify the processes of the system which may include:
  - A detailed description of the functions to be implemented by the system, including the description of algorithms, processing rules, decision criteria, etc.
  - The aggregation of such functions into groups building a foundation of system design.
  - The partitioning of the system into subsystems (if appropriate).
  - The way in which products (particularly application software) are used within the system.
  - The interfaces with other systems and between subsystems (data to be transferred, the way in which the interface will operate, triggering events).
  - Detailed definitions of the automated and manual processes which make up the system.
  - Each function/group/subsystem will have its own complete specification to allow it to be developed by separate individuals or groups within the team if required.

4.6.4.4 Specify Data Requirements

- Logical and Preliminary Data Models
- They define the data structures and contents, which form a basis for the system specification. This deliverable is made up of two parts:

- The Logical Data Model focusing on data object and its relations.
- The Physical Data Model primarily addressing non-functional requirements such as performance, maintainability, or compliance with IT-Strategy.

4.6.4.5 Detail System Specification

- Detailed System Specification
- To complete the specification of the system by adding facilities such as:
  - Processing algorithms: define the functional prescription of how data entered by the users will have to be processed.
  - Business transaction use cases: define the functionality of algorithms to be processed according to certain triggers (time of the day, existence of certain data in certain placed, etc.).
  - Performance: define the number of records which have to be processed in a given time interval according to the functional algorithms defined.
  - Back-up and contingency procedures: examine the impact on the business of partial or complete system failure or extended downtime and to identify, where possible, appropriate preventive or contingency measures. Define the data back-up requirements for the new system.
  - Error handling: as these facilities are essential parts of the system they may already be defined. The aim of this task is to ensure that nothing is missed and that the coherence and consistency of these facilities is reviewed at the system (rather than subsystem) level.
  - Ensure coverage of requirements in Information Security Framework.

4.6.4.6 Specify Data Conversion System

- Data Conversion System Specifications
- To specify the Data Conversion System, including automated and manual and align with necessary security controls required by applicable Information Security Frameworks. It is the means by which the data necessary for the proposed system will be obtained and made available in an appropriate form. Data conversion can encompass a wide range of activities, including:
  - Automated conversion of existing electronically stored data to new formats and/or storage technology.
  - Manual conversion of existing electronically stored data and re-entry into the new system.
  - Collection of data not currently held electronically and entry into the new system either through standard system facilities or through specially-written conversion processes.
  - Preparation of tables or directories for manual use.

Deliverable: System Delivery Specification

The System Delivery Specification is the definitive specification for all subsequent design and build activity. The system design will be developed based upon the information contained therein.

It includes information on functions, user interfaces, manual processes, automated processes, data structures, security, control, back-up and contingency requirements, as well as relevant error handling procedures.

This is then expanded to a full delivery specification with the addition of data conversion specifications, delivery, test and transition plans and Acceptance Test Strategy.

This is a particularly crucial document because:

- It is likely to be the last specification document reviewed by the users.
- Sign-off of this document effectively represents approval for the remainder of the system delivery cycle.
- The detailed plans and estimates for the remainder of the project will be based on its contents.

4.6.4.7 Create Security Documentation

Security documentation describes all security related functions and processes of the envisioned system. It is based on the Information Security Framework. It is exhaustive in scope and depth.

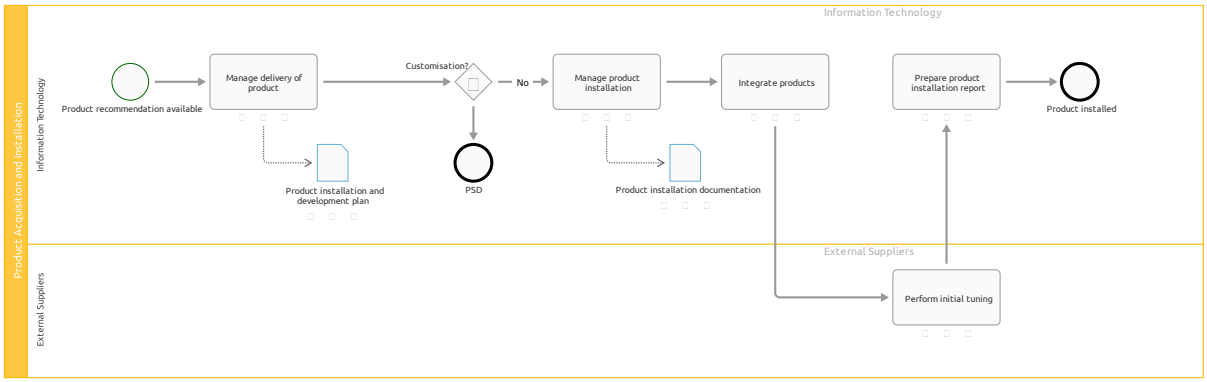
The security documentation must be handled according to the Information Security Framework.

4.6.5 Quality Keynotes

- User interface prototyping should be carried out effectively and efficiently.
- Change control procedures must be followed. It is important to capture changes and rationale during the process.
- Bi-directional traceability is maintained among System Requirements.
- Adequate attention must be given to the impact on computer operations and computer capacity. A separate capacity planning exercise may be required (cf. module AIP Architecture and Planning).
- All deliverables must be cleared adequately with Business and IT.
- Determine if there are any significant changes to the plans for the rest of the project.
- Determine if the plans affected by progress on parallel project and if the plans for phasing and transition are still valid. Determine if the route map is still valid.
- Determine if there are significant changes to the projected costs, benefits and risks.

4.7 Product Acquisition and Installation (PAI)

4.7.1 Module View



[6]

PAI RACI+S	Product Management	Information Technology	External Suppliers
Manage delivery of product	C	A, R	C
Manage product installation	C	A, R	C
Integrate products		A, R	C
Perform initial tuning		A, C	R
Prepare product installation report		A, R	

4.7.2 Module Description

4.7.2.1 Overview

This module deals with the acquisition and installation of products supplied by third parties. Products may be hardware or software. They may be part of the application system itself or belong to the technical environment required to develop or operate the system.

4.7.2.2 Prerequisites and Related Efforts

The prerequisite for this module is the Product Recommendation, which identifies the product(s) to be acquired and installed. This product recommendation is normally the output from the Product Evaluation and Recommendation (PER) module, although it is possible that this module will be used to acquire and install products for which no evaluation is necessary.

[6] Due to a rendering bug the gateway type symbol is not shown. This model contains only an exclusive gateway.

4.7.3 Key Considerations

The module is structured to cater for the possibility of acquiring and installing several products at once. It may also be necessary to perform this module several times within the scope of a single project.

Significant delays to projects s can be caused by late delivery or faulty installation of products, or even by administrative hold-ups in placing purchase orders. Therefore, care should be taken when planning the project that adequate elapsed time is allowed for these activities and that contingency plans are included to allow for possible delays.

4.7.4 Activities and Work Products

4.7.4.1 Manage Delivery of Products

- Product Installation & Development Plan
- This work product aims at monitoring the delivery, installation and acceptance of the product.
- Delivery and installation plan
- Responsibility for installation
- Training prerequisites
- Vendor training availability and schedule
- Product acceptance criteria
- Product Installation test

4.7.4.2 Manage Product Installation

- Product Installation Documentation
  - This deliverable consolidates the documentation regarding the installation and preliminary tests of technology and application products.
  - Actions taken to prepare participating staff and to perform the installation.
- Installation documentation:
  - For hardware or environment products record details such as location and cabling.
  - For software products record details such as the physical location, library names, and data base identification.
- Results of the vendor demonstrations / tests and product tests carried out and details of any errors or problems encountered.
- Documented procedures to support operation of the product during development and testing phases.
- To initiate procedures for the provision of maintenance and support by the vendor.

4.7.4.3 Integrate Products

To integrate the product within the current architecture and infrastructure.

4.7.4.4 Perform Initial Tuning

The action of configuring and optimizing the purchase system to match its performances with the requirements and specifications.

Deliverable: Product Installation Report

To perform the activities necessary to ensure that purchase orders are placed for all products involved. The responsibility for order placement may lie with the appropriate purchasing function: the role of project management is to ensure, as far as possible, that appropriate actions are taken and delays minimized.

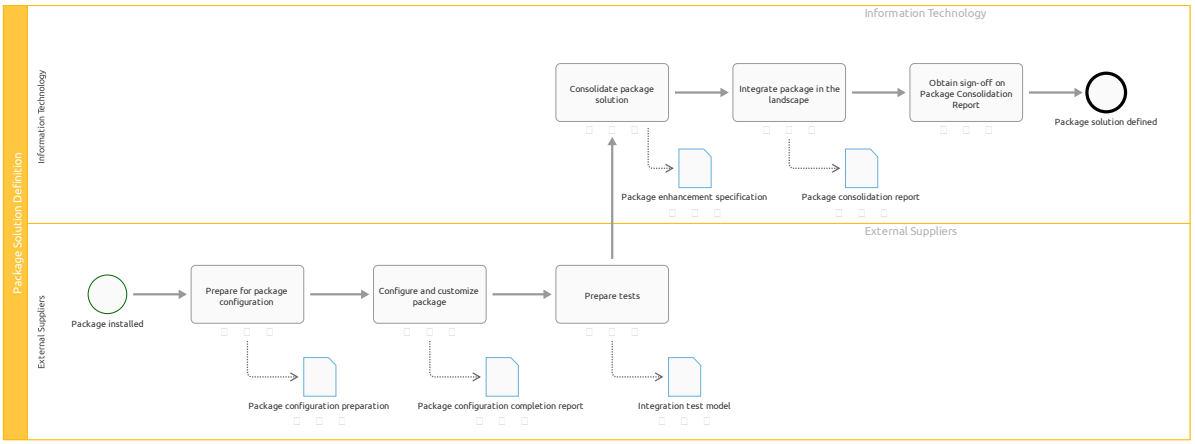
4.7.5 Quality Keynotes

- Products should be ordered in sufficient time for installation and commissioning before they are required for use.
- User staff should be allocated for timely installation and commissioning.
- Application packages need to be installed and tested before they are required in TPD.
- Relationships with the vendor should be optimal to ensure commitment, performance, and active support.
- The purchasing team will provide support within this module so that the organization purchasing policies and process are appropriately followed.

Internal

4.8 Package Solution Definition (PSD)

4.8.1 Module View



PSD RACI+S	Product Management	Information Technology	External Suppliers
Prepare for package configuration		A, C	R
Configure and customize package		A, C	R
Prepare tests	R	A, C	R
Consolidate package solution	R	A, R	
Integrate package in the landscape	C	A, R	C
Obtain sign-off on package consolidation report	S	A, R, S	

4.8.2 Module Description

4.8.2.1 Overview

The purpose of the Package Solution Definition module is to produce a detailed configuration of the package to be delivered (software and processes configuration). The PSD module is core to a package implementation. The module expands upon the user requirements and outline design documented in System Requirements.

The focus is still on user requirements; however, the emphasis throughout this module is on how the package is to be implemented and how a system of working methods can be built around the package to produce a solution that meets the users' needs.

4.8.2.2 Prerequisites and Related Efforts

Prerequisites for this module are the Business Requirements, System Requirements including the security requirements, containing the Implementation Solutions defined in PER. A trained team is required, installed package from PAI and code structures defined from associated

projects/development activities. The hardware must also have been installed and the IT infrastructure should exist.

4.8.3 Key Considerations

The involvement of users in this module is crucial. The users will be working within the team and their input is vital. The user members of the team must be committed to the package implementation and the senior users must have the authority to make business policy and processing decisions.

The team should be trained in the package functionality prior to working on the configuration.

It is crucial to have the input of a package specialist who may be an employee of the vendor. This input needs to be negotiated during PAI (or during PER as a part of the vendors proposals).

The package to be configured must be installed and fully available prior to work commencing. The new policies and working practices must be communicated to staff as soon as they are agreed to allow time for staff to familiarize themselves with them.

4.8.4 Activities and Work Products

4.8.4.1 Prepare for Package Configuration

Package Configuration Preparation

- The purpose is to plan and establish the environment and procedures for the package configuration.
- Prepare a detailed plan for the package configuration.
- Put in place appropriate standards and naming conventions for the package parameters. Specify the business processes that will be impacted by the package implementation.
- Describe the interfaces with the other systems which are part of the overall solution.
- Map the Data Models between the package and the proposed solution.
- Establish the technical framework required to support the configuration of the package.
- Plan test strategy and environment for the package configuration.

4.8.4.2 Configure and Customize Package

Package Configuration Completion Report

- The purpose is to confirm that the following configuration activities have been performed.
- Confirm the scope of the package parameter configuration.
- Configure the master and standing data required within the package.
- Configure the security and control parameters.
- Configure the package functionality to support the business processes.
- Configure reports / Inquiries facilities.
- Validate and verify the configuration by the use of test scenarios.
- Place the acquired system under Software Configuration Management.

4.8.4.3 Prepare Tests

Integration Test Models

4.8.4.4 Consolidate Package Solution

Package Enhancement Specification

The purpose of this work product is to develop detailed business specifications of any required enhancements to the package.

- Specify the functional requirements for the enhancement, once they have been reviewed by a package specialist. To request for vendor quote.
- Assess the costs/benefits and approve, if appropriate.
- Plan the enhancement activities and impact on the overall project .
- Perform the required system and acceptance test.

4.8.4.5 Integrate Package in Landscape

Deliverable: Package Consolidation Report

This deliverable aims at consolidating the outputs of the module.

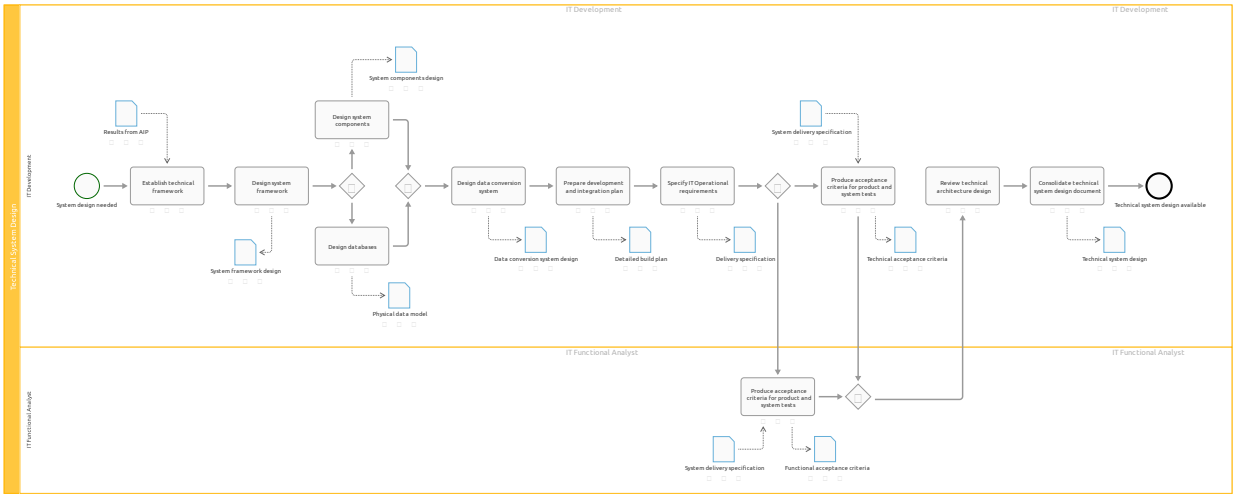
- Bring together all documentation related to the package configuration.
- Document any business policy changes required to support the operations of the package.
- Document the accepted enhancements in the product to support the business processes.

4.8.5 Quality Keynotes

- Is there a package specialist available either from within the organization or from the vendor?
- Is there communication between project managers from other relevant projects? Are relevant decisions discussed?
- Are change management procedures being followed?
- Is the philosophy not to have enhancements but to change Ways of Working?
- Is the emphasis on core processes?
- Is consideration of configuration alternatives being restricted to those that are realistic possibilities?
- Does the package, as specified, match the culture and capabilities of the users?
- Have any major organizational impacts been highlighted and addressed?
- Will it be necessary to specify enhancements to the package?
- Is the configuration correctly staged?

4.9 Technical System Design (TSD)

4.9.1 Module View



[7]

TSD RACI+S	IT Development	IT Operations	IT Functional Analysts
Establish technical framework	A, R		
Design system framework	A, R		
Design system components	A, R		
Design databases	A, R		
Design data conversion systems	A, R		
Prepare development and integration plan	A, R		
Specify IT Operational requiremetns	A, R	C	
Produce acceptance criteria for product and system tests (technical)	A, R		
Produce acceptance criteria for product and system tests (functional)			A, R
Review technical architecture design	I		
Consolidate technical system design document	A, R, S	S	S

4.9.2 Module Description

4.9.2.1 Overview

The Technical System Design describes how the proposed system is to be built, continuing from the System Delivery Specification document, which describes what the system is to do. The design is specific to the technical environment in which the system is to operate and is also constrained by the development environment and tools to be used in building the system. The output of this module drives the subsequent development of software and operational procedures; the level of detail required depends upon the development techniques and tools to be used (such as programming languages and code generators) and the quality and experience of the development staff.

[7] Due to a rendering bug the gateway type symbols are not displayed. This model contains only parallel gateways.

4.9.3 Key Considerations

Large system delivery projects can be divided into several concurrent and/or overlapping sub-projects/development activities.

The initial phases should be carried out for the whole project to ensure consistency and successful integration, and the concurrent, or overlapping sub-projects/development activities, should be confined to the SUMMIT phases Design, Build and Test, and Transition phases. Hence, Technical System Design could be carried out several times within a single project, each sub- project producing its own design specification.

4.9.4 Activities and Work Products

4.9.4.1 Establish Technical Framework

The purpose of this work product is to collate the various components of the technical design and consolidate them in order to produce a comprehensive and concise design.

- Complete definitions of the deployment artefacts (e.g. processes, services, container, war, ...), databases, brokers etc which comprise the system, ready for subsequent development and implementation
- Definition of the operational requirements, to support the preparation of operations procedures
- Design and operation procedures for the data conversion system
- Detailed test plans

4.9.4.2 Design System Framework

System Framework Design

- The system framework defines the overall shape of the system and breaks the system down into component parts for detailed design work. This includes:
- System partitioning into implementation units.
- Interfaces between these components (data format, source(s) and destination(s) including data and control information as well as application data).
- Interfaces to other systems (either currently operational or under parallel or planned development).
- Parameter settings for any application products, which form part of the system.

4.9.4.3 Design System Components

System Components Design

This work product consists of a detailed design for each implementation unit, which is subsequently incorporated into the system design document. In addition to the design elements described above it includes cross-references to the physical data model, which is designed in parallel with the implementation units.

- Detailed design of the user interfaces, with reference to the technical environment in which they are to operate and based on the definition of user interfaces in the System Delivery Specifications.

- Implementation unit partition into modules where necessary and detailed design of these modules.
- Specification of error handling procedures for each Implementation Unit.
- Design is reviewed and evaluated to ensure that it will not present problems in the future maintenance of the system. This activity is performed for each implementation unit.

4.9.4.4 Design Databases

Physical Data Model

The Physical Data Design contains complete definitions of the databases to be implemented, validated for consistency and performance:

- Physical data models.
- DBMS and view definitions.

4.9.4.5 Design Data Conversion System

Data Conversion System Design

If data conversion is necessary as a consequence of changed/extended data structures in the new release/system, it must be designed here.

This work-product provides the data conversion system design including:

- Technical environment in which data conversion is to take place.
- Detailed designs of the implementation units and modules.
- Specifications of the operations procedures to be followed during the conversion.
- Test plan for the conversion.

4.9.4.6 Prepare Development and Integration Plan

- Detailed Build Plan

4.9.4.7 Specify IT Operational Requirements

Delivery Specification

- Specifies the requirements which will be placed by the system on the computer operations function, enabling the impact of the system on operations resources, schedules and plans to be assessed and providing a basis for the development of operations procedures for the system.
- Schedule of operation for system components which are performed or available on a regular basis (rather than on demand).
- Security and control requirements, based on Information Security Framework and defined during the SRA and AIP have to be considered.
- Impact of the proposed system on the existing contingency plans (When necessary, a contingency planning exercise should be initiated).

- Reviewed impact on the Capacity Plan to ensure that the installation of the proposed system and its subsequent growth will not exceed the available capacity.
- Resources and skills required to install and operate the system in terms of operations staff, equipment (ancillary rather than primary hardware), consumables (such as stationery).

4.9.4.8 Produce Acceptance Criteria's for Product and System Tests

- Functional Acceptance Criteria
- Technical Acceptance Criteria
- Security Acceptance Criteria

4.9.4.9 Technical Design Architecture Review

- Architecture team can review the Technical Design

4.9.4.10 Package Technical System Design Document

Deliverable: Technical System Design

The purpose of this work product is to collate the various components of the technical design and consolidate them into a single document:

- Complete definitions of the processes and databases which comprise the system, ready for subsequent development and implementation.
- Definition of the operational requirements, to support the preparation of operations procedures.
- Design and operation procedures for the data conversion system.
- Detailed test plans.

4.9.5 Quality Keynotes

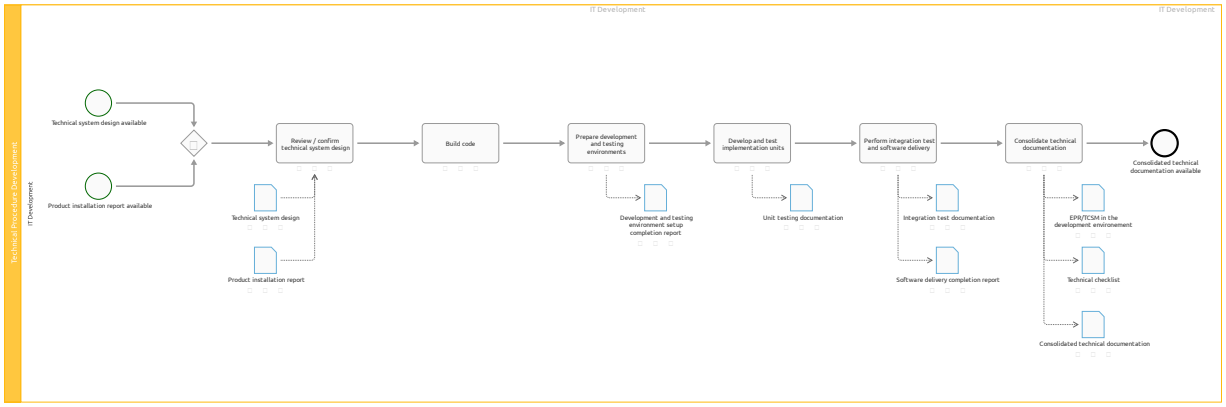
- The design standards adopted should be appropriate to the development environment (particularly where a package is being modified or fourth generation tools are in use).
- Change management procedures should be followed.
- Control should be maintained on work being undertaken by a 3rd party supplier.
- Proper attention should be given to computer operations requirements, computer capacity requirements, and contingency planning, with the appropriate experts involved.
- The level of detail in specifications should reflect the degree of novelty of the function (i.e. very brief for standard functions but with more explanation of unique functions).
- The designers should produce a system, which will be easy to maintain, enhance and reuse.
- Any significant changes to the plans for the rest of the project required should be determined and if the plans are affected by progress on parallel projects/development activities.

- Final assurance should be sought that the user requirements set out in the early stages of the project will be supported by the system as designed.
- Traceability of user requirements is maintained through the BRA, SRA, PER, SDS and TSD activities.

Internal

4.10 Technical Procedure Development (TPD)

4.10.1 Module View



[8]

TPD RACI+S	IT Development
Review / confirm technical system design	A, R
Build code	A, R
Prepare development and testing environments	A, R
Develop and test implementation units	A, R
Perform integration test and software delivery	A, R
Consolidate technical documentation	A, R

4.10.2 Module Description

4.10.2.1 Overview

The Technical Procedure Development module deals with the development, unit testing and integration testing of application software and operational software and procedures. Any special software or procedures required for data conversion is also developed and tested.

4.10.2.2 Prerequisites and Related Efforts

The prerequisite for this module is the SUMMIT work product Technical System Design report, which contains detailed specifications for all the software and procedures to be developed. Other inputs include specifications and installation reports for any products involved, either as part of the system itself or to support the development of the system.

4.10.3 Key Considerations

This module is usually carried out in parallel with the development of user procedures (INR module) and test case scenarios. Communication between the two development processes is

[8] Due to a rendering bug the gateway type symbols are not displayed. This model contains only parallel gateways.

essential to ensure that technical procedures and user procedures are consistent, particularly at the user interfaces.

The introduction of fourth generation and software engineering techniques and tools means that the software development process is becoming increasingly automated, shifting the emphasis away from traditional programming and testing. Preparation of environments and operations procedures will continue to be of vital importance to a successful system delivery. The timely installation of products is also likely to be critical to the success of a delivery project.

The processes of development and testing of data conversion software are no different from those required for the application software and no distinction is made at the activity level.

4.10.4 Activities and Work Products

4.10.4.1 Review/Confirm Technical System Design

4.10.4.2 Build Code

4.10.4.3 Prepare Development and Testing Environment

Development and Testing Environment Setup Completion Report

This work product certifies the production readiness of the development and testing environment according to the required specifications. Possible deviations from the required specifications have to be specified.

- Verification of the description of the test environment.
- Consider security and control requirements found in the Information Security Framework.
- Used/implemented software versions.
- Used data and infrastructure.
- Completion notification.

4.10.4.4 Develop and Test Implementation Units

Unit Testing Documentation

Unit Test cases planning, documentation and completion report to outline the errors encountered during testing and the modifications applied to the original source. This will facilitate regression testing.

- Unit Test Approach and Test Cases:
  - Refined unit test plan.
  - Completed Unit Test case specifications.
- Unit Test Report:
  - To describe the way the tests have been conducted, the way they were controlled and the meaning of the results, i.e. why such results).
- Unit Test Completion Report:
  - Statement about the actual state of target achievement.

- Completion notification.

4.10.4.5 Perform Integration and Product Tests and Software Delivery

Integration Test Documentation

Integration Test cases planning, documentation and completion report to outline the errors encountered during testing and the modifications applied to the original source. This will facilitate regression testing.

- Integration Test Approach and Test Cases:
  - Refined Integration Test plan.
  - Completed integration test case specifications.
- Integration Test Report:
  - To describe the way the tests have been conducted, the way they were controlled and the meaning of the results, i.e. why such results).
- Integration Test Completion Report:
  - To be provided as a completion notification by a sentence in an appropriate status report. This statement confirms the successful completion of the integration test. It should include a statement about the realization of former targets issues and issues assessment (findings).
  - Fixing rate statistics (status report).
  - Statement about the actual status of the target achievement.
  - Completion notification.
- Software Delivery Completion Report
- The work product is the confirmation of the physical delivery of the software.
  - Specify if testing is completed for each test item.
  - Determine the result of each test item.
  - Define the impact of unsatisfactory test results.
  - Communicate to the Release Manager or Project that the piece of code can be transferred for System Testing.
  - Release the code for System Testing (Software Configuration Management Team).
  - Track the release of the code under the appropriate Software Configuration Management tool.
  - Technical documentation about the code is recorded within the code or safely stored through the Software Configuration Management tools.

4.10.4.6 Consolidate Technical Documentation

- Code under configuration management in the Development Environment using defined method
- Technical Checklist

Deliverable: Consolidated Technical Documentation

This is the consolidation of all the technical documentation produced in this module. It is the responsibility of the project manager to ensure that this documentation is produced and stored in the appropriate location.

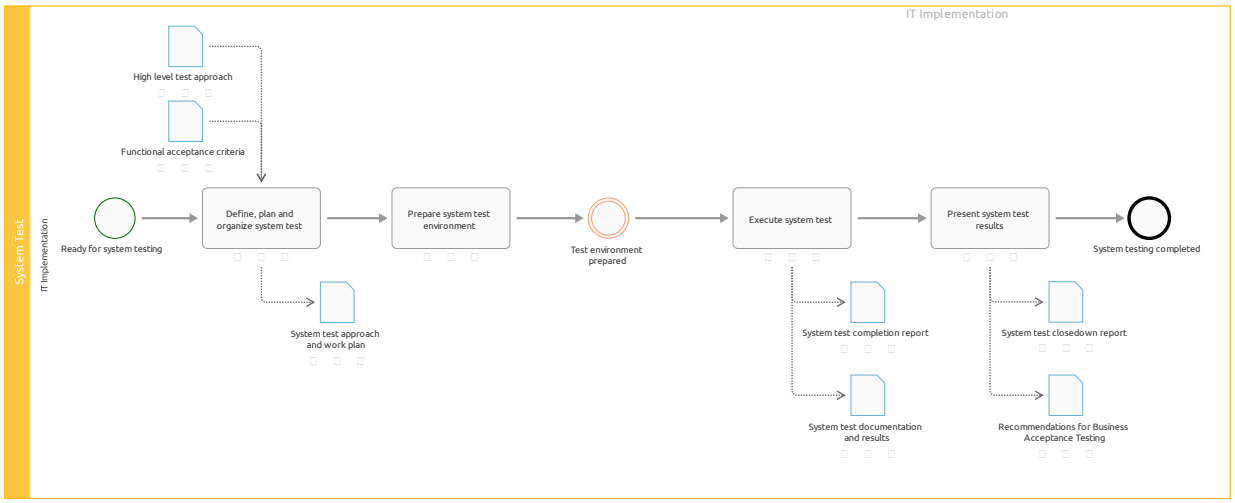
4.10.5 Quality Keynotes

- The programming standards adopted should be appropriate to the development environments. They should meet the needs for future maintenance and for introducing new releases of packages and should be enforced.
- Change control procedures should be followed.
- Configuration management procedures should be in place.
- Advantage should be taken of opportunities to coordinate integration testing with subsequent systems testing.

Internal

4.11 System Test (STT)

4.11.1 Module View



STT RACI+S	Change Management	IT Development	IT Implementation	IT Test
Define, plan and organize system test	C	C	A, R	
Prepare system test environment	C	C	A, R	R
Execute system test	C	C	A, R	
Present system test results	C, S	C, S	A, R	

4.11.2 Module Description

4.11.2.1 Overview

Although individual system elements underwent unit testing and integration testing in Technical Procedure Development (TPD), the system as a whole is now subjected to testing. In addition, to verify that the system correctly performs all required business functions, it is essential to examine technical features such as the start-up and shut-down procedures, interfaces between software and hardware components and the back-up and recovery procedures. It must be demonstrated that the system can process the required volume of transactions and perform within the specified parameters of speed and memory. The behaviour of the system must be examined under extreme and worst-case case conditions. Furthermore, user and operation procedures must be tested along with the software and hardware.

The module System Test is the responsibility of the IT project manager. There are essentially two core tasks, which must be considered during test preparation:

- That the Product Test successfully verifies the new features documented within the modules System Requirements Analysis (SRA) and System Delivery Specification (SDS). The Product Test can be carried out as a pure application test or within the system environment in which the system is embedded (cross system test or end-to-end-test) or both test procedures will be executed one after the other.

- The Regression Test successfully verifies the collectivity of features described in the complete system documentation. The major focus is that new and old features can run together according to the required manner. The regression test can be carried out as a pure application test or within the system environment in which the system is embedded (cross-system test or end-to-end test) or both test procedures will be executed one after the other.

4.11.2.2 Prerequisites and Related Efforts

Prerequisite is the software (and the operations procedures for executing them) that optionally have already passed Unit and/or Integration testing in Technical Procedure Development (TPD). A full set of test approaches, detailed test models inclusive of detailed specifications of each test cycle, is required for System Testing and will be provided in parallel to Technical System Design (TSD) and Technical Procedure Development (TPD). The Overall Test Strategy is initially developed in SDS, with the details provided in TSD. The Architecture and Infrastructure Planning (AIP) module is initiated to define the requirements of the system testing and production environments.

4.11.3 Key Considerations

To enable efficient processes (and to ensure the required level of quality is achieved) project management and test management have to decide in terms of the topics stated below. Their decisions have to be documented and reasoned in the respective test approaches.

- Depending on scope and system landscape complexity, Product and Regression Test can be conducted as an isolated test with the respective application(s) and/or together with the system environment.
- The necessity of a separate system environment tests (cross-system test or end-to-end test) to be carried out independently from Product Test and Regression Test has to be evaluated. The target of this test is the correct operation of interfaces, the demonstration of seamless interactions and the stability of the entire participating system landscape.
- The combination of the Product Test and the Regression Test as one test execution procedure has to be evaluated regarding the release scope.
- The scope of Product Test and Regression Test include functional aspects (functions, timings, behaviour, application design, etc.) as well as technical aspects (performance, recoverability, restart ability, disaster recovery, benchmarking, hardware sizing, installation, configuration, compatibility, sizing guidelines etc.). The selection of the appropriate test aspects is the responsibility of project management.
- The necessity of a Regression Test has to be evaluated. In any case a Regression Test can be skipped if a completely new product will be introduced.
- The Build & Test part of the project will require support of one or more persons providing deepest functionality know-how ("functional architect") to represent the business point of view. It is of major importance to have both, the specification documents (during SRA and SDS) and the test preparation material (test models and test cases) reviewed by these people, in order to detect any different interpretation of the specification as soon as possible.

Internal

4.11.4 Activities and Work Products

4.11.4.1 Define, Plan and Organize System Test

- System Test Approach and Work Plan
- The purpose of the System Test is to validate the new release against the defined requirements documented in the module System Requirements Analysis (SRA) and System Delivery Specification (SDS). The goal of the System Test is to test the actions of the new or modified features and functions of the application. If the new release encompasses several applications then a System Test will be processed for each application. The System Test could also incorporate cross-system tests or end-to-end tests. The System Test Approach describes the test strategy and processes.
- Content:
  - Test strategy:
    - Objectives
    - Assumptions
    - Scope, detailed approach what will be validated including security functions e.g. Need to know and data classification
    - Out of scope
    - Entry and exit criteria
    - Determination of product test completion criteria
    - Overview of functional and technical test cycles
    - Requirements for and determination of test methods
    - Requirements for and determination of test tools
    - Requirements for test environment and test database
    - Determination for fix-it criteria
  - Detailed test schedule
  - Roles and responsibilities of participating staffing
  - Test issue tracking
  - Requirements for test documentation and reporting

4.11.4.2 Prepare System Test Environment

- System Test Environment

4.11.4.3 Execute System Test

Pre-acceptance tests are to be performed during STT, in order to detect issues as early as possible.

- System Test Documentation and Results
- The work product records each test step of each test condition with the effective result. Furthermore, test issue identification and processing (issue description, escalation

respectively the outcome of the escalation e.g. according to issuetracking procedures) must be traceable and clearly documented.

- Content:
  - Test cycle control sheet for each execution (signed by test responsible and executing persons)
  - Filled test condition and test step sheets
  - Corresponding system output (e.g. in the form of window copies or reports)
  - Test issue processing documentation
  - Documentation of open issues
- System Test Completion Report
- The System Test Completion Report serves as a declaration of agreement about the completion of the System Test according to the requirements and guidelines defined in the System Test Approach and Work Plan. The document will be signed by the relevant and responsible persons to execute the product test activities as well as by project management.
- Content:
  - Declaration of agreement
  - Eventually caveats

4.11.4.4 Present STT Results

- Recommendations for BAT

Deliverable: STT Closedown Report

4.11.5 Quality Keynotes

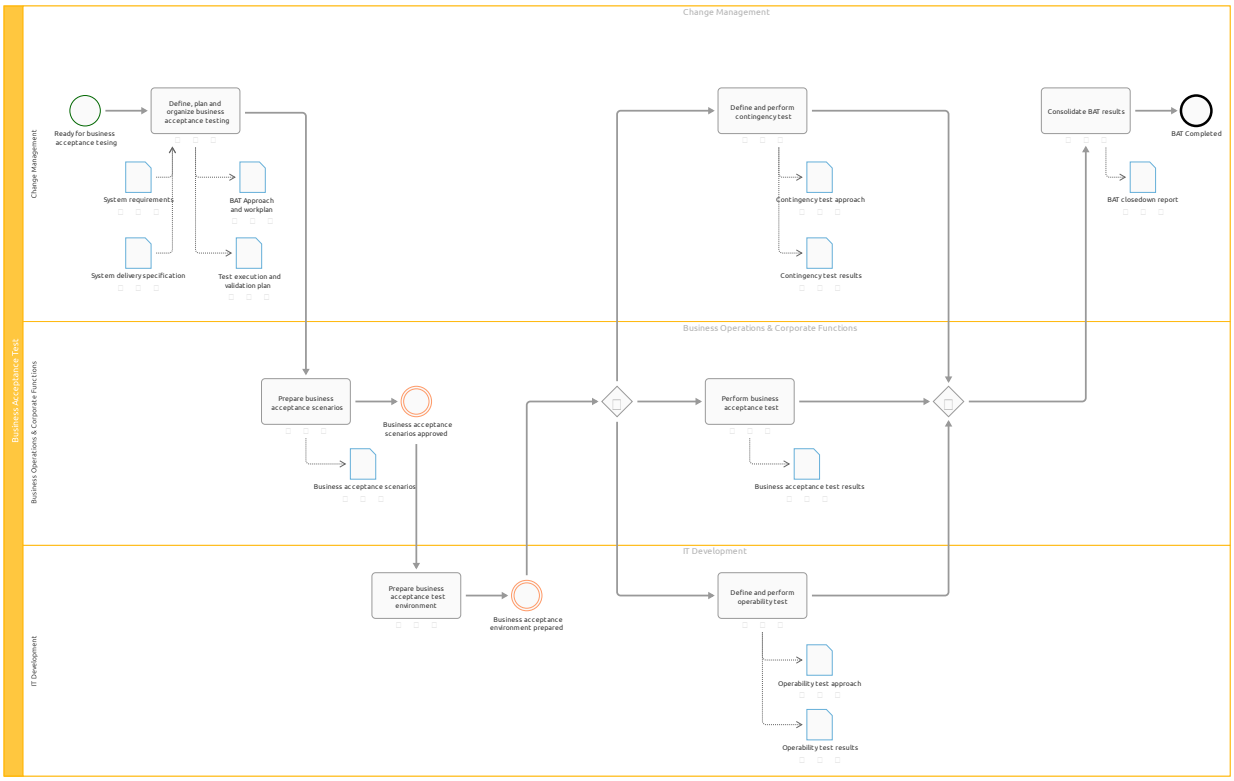
- The personnel with prime responsibility for STT should be sufficiently objective in determining scope (generation of test approach and test model), preparing test data and checking results. To avoid interest conflicts people should also be participating in the specification module (SDS). To avoid any loss of time the people should start immediately after the finalization of the System Delivery Specification (SDS).
- In parallel to the running off STT activities, a fix-it team originated from the development resources must be established. The Application Development team is responsible for resolving software issues assessed and acknowledged prior to determined processes according to the defined criteria. Rational decisions should be taken concerning when to repeat tests or to proceed.
- The execution of the Regression Test should not be started until the application shows a satisfied stability within the Product Test. The definition of stability has been stipulated by means of quantified metrics before product test starts. Product test stability needs to be demonstrated by the test documentation that will enable a decision process to go or not to go.
- Adequate IT and Business resources, if needed, should be available.
- Relevant for the start of Acceptance Testing is the contractually agreed date of the software delivery and the quality of the software at the end of the STT. STT has to

ensure that the quality of the software is sufficient for SW delivery. The definition of satisfactory results has been stipulated by means of quantified metrics before acceptance testing starts.

Internal

4.12 Business Acceptance Test (BAT)

4.12.1 Module View



[9]

BAT RACI+S	Change Management	Business Operations & Corporate Functions	Product Management	IT Development	IT Functional Analysts	IT Test
Define, plan and organize business acceptance	A, R		C	C	C	R
Prepare business acceptance scenarios	A, C	R	R	C	I	R
Prepare business acceptance test environment	C			A, R		C
Perform business acceptance test	A, C	R	R	C	R	R
Define and perform operability test	A, R	R	C	R	I	R
Define and perform contingency test	A, R	R	C	R	I	R
Consolidate BAT results	A, S	R, S	R, S	R, S		

[9] Due to a rendering bug the gateway type symbols are not displayed. This model contains only parallel gateways.

4.12.2 Module Description

4.12.2.1 Overview

The Acceptance Test module describes activities performed to prove that the requirements documented in the specification are fulfilled by the new release. Contrary to the System Test module running under the responsibility of IT, the Business Acceptance Test (BAT) module will be driven by units representing the Business. Therefore, a handover of responsibility between System Test and Business Acceptance Test takes place.

The Business Acceptance Test validates the functional requirements described in the SRA and/or the SDS to the level specified in the BAT approach. The focus is on whether the new release works as specified. Non-functional requirements, if not tested in System Test or Technical Acceptance Test should also be considered here.

4.12.2.2 Prerequisites and Related Efforts

Prerequisite is the software (and the related operations execution procedures) that optionally already passed Product and/or Regression testing in System Test (STT). A full set of test specifications, containing test approaches, detailed test models inclusive of detailed specifications of each test cycle, is required for Business Acceptance Testing.

A number of entry criteria related to the following domains also have to be fulfilled:

- Planning & Definition - Test Strategy
- Planning & Definition - Resources
- Planning & Definition - Preparation
- Software Quality - Code Complete
- Software Quality - Test Coverage
- Software Quality - Test Results

4.12.3 Activities and Work Products

4.12.3.1 Define, Plan and Organize Business Acceptance

- Business Acceptance Test Approach and Work Plan

The purpose of the Business Acceptance Test (BAT) is to ensure that the new release performs in accordance with the business requirements and functional specifications. While the Technical Acceptance Test focuses on non-functional requirements and technical contingency, the scope of the BAT focuses on functional requirements.

- Content:
  - Test approach:
    - Objectives
    - Assumptions
    - Scope (detailed approach of what will be validated)
    - Entry and exit criteria

- Overview of functional test cycles
- Requirements for and determination of test methods
- Requirements for and determination of test tools
- Requirements for test execution
- Determination for fix-it criteria

- Detailed test schedule (prepared by IT)
- Roles and responsibilities of participating resources
- Test issue and event tracking
  - Requirements for test documentation and reporting
  - Test Execution and Validation Plan

4.12.3.2 Prepare Business Acceptance Scenario

- Business Acceptance Criteria

4.12.3.3 Prepare Business Acceptance Test Environment

4.12.3.4 Perform Business Acceptance Test

- Business Acceptance Test Results

The work product records the test coverage and success rate. Furthermore, test issue identification and processing (issue description, escalation respectively the outcome of the escalation, e.g. according to issue tracking/workflow tool procedures) must be traceable and clearly documented.

4.12.3.5 Define and Perform Operability Test

- Operability Test Approach
- Operability Test Results

Responsibility for the Operability Test is described in the Test Approach. The aim is to validate a normal business day following the 24-hour timeline and to identify that the new functionality and the related business applications changes perform all operational activities within deadlines.

Per project and/or release, the definition of the Operability test objectives & requirements and the issuance of the related test scenarios will be performed as described in the Operability Test Approach and in accordance with the organizational structure relevant to the project. For ICSD this is IT Test Management and Change Management.

It must be ensured that all business users who are designated participants of an Operability Test are ready for the tests i.e. ensure new functionality is understood, scope & objectives of tests are understood and that Operability Test preparation is secured. Such preparation must include detailed end user training, documentation of test scenarios and the availability of draft work flows or procedures.

4.12.3.6 Define and Perform Contingency Test

When a dedicated Contingency Test cycle is run in parallel with BAT, a closedown report dedicated to Contingency Test is produced. Otherwise, Contingency Test Results are included in BAT Closedown Report.

- Contingency Test Approach
- Contingency Test Results

4.12.3.7 Present BAT Results

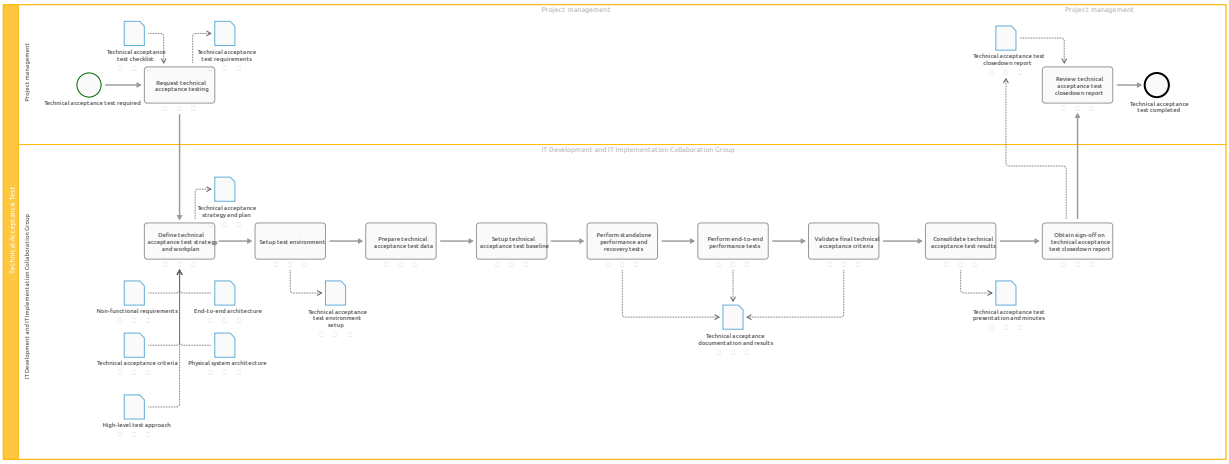
Deliverable: Business Acceptance Test Closedown Report

4.12.4 Quality Keynotes

- The personnel with prime responsibility for BAT should be sufficiently objective in determining scope (generation of the test approaches), preparing test data and checking results. To avoid interest conflicts, they should also be responsible for or participating in the specification activities. To avoid any lost time, they should start immediately after finalization of the specification.
- In parallel to running BAT activities the Application Development team that originated from the development resources and already established for STT is responsible for solving issues assessed and acknowledged in the prior determined processes and according to defined criteria.
- Adequate IT and Business resources should be available. Confirmation of whether resources assigned to BAT activities are available and not allocated to STT or elsewhere is required, especially for BAT preparation.
- Future internal users should be properly involved in the Business Acceptance Testing. The same applies to future operators due for operation of the new release.
- Issues identified within the STT module must be re-tested intensively in BAT.

4.13 Technical Acceptance Test (TAT)

4.13.1 Module View



TAT RACI+S	Project Management	Product Management	IT Development & IT Implementation Collaboration Group	IT Operations	IT Infrastructure	IT Test
Request technical acceptance testing	A, R					
Define technical acceptance test strategy and workplan			A, R	C	C	R
Setup test environment			A, R			
Prepare technical acceptance test data			A, R			
Setup technical acceptance test baseline			A, R		C	R
Perform standalone performance and recovery tests			A, R		C	R
Perform end-to-end performance tests			A, R		C	R
Validate final technical acceptance criteria			A, R			
Consolidate technical acceptance test results			A, R			
Obtain sign-off on technical acceptance test closedown report			A, R, S	S	S	
Review technical acceptance test closedown report	A, R	S				

4.13.2 Module Description

4.13.2.1 Overview

TAT is necessary if there is a significant impact on security, performance, recovery or changes in relation to volumes or response time. Consequently, technical changes with no such impact, do not require the use of the TAT module, unless e.g. annual penetration testing is due. In these case, the test approach should be identified in the SDS.

The Technical Acceptance Test (TAT) is intended to validate non-functional requirements with a focus on both, service levels and non-functional security requirements. Further it ensures the new release software can be introduced into production without taking unforeseeable risks.

The purpose of the Technical Acceptance Test is to verify that the delivered code meets the Technical Acceptance Criteria (TAC). This may be achieved by execution of test types like reliability, availability, maintainability, recoverability, performance, stress/volume and benchmark

testing (repetitive processing of the same input over different logical configurations to determine the optimum configuration).

The selection of the appropriate test types is the responsibility of IT. The Technical Acceptance Test is especially focused in verifying the trouble-free application of operational activities and procedures to ensure the proper integration of the architecture components. Therefore, IT is responsible for running the TAT and deciding on the technical acceptance.

4.13.2.2 Prerequisites and Related Efforts

Prerequisites are that the software has already passed unit and/or integration testing in Technical Procedure Development (TPD) respectively the Product and/or Regression Testing in System Test (STT), the TACs and the AIP deliverables (End-to-End Architecture and Physical System Architecture).

A full set of test specifications, containing test approaches, detailed test models inclusive of detailed specifications of each test cycle, is required for Technical Acceptance Testing and will be generated in parallel to Technical System Design (TSD) and Technical Procedure Development (TPD).

The Overall IT Test Strategy is produced as part of System Delivery Specification (SDS). The Architecture and Infrastructure Planning (AIP) module will be initiated to define the requirements of the system testing and production environments. The manual functions are required to complete the test specifications, and fully-trained users and operators are needed to perform acceptance testing.

4.13.3 Activities and Work Products

4.13.3.1 Define TAT Strategy and Work Plan

- TAT Strategy and Plan
- Complementary to the Business Acceptance Test, the technical aspects are to be tested within the Technical Acceptance Test: performance, recoverability, restart ability, disaster recovery, benchmarking, hardware sizing, etc. The Technical Acceptance Test Approach involves the test strategy and processes of the Technical Acceptance Test.
- Content:
  - Test strategy:
    - Objectives.
    - Assumptions.
    - Scope (detailed approach what will be validated).
    - Out of scope (items not covered).
    - Refinement of acceptance criteria.
    - Overview of technical test cycles.
    - Requirements for and determination of test methods.
    - Requirements for and determination of test tools.
    - Requirements for test environment and test database.

- Determination for fix-it criteria.

- Detailed test schedule:
  - Roles and responsibilities of participating staffing.
  - Test issue tracking.
  - Requirements for test documentation and reporting.

4.13.3.2 Set Up TAT Environment

- TAT Environment Set Up

The Technical Acceptance Test Environment Set Up is complete and the test environment can be operated for Technical Acceptance Testing as defined in the Technical Acceptance Test Approach and Work Plan (checkpoint).

4.13.3.3 Prepare TAT Data

4.13.3.4 Set up TAT Baseline

4.13.3.5 Perform Standalone Performance and Recovery Tests

4.13.3.6 Perform End-to-End Performance Tests

4.13.3.7 Final TAC Validation

Run and execute all TACs with the latest release package prior to the transition test.

- Technical Acceptance Documentation and Results
- The work product records each test step of each test cycle with the effective result. Furthermore, test issue identification and processing (issue description, escalation respectively the outcome of the escalation e.g. according to issue tracking procedures) must be traceable and clearly documented.
- Content:
  - Test cycle control sheet for each execution (signed by test responsible and executing persons).
  - Filled test condition and test step sheets.
  - Corresponding system output (e.g. in the form of window copies or reports).
  - Test issue processing documentation.

4.13.3.8 Present TAT Results

- TAT Results Presentation and Minutes
- TAT team presents to IT project management and Line Management, the TAT results which are documented in the TAT Closedown Report.
- It is essential to minute the workshop and record the attendees' agreement on the completion of the TAT according to the guidelines and requirements defined in the TAT Strategy and Plan. Any caveat and/or work commitment will be listed as this agreement is a necessary condition for production launch.

- These minutes are to be considered as formal sign off for agreement of the TAT Closedown Report.

Deliverable: TAT Closedown Report

The TAT Closedown Report summarizes the TAT objectives and scope as defined in the TAT Strategy and details testing results per Technical Acceptance Criteria. These results are presented to Project and Line Management in a dedicated workshop.

Issues if any are documented and will be subject to recommendations from the TAT team.

4.13.3.9 Project Management Activities

(1) Request TAT

With the TAT team support, project management has to define clearly the project/release requirements.

The TAT will then be able to estimate the necessary resources and to prepare the TAT Strategy and Plan.

- Input: TAT Checklist
- The TAT Checklist helps project management to define the TAT requirements prior the preparation of the TAT plan & strategy.
- Output: TAT Requirements
- The TAT Checklist helps project management to define the TAT requirements prior the preparation of the TAT plan & strategy.

(2) Review TAT Closedown Report

A dedicated workshop is organized by the TAT team, to present the TAT results to Project and Line Management.

- Input: TAT Results Presentation
- TAT team presents to IT Project and Line Management, the TAT results which are documented in the TAT Closedown Report.
- It is essential to minute the workshop and record the attendees' agreement about the completion of the TAT according to the guidelines and requirements defined in the TAT Strategy and Plan. Any caveat and/or work commitment will be listed as this agreement is a necessary condition for production launch.
- These minutes are to be considered as formal sign off for agreement of the TAT Closedown Report.
- Output: ATBTS and Work Commitment
- ATBTs and Work Commitment to address recommendations resulting from the TAT completion.

4.13.4 Quality Keynotes

- The personnel with prime responsibility for TAT should be sufficiently objective in determining scope (generation of the test approaches), preparing test data and checking

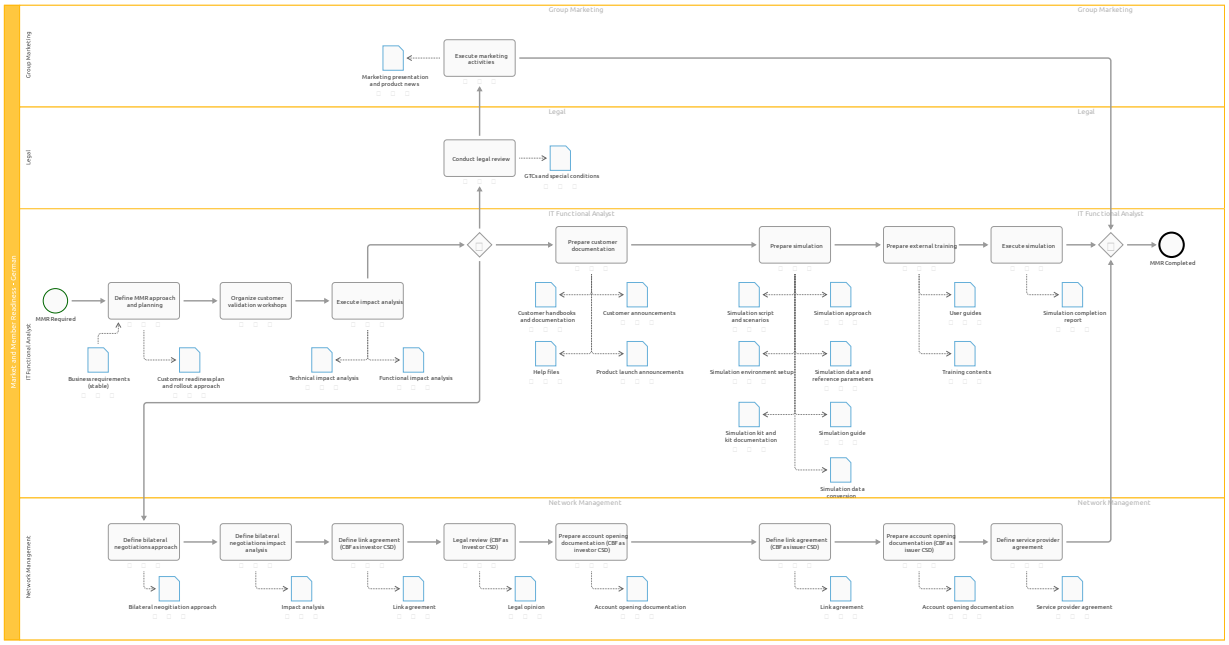
results. To avoid any loss of time they should start immediately after finalization of the specification.

- In parallel to running TAT activities, the fix-it team originated from the development resources and already established for STT, is responsible for resolving issues assessed and acknowledged in the prior determined processes and according to defined criteria. Rational decisions made by the test management should be taken concerning when to repeat test or proceed.
- Adequate computer and human resources should be available.

Internal

4.14 Market and Member Readiness (MMR)

4.14.1 German Domestic Market Module View

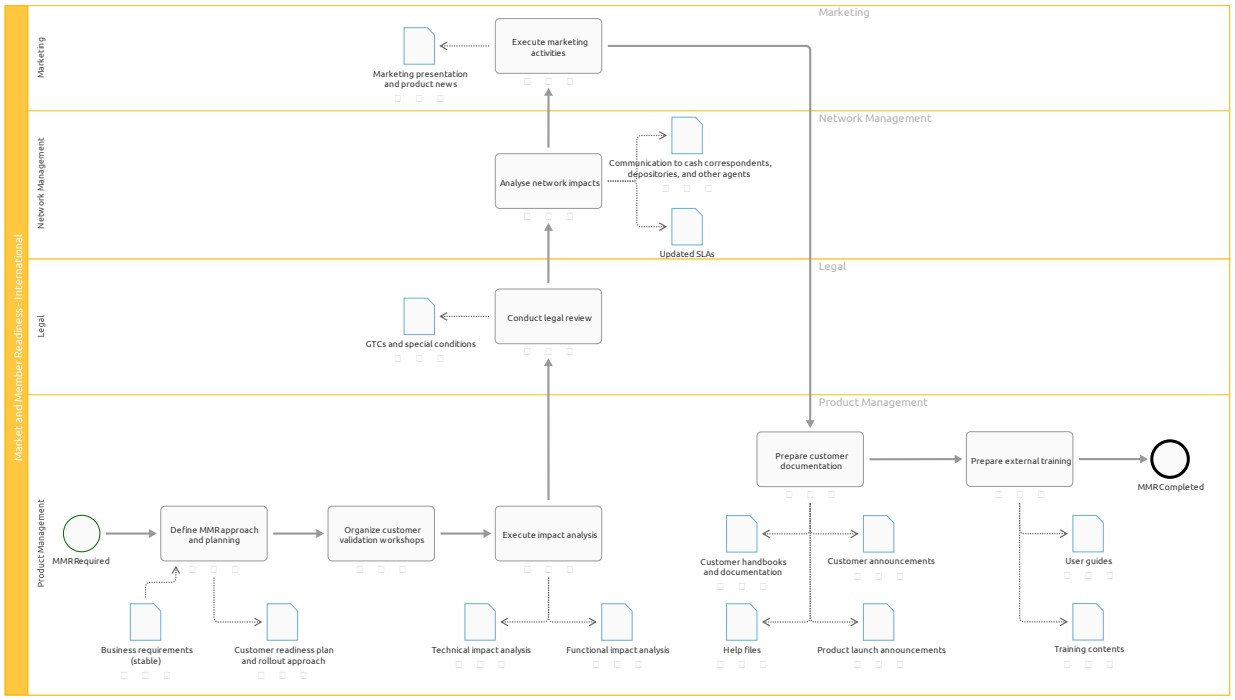


[10]

[10] Due to a rendering bug the gateway type symbols are not displayed. This model contains only parallel gateways.

MMR German RACI+S	Legal	Business Operations & Corporate Functions	Customer Service	Relationship Management	Network Management	Group Marketing	Product Management	IT Functional Analysts	IT Development
Define MMR approach and planning		S	C, S	C, S	C			A, R, S	C
Organize customer workshops								A, R	
Execute impact analysis		S	C, S	C, S	C			A, R, S	C, S
Prepare customer documentation			C	C	C, R			A, R	
Prepare simulation			A, R	C				R	C
Prepare external training			C, R	C, R				A, R	C
Execute simulation			A, R	C				R	C
Conduct legal review	A, R, S								
Execute marketing activities			C			A, R	C	C	
Define bilateral negotiations approach	S	S		C, S	A, R, S			S	
Define bilateral negotiations impact analysis	S	C, S		C, S	A, R, S			C, S	
Define link agreement (CBF as investor CSD)	R	C		C	A, R, S			C	
Legal review (CBF as investor CSD)	R				A, C			C	
Prepare accont opening documentation (CBF as investor CSD)	R	C			A, R			C	
Define link agreement (CBF as issuer CSD)	R	C, S		A, R, S	C			C	
Prepare account opening documentation (CBF as issuer CSD)		C		A, R				C	
Define service provider agreement	R	C			A, R, S			C, S	C

4.14.2 International Market Module View



Internal

MMR International RACI+S	Legal	Marketing	Network Management	Quality Management, Controls & Change Management	Product Management	Information Technology
Define MMR approach and planning		C	C	C	A, R, S	
Execute impact analysis		C	R	R	A, R, S	R
Legal review	A, R					
Analyse network impacts			A, R			
Execute marketing activities		A, R			R	
Prepare customer documentation		C	R	R	A, R	
Prepare external training				R	A, R	

4.14.3 Module Description

4.14.3.1 Overview

The Member and Market Readiness (MMR) module focuses on the external facing aspects of the new release. It identifies the impacts on Users (Members, Market, Customers, Vendors, etc.) and network aspects (Cash Correspondents, Depositories, Agents, etc.). It addresses the Marketing, information preparation, communication, publication and training required to prepare for using the new release from the initial launch. The purpose is to assist, support and track the technical, functional and organizational readiness of the users.

4.14.3.2 Prerequisites and Related Efforts

The planning and analysis for the MMR stage must begin as early as possible in the project timeline. Depending on the complexity of the project and severity of any impact to existing Customers or Members, a release notification/customer announcement may be required to the Market 6 months or more prior to implementation to allow for preparation on the customer's side. The initial input for planning and analysis is the PPI phase and the Business Requirements, once they are stable. It might be necessary to validate the release content and scope during PPI, BRA/SRA/and SDS phase with customers, either via bilateral meetings or via customer validation workshops.

The detailed Marketing and Customer information (Product Launch Announcement) will be developed based on SDS.

Close cooperation with all organizational areas (Business, Business Operations, IT Development and Delivery, Product Management & Design, functional area ownership (GSF, ISF, etc.), Change Management, Marketing, Relationship Management and Corporate Centre, Legal, Customer Service) that will be affected is essential to ensure that all input from analysts regarding impacts are included, issues addressed and communicated both internally and externally.

4.14.4 Key Considerations

Many areas of the organization will be involved in the MMR activities and the related activities must be carefully coordinated and communicated both internally and externally.

The project manager has overall responsibility for the MMR activities and should assign resource(s) to coordinate feedback from the relevant parties.

MMR is critical to ensuring that both the internal organization and external stakeholders are ready to begin using the new release as soon as it is implemented in production.

4.14.5 German Domestic Market Activities and Work Products

4.14.5.1 Define MMR Approach and Planning

- Customer Readiness Plan and Rollout Approach
- Defines the readiness conditions, content required and schedule of information to be delivered:
  - Marketing & Sales Approach:
    - Positioning of the new release Customers/Members.
    - Measurement criteria.
    - List of marketing activities (brochures, press releases, press conferences, information bulletins, product presentations, etc.).
    - Pricing plans.
  - Communication Plan:
    - Overview of all communication media and events used for communicating the release content and impact to customers.
  - Training Approach:
    - Specifies the general training concepts, scope, external and internal target groups, schedule, overview concept and content of training material.

4.14.5.2 Organize Customer Validation Workshops

Activity to validate/fine tune the envisaged functionality, technical and operational changes on the customer side to be held at the end of the SRA beginning of the SDS phase.

4.14.5.3 Execute Impact Analysis

- Functional and Technical Impact Analysis
- Document outlining all customer and network impacts identified from the Business Requirements.

4.14.5.4 Legal Review

- General Terms and Conditions and Special Conditions
- Review and potential update of those documents

4.14.5.5 Prepare Customer Documentation

- Customer Announcements
- Announcements to Customers about the upcoming release.
- Product Launch Announcements
- Identification of Customer facing changes from current release.

- Help Files
- Updates to Customer help text on user interfaces.
- Customer Handbooks and Documentation
- Updates to Customer documentation.

4.14.5.6 Execute Marketing Activities

- Marketing Presentation and Product News
- Specific documentation as identified in the Marketing & Sales Approach.

4.14.5.7 Prepare Simulation

- Simulation Approach
- This document describes the simulation approach including objectives, scope, concepts & methodologies, organizational aspects, dependencies and risks.
- The document will serve as a basis for the simulation team to plan, set-up and conduct all the work to be done for the simulation.
- The document contains fundamental planning and implementation information in the following areas:
  - Framework of the simulation
  - Time schedule for planning and execution
  - Participation in the simulation
  - Technical Set up of Simulation environment
  - Support from outside of the team
  - Responsibilities
  - Risk

Simulation Data and Reference Parameter

- The work product details the simulation data and parameter to be fed in, modified and maintained. It describes accurately the required data quality and the measures for quality control. To update the data basis correctly a frozen zone before simulation starts has to be initiated. Business Operations are responsible for entering data, whereas IT is responsible for converting data according to the specification.
- Content:
  - Detailed list of required simulation data.
  - Requirements for data quality.
  - Measures for quality control.
- Simulation Guide

This document provides key information for the external simulation participants and guidance instructions for planning simulation activities, and the simulation scripts for its execution:

- Simulation model and time frame

  - Registration form
  - Technical setup of the environment
  - Technical testing for preparation
  - Guidelines for guided and open simulation
  - Support by CBF prior to and during simulation for external participants
- Simulation Script and Scenarios
- Specific scripts or scenarios announced in the Simulation Guide which need to be executed on member side will be described and provided to the members.
- Content:
  - Simulation Scripts (if necessary).
  - Member activities for preparation and execution of the simulation.
- Simulation Environment Setup
- Preparation readiness of the simulation environment (infrastructure and software) as defined in the Simulation Approach and the Data Conversion Approach.
- Simulation Kit and Kit Documentation
- The Simulation Kit supports the installation of the front-end software on member-side as well as on internal systems. In addition, the simulation kit is supplied with an installation guide or manual. The activity is finalized after successful delivery of the software to all target receivers.
- Content:
  - Simulation front-end software.
  - Simulation kit software.
  - Installation Manual.
  - Delivery statistics.
- Simulation Data Conversion

Required reference data is complete and error-free and is converted according to the requirements and guidelines defined in the Simulation Approach and the Data Conversion Approach.

4.14.5.8 Prepare External Training

- User Guides
- Functional guide to introduce the Users to the new release including overview, access to system, description of input/output functions, procedures, tips and tricks, security advice and glossary of terms.
- Training Contents
- Overview, manuals, front-end software kit and associated manuals, technical training manual.
- E-learning: Availability of on-line demonstration of functionality to be launched.
- Customer Simulation Environment

- Availability of an environment for Customer Testing.
- There is an alternative to External Trainings which is a customer workshop or a customer presentation (during regular bilateral meetings/ customer user groups) might be held.

4.14.5.9 Execute Simulation

- Simulation Completion Report
- The Simulation Completion Report serves as declaration of agreement on the completion of the simulation based on the requirements and guidelines defined in the Simulation Approach. The document will be signed by the relevant and responsible people to execute the activities as well by the people responsible for operation of the production.
- Content:
  - Declaration of agreement.
  - Eventually caveats.
  - Simulation statistics (participants, irregularities, volumes, etc.).

4.14.5.10 Define Bilateral Negotiations Approach

- Bilateral Negotiations Approach

The purpose of this work product is to set the direction for the bilateral negotiations (e.g. between CSDs).

This involves consideration of alternative approaches to satisfy the requirements.

4.14.5.11 Define Bilateral Negotiations Impact Analysis

- Impact Analysis

Document outlining all impacts on entities part of the bilateral negotiations, impacts identified from the Business Requirements like:

  - Analysis of the ECB assessment
  - Eligibility of collateral

4.14.5.12 Define Link Agreement (CBF as Investor CSD)

- Link Agreement

This document is a new or an update of the agreement letter between the CSDs. It contains legal and operational sections.

4.14.5.13 Legal Review (CBF as Investor CSD)

- Legal Opinion

The Legal department will provide support to complete this deliverable, they will assess legal aspects related the creation or update of bilateral link. This can also include compliance matters and concerns, preparation and negotiation of new legal documentation, regulatory requirements, as well as requirements coming from other outside stakeholders.

4.14.5.14 Prepare Account Opening Documentation (CBF as Investor CSD)

- Account Opening Documentation

In case CBF is the investor CSD there could be a need to open new accounts with the CSDs.

4.14.5.15 Define Link Agreement (CBF as Issuer CSD)

- Link Agreement

This document is new or an update of the agreement letter between the CSDs.

4.14.5.16 Prepare Account Opening Documentation (CBF as Issuer CSD)

- Account Opening Documentation

In case CBF is the issuer, the CSD agreement will be driven by the existing relationship and agreement between the entities (similar to the creation of a new customer).

4.14.5.17 Define Service Provider Agreement

- Service Provider Agreement

In case additional service providers are required an agreement will be concluded with external parties such as clearing agents.

4.14.6 International Market Activities and Work Products

4.14.6.1 Define MMR Approach and Planning

- Customer Readiness Plan and Rollout Approach

Defines the readiness conditions, content required and schedule of information to be delivered:

- Marketing & Sales Approach:
  - Positioning of the new release Customers/Members.
  - Measurement criteria.
  - List of marketing activities (brochures, press releases, press conferences, information bulletins, product presentations, etc.).
- Pricing plans.
- Training Approach: Specifies the general training concepts, scope, external and internal target groups, schedule, overview concept and content of training material.

4.14.6.2 Execute Impact Analysis

- Functional and Technical Impact Analysis
- Document outlining all customer impacts identified from the Business Requirements.

4.14.6.3 Legal Review

- General Terms and Conditions

4.14.6.4 Analyze Network Impacts

- Communication to Cash Correspondents, Depositories and Other Agents
- Updated SLAs

4.14.6.5 Execute Marketing Activities

- Marketing Presentation and Product News
- Specific documentation as identified in the Marketing & Sales Approach.

4.14.6.6 Prepare Customer Documentation

- Customer Announcements
  - Announcements to Customers about the upcoming release.
- Release Notes
  - Identification of Customer facing changes from current release.
- Help Files
  - Updates to Customer help text on the user interfaces.
- Customer Handbook and Creation Market Guide
  - Updates to functional and Market related information in Customer documentation.

4.14.6.7 Prepare External Training

- User Guides
- Functional guide to introduce the Users to the new release including overview, access to system, description of input/output functions, procedures, tips and tricks, security advice and glossary of terms.
- Training Contents
  - Overview, manuals, front-end software kit and associated manuals, technical training manual.
  - E-learning: Availability of on-line demonstration of functionality to be launched.
- Customer Simulation Environment
  - Availability of an environment for Customer Testing.

4.14.7 Quality Keynotes

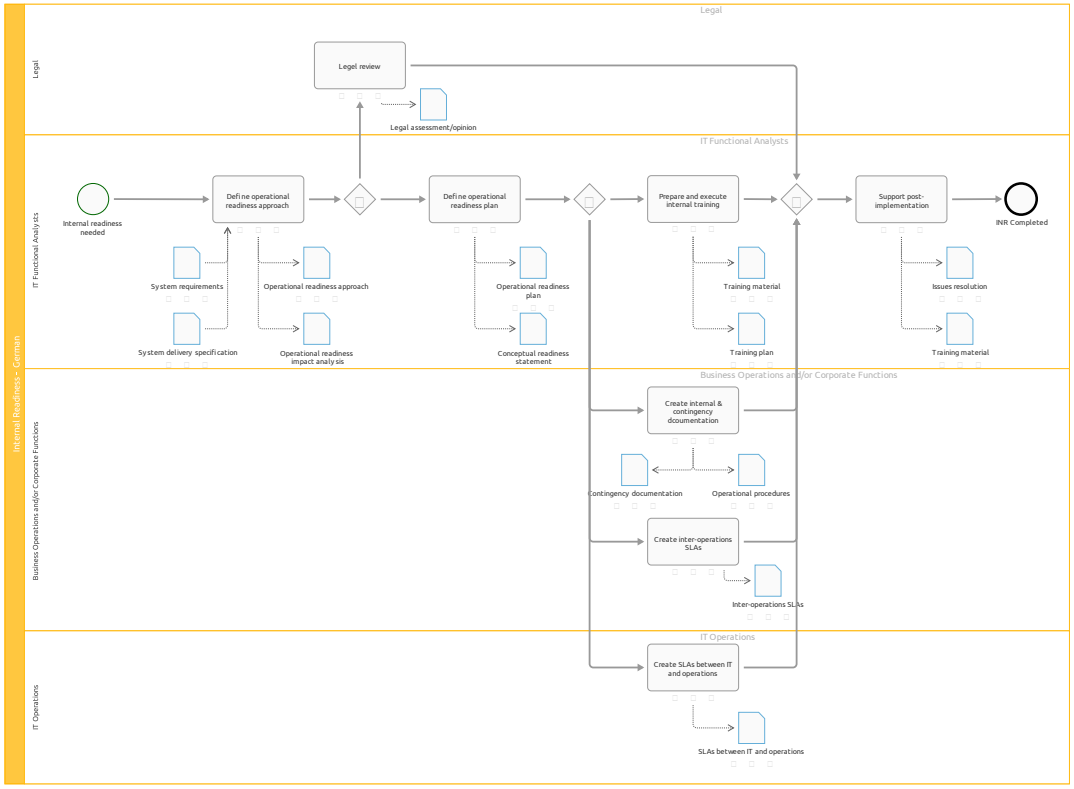
- The Market and Member activities must start as early as possible and be as complete as possible.
- Team working on this module must be able to understand the Members/Customers and be able to communicate with them at all levels and have a good understanding of the functional/operational changes (SDS document).
- Reinforce the acceptance of the release by completing marketing measurements (roadshows, Member/Customer visits, fair participation, specific give-away, management contracts, press releases, website updates, etc.)

- Reinforce the acceptance of the release by defining a comprehensive communication plan using existing customer communication channels and being upfront aligned with all relevant customer facing areas
- Member/Customer should show sufficient participation and commitment to training and preparation. Measurements to quantify and assess participation should be established.

4.15 Internal Readiness (INR)

This module is tailored to Clearstream Banking Frankfurt and Clearstream Banking Luxembourg and the role names used reflect that tailoring. Elements of the module may still be applicable to other entities' projects though specifics about roles and activities may require adaptation.

4.15.1 German Domestic Market Module View



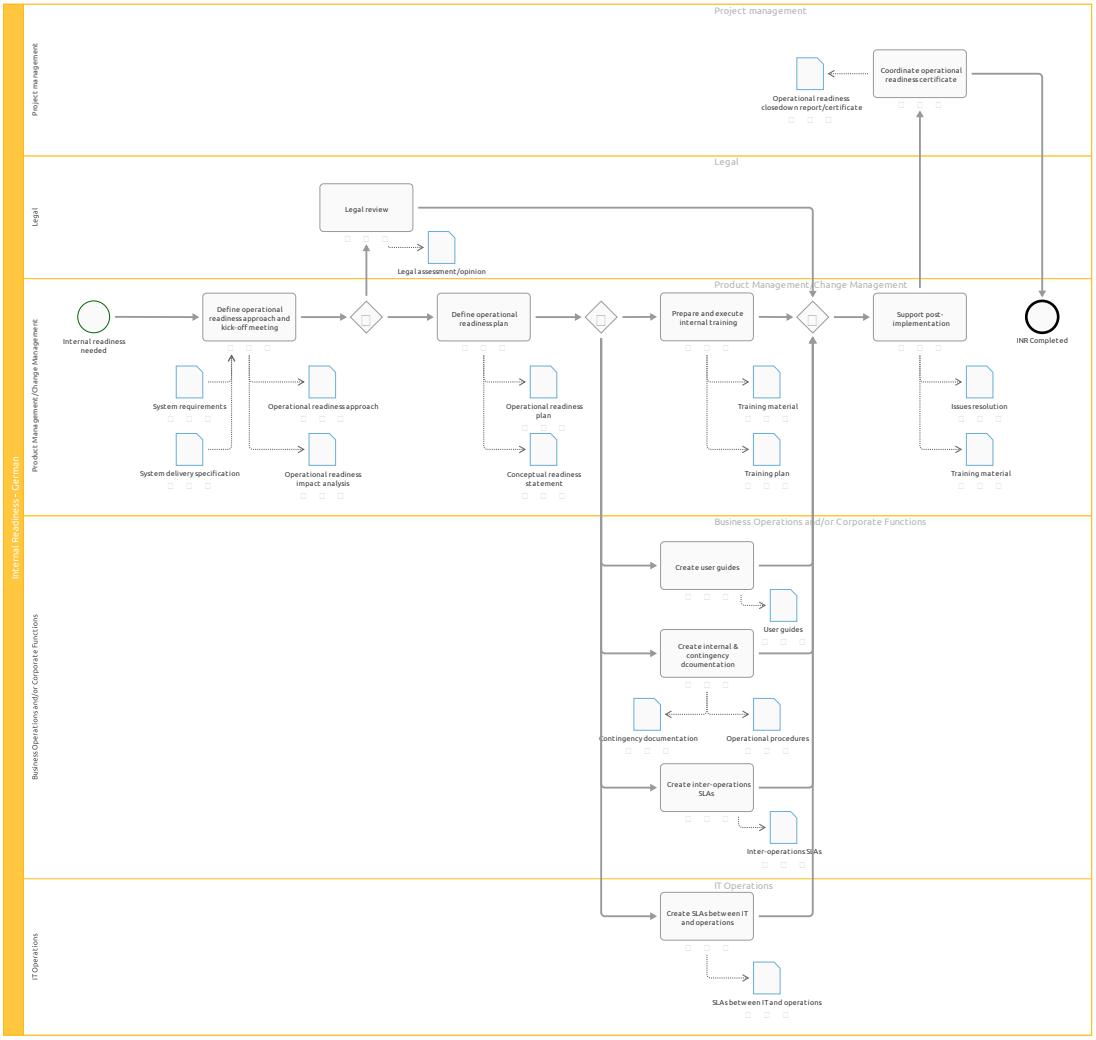
11

[11] Due to a rendering bug the gateway type symbols are not displayed. This model contains only parallel gateways.

INR International RACI+S	Legal	Change Management (for Core Product Projects)	Change Management (for IFS, GSF and Corp. Center projects)	Business Operations & Corporate Functions	Product Management (for Core Product Projects)	Product Management (for IFS, GSF and Corp. Center projects)	Information Technology	Project Management
Define operational readiness approach and kick-off meeting		A, R	C	C	C	A, R	C	
Define operational readiness plan		A, R	C	C	C	A, R		
Create internal & contingency documentation		R	C	A, C, S	C	R	C	
Create inter-operations SLAs		R	C	A, R, S	C	R	C	
Create SLAs between IT and Operations		R	C	C	C	R	A, R	
Prepare & execute internal training		A, R	C	C	C	A, R		
Create user guides				A, R			C	
Legal review	R			A, S	C	C	C	
Support post-implementation		A, R	A, C	C	C	R	R	
Coordinate operational readiness certficiate		S	S	S	S	S	S	A, R

Internal

4.15.2 International Market Module View



12

[12] Due to a rendering bug the gateway type symbols are not displayed. This model contains only parallel gateways.

INR International RACI+S	Legal	Change Management (for Core Product Projects)	Change Management (for IFS, GSF and Corp. Center projects)	Business Operaitons & Corporate Functions	Product Management (for Core Product Projects)	Product Management (for IFS, GSF and Corp. Center projects)	Information Technology	Project Management
Define operational readiness approach and kick-off meeting		A, R	C	C	C	A, R	C	
Define operational readiness plan		A, R	C	C	C	A, R		
Create internal & contingency documentation		R	C	A, C, S	C	R	C	
Create inter-operations SLAs		R	C	A, R, S	C	R	C	
Create SLAs between IT and Operations		R	C	C	C	R	A, R	
Prepare & execute internal training		A, R	C	C	C	A, R		
Create user guides				A, R			C	
Legal review	R			A, S	C	C	C	
Support post-implementation		A, R	A, C	C	C	R	R	
Coordinate operational readiness certficiate		S	S	S	S	S	S	A, R

4.15.3 Module Description

4.15.3.1 Overview

The module INR covers the planning, organization, control and execution of the internal service implementation. This module contains the identification of changes for Business Operations, Central Functions and Customer Services, the information of the relevant line and project organizations about the changes to conduct and the support or tracking of the change implementation. The operational readiness for the units and processes directly related to or responsible for the production execution of the system are also amongst the major tasks to be fulfilled, with a focus on Business Operations day-to-day workflows. Afterwards, the hand-over of the system to the operational and maintenance units will be realized.

4.15.3.2 Prerequisites and Related Efforts

In parallel and subsequent to the development of the software its "environment" has to be established. The Starting point are the Business Requirements.

Internal Readiness is established by working closely with the entire applicable internal organization and in coordination with the planning and set-up activities of Market and Member Readiness (MMR).

4.15.4 Key Considerations

To ensure a smooth and frictionless coordination between the involved parties and to enable an early incorporation of the staff responsible to operate the new Business function in production an Internal Readiness planning, coordination and management team has to be established within the project . In dependence of scope and size this team could be unified together with the institution responsible to manage the Market and Member Readiness. It is recommended that the scope is always defined in detail after the set up.

4.15.5 German Domestic Market Activities and Work Products

4.15.5.1 Define Operational Readiness Approach

- Operational Readiness Approach
- Operational Readiness Impact Analysis

- Based upon the review of all relevant project documentation (BRAs, SRAs, SDS etc.), Operational Readiness will perform the operational readiness impact assessment.
- The impact analysis will be delivered by the end of the specification phase.

4.15.5.2 Define Operational Readiness Plan

- Operational Readiness Plan

  This plan will document, in terms of time and efforts, all the activities to be performed in order to deliver and implement the Operational Readiness approach. The plan will have to be synchronized with and integrated into the overall business plan held by the Project Manager and will be the basis for the progress reporting to the Project Manager via different Committees. It contains a task list for the impacted Business Areas. Triggered by changes identified during the implementation it might be necessary to update this plan and its content.

- Conceptual Readiness Statement

  A Conceptual Readiness Statement may be used to align the impact and the resulting conceptual approach and tasks with the impacted Business Areas. The document will be signed by the persons responsible for operating the impacted Business function in production of the new release.

4.15.5.3 Legal Review

- Legal Assessment / Opinion

  This includes all internal and external legal documents that need to be drafted, amended and reviewed and are not covered in the MMR area (e.g. MMR documents: customer documentation like GTCs, CSD network documentation; e.g. not covered in MMR and therefore in INR responsibility: bilateral contracts with NCBs, non-IT service providers).

4.15.5.4 Create Internal and Contingency Documentation

- Operational Procedures

  Operational Readiness will involve impacted Business areas as early as possible in the process of the update of procedures and processes. Those Business areas will provide an inventory of the existing procedures and processes with the relevant information such as status (complete / incomplete) and the date of the last update. For all procedures/processes and the related documents Operations defines the owner.

  That information will be discussed with the relevant Business areas that should be able to make a first assessment of the procedures that will need to be updated, respectively be created for the upcoming change of functionality. This inventory will allow Operations / impacted Business areas to assess the number of resources needed to update or create such procedures in time for the launch of the new functionality and to plan this activity accordingly.

  Operational Readiness will provide any assistance needed by Operations / impacted areas to fully understand the functional change in order to achieve the timely update or the creation of such procedures and processes. Internal handbooks, manuals and documentation is however to be updated by the impacted Business areas themselves.

- Contingency Documents

Based upon adequate information received from the Project Manager on the system architecture, application interfaces and any production support arrangements agreed with IT Operations or external suppliers, Operational Readiness will update or lead the creation of Contingency Documentation, subject to final Business approval. Those Contingency Documents might need to be aligned with and communicated to external parties like NCBs, customers.

4.15.5.5 Create Inter Operations SLAs

- SLAs Inter Operations

  As a result of the scope debriefing session, Operations should be able to make a first assessment of the SLAs that will need to be updated, respectively be created for the upcoming change of functionality. This inventory will allow Operations / Business impacted areas to assess the number of resources needed to update or create such SLAs in time for the launch of the new functionality and to plan this activity accordingly.

  Operational Readiness will provide any assistance needed by Operations / impacted areas to fully understand the functional change in order to achieve the timely update or creation of such SLAs.

4.15.5.6 Create SLAs between IT and Operations

- SLAs between IT and Operations

  The owner for the update and maintenance of these SLAs is IT Account Management.
- Operational Readiness will act on behalf of Business Operations for reviewing updates made by IT to the SLAs, in line with the new functionality to be delivered and in consultation with the project management stream in charge of the non-functional requirements.
- Operational Readiness will be included in the reviewers of the existing SLAs and will when relevant, raise the need for new SLAs.
- This responsibility is dependent on the availability of updated SLAs that reflect the latest system status and services.

4.15.5.7 Prepare and Execute Internal Training

- Training Plan and Material

The training addresses Operations or other impacted Business areas, customer services, sales and management, e.g. Business Duty Manager. The impacted Business Areas are responsible for the Training. The team supports in defining the changes triggered by the project and supports the impacted Business Areas in identifying training needs and training scope. The project team offers on request Information Sessions to the impacted Business Areas. The Information Session focus on the changes triggered by the project.

After verifying the training requirements, training resources have to be allocated and the operation procedures are updated in relation to the release features. The content of the training material is edited according to didactical requirements, e.g. should incorporate exercises, examples, diagrams, etc.

Content:

- Training Plan:
  - Basic approach and scope
  - Training Schedule
  - Roles & Responsibilities
- Training Material:
  - Overview presentation slides (focus on the changes)
  - Functional training manual
  - Technical training manual (including installation and configuration processes)

4.15.5.8 Support Post-Implementation

IT Test Management and Change Management/Product Management and Design are responsible for organizing the post implementation support (on-site support, location of support & phone numbers, etc...), in consultation with the Project Manager.

- Issues Resolution

The issue log maintained by Change Management/Product Management and Design and IT Test Management will categorize incidents that have occurred post implementation, allowing Operational Readiness to identify any issues that relate to their activity (mainly related to lack of knowledge in using the new functionality properly). Operational Readiness will then take the necessary actions to remedy the shortcomings.

- Training Material
- The responsibility of knowledge transfer remains with Operational Readiness until one month after the implementation.
- Training requests coming after that date will be handled by Quality Management, Controls & Change Management - "internal training".
- Operational Readiness will provide the internal trainer with the existing training material. The internal trainer will be invited to attend the available awareness / training sessions in view of own preparation for coming trainings.

Deliverable: Operational Readiness Closedown Report or Implementation Readiness Statements

In addition to an overview of the project run and outcome and their compatibility with the original objectives, the work product supplies a list of open activities. The Operational Readiness Closedown Report or all Implementation Readiness Statements together serve as a final milestone to transfer responsibilities of Business functions to Operations. The document will be signed before launch by the team responsible for operating the impacted Business function in production of the new release.

4.15.6 International Market Activities and Work Products

4.15.6.1 Define Operational Readiness Approach and Kick-Off Meeting

- Operational Readiness Approach
- Kick-Off Meeting

- Together with all stakeholders involved, a Kick-Off meeting is organized in order to start the impact analysis.
- Operational Readiness Impact Assessment

Based upon the review of all relevant project documentation (BRAs, SRAs, SDS etc.), Operational Readiness will perform the operational readiness impact assessment.

The high-level impact analysis will be delivered at the end of the concept phase (if Operational Readiness is involved in due time) and the refined impact analysis by the end of the specification phase.

To support the Operability Test, efforts should ideally be coordinated with IT and Change Management/Test Management in view of providing an overall impact assessment to the impacted units / project team and should also be in line with the delivery of the IT Contract Offer.

4.15.6.2 Define Operational Readiness Plan

- Operational Readiness Plan

This document will describe, in terms of time and efforts, all the activities to be performed in order to deliver and implement the Operational Readiness approach. The plan must be synchronized with and integrated into the overall business plan held by the Project Manager and will be the basis for the progress reporting to the Project Manager to different Committees.

4.15.6.3 Create Internal and Contingency Documentation

- Operational Procedures

Operational Readiness will involve Business Controls as early as possible in the process of the update of procedures. Controls will provide an inventory of the existing procedures with the relevant information such as status (complete / incomplete) and the date of the last update.

That information will be discussed with the relevant Business areas that should be able to make a first assessment of the procedures that will need to be updated, respectively be created for the upcoming change of functionality. This inventory will enable Operations / impacted Business areas to assess the number of resources needed to update or create such procedures in time for the launch of the new functionality and to plan this activity accordingly.

Operational Readiness will provide any assistance needed by Operations / impacted areas to fully understand the functional change in order to achieve the timely update or the creation of such procedures.

- Contingency Documents

Based upon adequate information received from the Project Manager on the system architecture, application interfaces and any production support arrangements agreed with Operations or external suppliers, Operational Readiness will update or lead the creation of Contingency Action Plans (CAPs), subject to final Business approval.

This responsibility is dependent on the availability of updated CAPs that reflect the latest system architecture and interfaces.

If Unit Recovery Plans (URP) are impacted, Operational Readiness members will inform Operation Readiness members that any new application / process needs to be consider and added to the existing URPs.

4.15.6.4 Create User Guides

- User Guides

User guides are under the responsibility of IT.

In case of a new application, a user guide should ideally be provided by the Project Manager to the project/development team. This user guide should either be written by the developers responsible for the GUI design, and / or the external party having developed the new GUI.
In the case of existing applications, the Project Manager will perform an inventory of the existing user guides. The user guides will be updated by the project/development team. The Project Manager, the impacted business line and Operational Readiness will assess the need / impact on resources to update any User Guides that are out of date.

4.15.6.5 Create SLAs

- SLAs Inter Operations

As a result of the scope debriefing session, Operations should be able to make a first assessment of the SLAs that will need to be updated, or created for the upcoming change of functionality. This inventory will allow Operations / Business impacted areas to assess the resources needed to update or create such SLAs in time for the launch of the new functionality and to plan this activity accordingly.

Operational Readiness will provide any assistance needed by Operations / impacted areas to fully understand the functional change in order to achieve the timely update or creation of such SLAs.

- SLAs between IT and Operations

SLAs to be verified and updated if required.

Operational Readiness will act on behalf of Business Operations for reviewing updates made by IT to the SLAs, in line with the new functionality to be delivered and in consultation with the project management stream in charge of the non-functional requirements.

Operational Readiness will be included in the reviewers of the existing SLAs and will when relevant, raise the need for new SLAs.

This responsibility is dependent on the availability of updated SLAs that reflect the latest system status and services.

4.15.6.6 Prepare and Execute Training

Training Plan and Material

The training addresses Operations or other impacted Business areas, customer services, sales and management, e.g. Business Duty Manager.

After verifying the training requirements, training resources have to be allocated and the operation procedures are updated in relation to the release features. The content of the training material is edited according to didactical requirements, e.g. should incorporate exercises, examples, diagrams, etc.

Content:

- Training Plan:

- Basic approach and scope
- Training Schedule
- Roles & Responsibilities

- Training Material:
  - Overview presentation slides (focus on changes)
  - Functional training manual
  - Technical training manual (including installation and configuration processes)

4.15.6.7 Support Post-Implementation

IT Test Management and Change Management are responsible for organizing the post implementation support (on-site support, location of support & phone numbers, etc...), in consultation with the Project Manager.

- Issues Resolution

The issue log maintained by Change management and IT Test Management will categorize incidents that occur post implementation, enabling Operational Readiness to identify any issues that relate to their activity (mainly related to using the new functionality incorrectly because of lack of training). Operational Readiness will then take the necessary actions to remedy the shortcomings.

- Training Material

The responsibility of the knowledge transfer remains with Operational Readiness until one month after the implementation.

Training requests coming after that date will be handled by Quality Management, Controls & Change Management - "internal training".

Operational Readiness will provide the internal trainer with the existing training material. The internal trainer will be invited to attend the available awareness / training sessions in view of own preparation for coming trainings.

Deliverable: Operational Readiness Certificate.

In addition to an overview of the project run and outcome and their compatibility with the original objectives, the work product supplies a list of open activities. The Operational Readiness Closedown Report serves as a final milestone in the transference of responsibilities from Business functions to Operations. The document will be signed by the relevant responsible person who will carry out the project , as well as by the team responsible for operating the Business function in production of the new release.

4.15.7 Quality Keynotes

- The activities for Internal Readiness should start as early as possible with sign-off of the SDS.
- Parts of the project team members working on this module should have a deep understanding of the internal organization and processes, especially which processes belongs to which organization unit. The INR project team members should be self-reliant and able to produce the INR Approach and Impact Analysis.

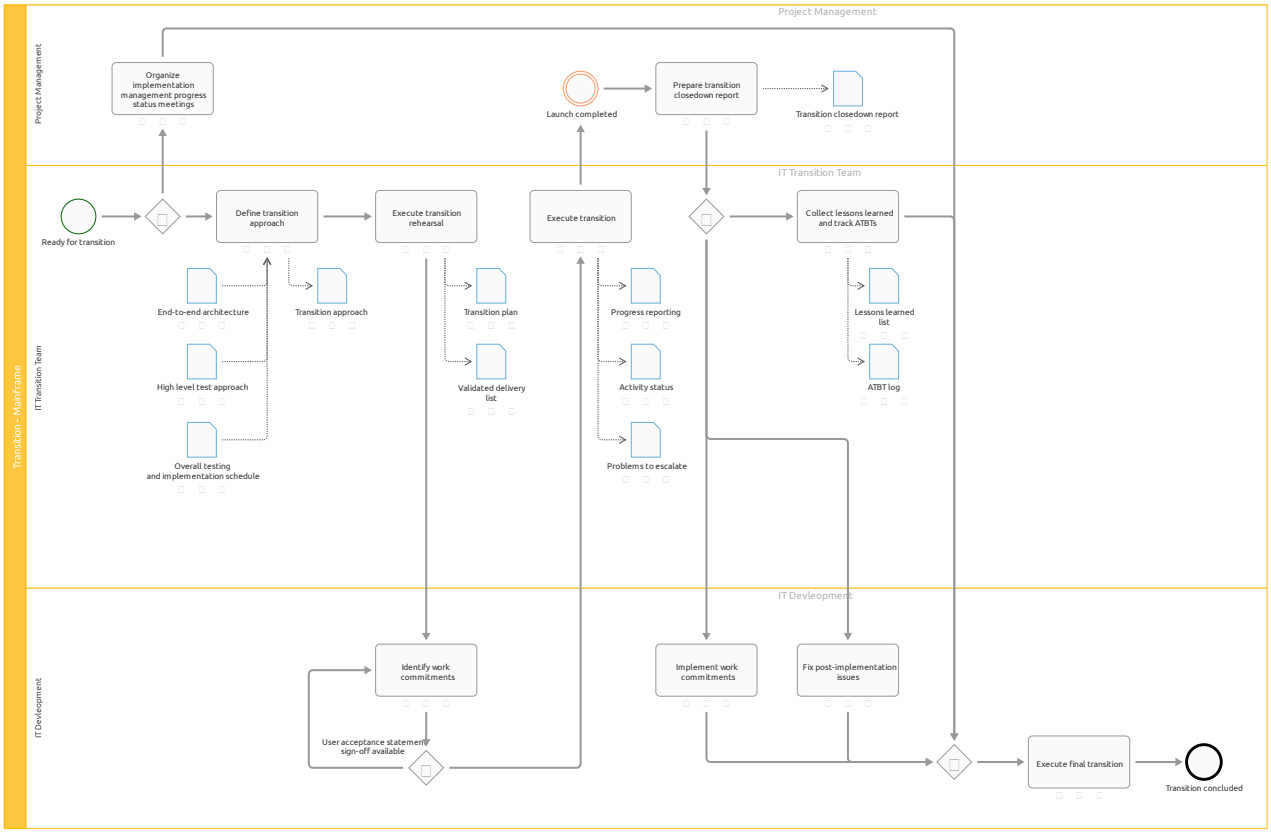
- It is recommended that staffing of the management unit due to coordinate the Internal Readiness is not actively involved in modules running in parallel (SDS, TSD, TPD, BAT and TAT).
- To reinforce the commitment of internal stakeholders to the new release and the activities required to introduce it, complementary communication activities have to take place to assure the acceptance of the entire company.
- To avoid any schedule and resource overlaps, the responsible person for Internal Readiness should establish a direct communication channel to other parallel running projects to be continually updated about any relevant impacts.
- Fixing of deficiencies of the new release is not in scope of the Internal Readiness. Internal Readiness focuses purely on the internal environment and structure in which the new release is embedded in future.
- Before authorizing the start of the production introduction, the system should be adequate in functionality and reliability and the operation departments should be able to run the new release. These matters should be assessed objectively on the necessary highest decision level.
- A sufficient time period should be provided for production introduction to ensure enough contingency time in case of irregularities (a weekend should be reserved for major releases). Also, a generous number of stand-by resources should be planned for production introduction and for a reasonable time period afterwards.
- Before handling over the new release to the operation staff, the documentation should be adequate, all outstanding points be documented and the operation staff capable and sufficient in number.
- A date should be scheduled for a post-implementation review.

Internal

4.16 Transition (TRA)

This module is tailored to Clearstream Banking Frankfurt and Clearstream Banking Luxembourg and the role names used reflect that tailoring. Elements of the module may still be applicable to other entities' projects though specifics about roles and activities may require adaptation.

4.16.1 Mainframe Module View

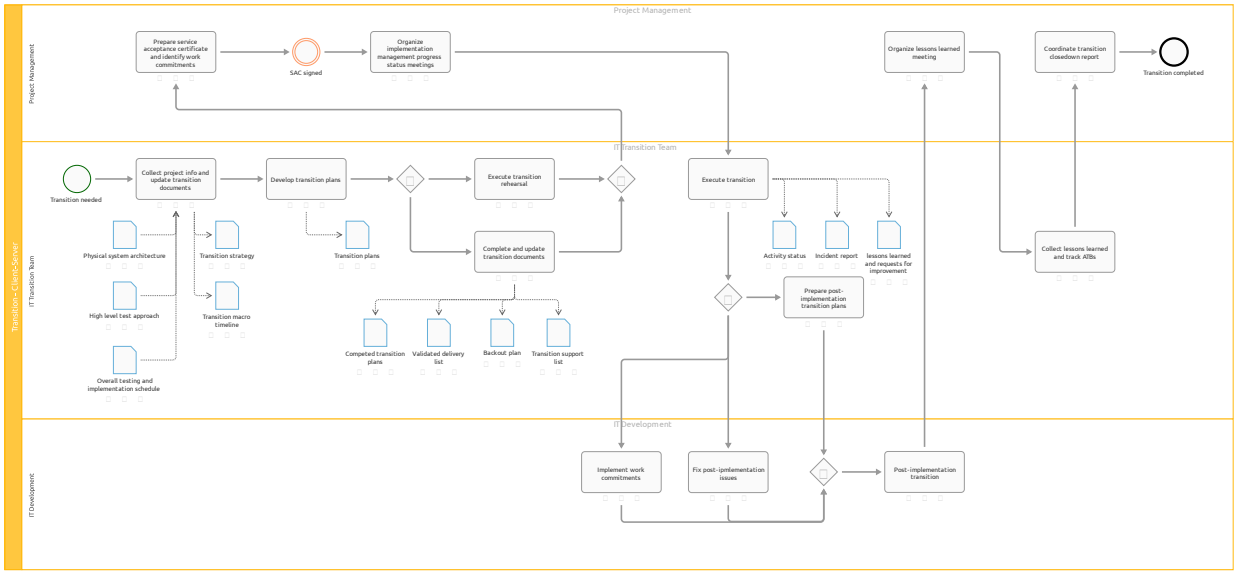


13

[13] Due to a rendering bug the gateway type symbols are not displayed. This model contains only parallel gateways except for the exclusive gateway after the activity "Identify work commitments."

TRA Mainframe RACI+S	Business Operations & Corporate Functions	Product Management	IT Implementation	IT Transition Team	IT Development	IT Operations	Project Management
Define transition approach	S	C, S	C, S	A, R, S	C, S	S	
Execute transition rehearsal	R	R	R	A, R, S	R	C	
Identify work commitments		C			A, R	C	
Execute transition	R	R	C, S	A, R, S	C	R	
Implement work commitments		C			A, R		
Fix post-implementation issues		C			A, R		
Collect lessons learned and track ATBTs		C	C	A, R, S	C	C	
Execute Final Transition	R	R			A, R	C	
Organize implementation management progress status meetings							A, R
Prepare transition closedown report			S				A, R

4.16.2 Non-Mainframe Module View



14

[14] Due to a rendering bug the gateway type symbols are not displayed. This model contains only parallel gateways.

TRA Non-Mainframe RACI+S	Change Management	Business Operations & Corporate Functions	Product Management	IT Implementation	IT Transition Team	IT Development	IT Operations	Project Management
Collect project info and update transition documents			C	C	A, R	C		
Develop transition plans			C		A, R	C		
Execute transition rehearsal	R	R	R	R	A, R	R	C	
Complete and update transition documents					A, R	C		
Execute transition	R	R	R	C	A, R	C	R	
Implement work commitments			C			A, R		
Fix post-implementation issues			C			A, R		
Prepare post-implementation transition plans			C		A, R	C		
Post-implementation transition	R	R	C			A, R	C	
Collect lessons learned and track ATBTs			C	C	A, R	C	C	
Prepare service acceptance certificate and identify work commitments								A, R
Organize implementation management progress status meetings								A, R
Organize lessons learned meeting								A, R
Coordinate transition closedown report				S				A, R

4.16.3 Module Description

The going live date of a project is a definitive deadline which can only be changed with considerable impacts on cost, delay and loss of credibility for the project team. This is the reason why a specific procedure has been put in place in order to ensure the final implementation.

This procedure is composed of the following phases:

- Implementation strategy definition, including cut over design, development and testing.
- Preparation phase, including planning, documenting and rehearsal of transition activities in a simulation environment.
- Execution phase, applying the modification on the production systems (usually on a weekend).
- Closedown phase, handling some post implementation activities.

4.16.4 Key Considerations

The necessity of a transition rehearsal has to be evaluated beforehand. The transition rehearsal verifies the necessary activities and their sequence of the introduction cutover process. The transition rehearsal has a technical focus mainly; nevertheless, the procedure and organization of the transition test have to be closely coordinated with the general transition plan documented in the transition approach.

4.16.5 Inputs

- Overall Testing and Implementation Schedule

Schedule maintained by IT needs to be agreed with all parties.

- Project Master Plan

Internal

First elements collected for the transition documents are:

- Project Code and Name (only for projects)
- Phases, main deliverables and milestones
- Target Launch Date
- Project Manager and direct reports

- PMM Minutes

Project Management Meeting Minutes. (only for projects)

It is recommended that the Project Manager invites a representative from the transition team to the PMM, or at least provide the minutes of PMM. The transition team should be aware of any changes in the project context.

PMM minutes are available in the official project folder.

- High Level Transition Plan

Transition Plan produced through System Delivery Specification:

- Cut Over time (end of day, night, etc.)
- High level view of systems impacted
- Functional Dependencies on impacted systems regarding cut over position
- List of environment changes
- List of impacted applications

Cut-over position reviewed and agreed by development and transition team.

It is recommended that the Project Manager dedicates a Project Management Meeting to review the cut over position with all parties involved. This should be done when SDS is being completed.

- Physical System Architecture

Infrastructure/hardware changes (hardware, security or infrastructure changes directly linked to the project).

- Technical Security Design

Includes all information security related documents mandated by the Information Security Framework.

- Technical Checklist

For each system impacted:

- People involved.
- Directory, disks, system requirements.
- DB hardware.
- Physical Data Model Changes.

- DB initial load.
- Binary /Job check.
- Application restart.
- Fall back.
- Specific verifications to validate changes.

A template is available for each platform.

- Updated Technical Checklist (for each system impacted)

Reviewed and updated after System Test is performed.

- Final Cut-Over Documents (for each system impacted)

Final version of the checklists which are published in addition with final transition plan.

- STT Recommendations

Feedback from the first cut-over performed for the System Testing.

Following information is complete and refined:

- List of infrastructure changes.
- List of impacted applications.
- Cut Over and initial load procedures.

Transition major tasks dependencies.

- TAT Recommendations

Feedback from cut-over collected during Technical Acceptance Testing.

Following information is refined:

- List of production configuration changes.
- List of tasks and dependencies.
- Cut-over procedures timing.
- Initial Load procedures timing.

- BAT Recommendations

Feedback from cut-over collected during Business Acceptance Testing.

Following information is refined:

- Complete list of infrastructure changes.
- Complete list of tasks and dependencies.
- Cut-over procedures timing.
- Initial Load procedures timing.
- Development & Business Verifications tasks & timings.
- Check procedures.

- Assigned Support Resources

Each unit concerned with the implementation of the project has to provide a list of resources, on site and on call, which will participate in the transition week-end.

4.16.6 Mainframe Activities and Work Products

4.16.6.1 Define Transition Approach

- Transition Approach

The key elements defining the strategy are the following:

  - The cut-over point of each application

This is the exact point where each application will be stopped in order to apply the migration to the new level of code. It can be different from one application to another, but must remain consistent and guarantee the overall integrity of the systems. This cut-over point is usually defined during the WE, either on Friday evening, or over the Friday night, or even on Saturday morning after all batch processing has been completed.

  - The point of no-return for the release

This is the exact point where a roll back is no longer possible.

  - The verification method of the transition

In order to insure technical and functional verification, some production activities or specific business inputs or other processing will be planned at out of working hours during the week-end. All this must be correctly validated and agreed with business users and eventually external counterparties.

  - The fall back possibilities and the plan at several steps of the transition

At several steps of the transition, severe problems may occur. In major cases those will be fixed and the transition will continue. In case of any show-stopper issues, all fall back possibilities must be anticipatively evaluated and planned.

  - The customer inputs and reports deliveries.

The implementation of a project will interrupt customer input and reports delivery for several hours. The generation of some reports may also be suspended in order to re-enable them after the migration, using the new level of code, either for verification purposes or simply for business requirements. The close and re-opening of the input and reporting channels will be part of the strategy and the plan.

4.16.6.2 Execute Transition Rehearsal

The items addressed during transition rehearsal are: risks/timing and readiness of the Transition Execution staff.

- Transition rehearsal takes place to minimize the risks associated with the introduction of new or modified components into the production environment.
- In addition, the time window available for applying those modifications is limited in order to reduce to a minimum the interruption of the normal production operations.

- The Transition Plan is assessed and adjusted through several test iterations against the following aspects:
  - Timing of each activity
  - Dependencies on external events
  - Transition tasks cross dependencies.
  - Feasibility of the ability of activities working in parallel.
  - End-to-End timing of the Transition.
  - Resourcing of the activities.
- Transition Plan

The Transition Plan is created at the same time as the Time Line. They are linked. It consists of:

- Global Transition Schedule (detailed description of transition and production activities):
  - Start Task ID
  - Last Task ID
  - Predecessor Task ID List
  - Activity Label
  - Ownership
  - Estimated Available Time/Start Time/End Time
  - Status
  - Actual Start Time/End Time
- Activity Description - Additional notes
  - Detailed Transition Schedule per system/application/team (detailed description of transition activities/checks to be performed)
  - For each transition activity
    - Description of transition tasks
    - System/application impacted or team
    - Task ID
    - Predecessor Task ID List
    - Activity Label
    - Ownership
    - Start Time
    - End time
    - Duration
    - Activity Description - Additional notes

It is the key reference for all participants in the transition weekend.

- Validated Delivery List

Validation of code to be delivered to production based on last deliveries received from Development teams.

4.16.6.3 Identify Work Commitments

Identify work commitments for either functionalities or issue corrections which are not incorporated in the application for the launch date. Each functionality or issue is reviewed prior to the launch and an agreement is found between Business and IT to incorporate these changes at a later stage (e.g. during post-implementation or in a subsequent release).

4.16.6.4 Execute Transition

It covers all activities occurring on the production environment in order to bring it to the correct level defined by the project.

Testing teams support the readiness of IT Operations and the transition execution, to ensure an optimal handover to IT Operations and Business Teams.

The execution team registers the start and end time of each transition activity. The transition team validates the entries and monitors the progress.

Any delay in a task may have impacts to be assessed in order to keep safe the overall Transition.

In case of critical delays, the status and issues are reported to the on-site management team in order to take appropriate decisions.

- Activity Status

Information that a transition activity in the transition plan is still to be done, ongoing or completed.

Actual end time is recorded in the Transition Plan.

- Progress Reporting
- Problems to Escalate

4.16.6.5 Implement Work Commitments

Implementation of work commitments planned to be released in the post-implementation.

4.16.6.6 Fix Post-Implementation Issues

Implementation of defects fixes, for defects that have been met in post-implementation tests.

4.16.6.7 Collect Lessons Learned and Track ATBTs

To aggregate and filter the different lists of lessons learned coming from any Transition Participants.

Once validated by the implementation management, a list of ATBTs is created and distributed to the relevant teams. The Transition team is responsible for tracking the ATBTs statuses.

- Lessons Learned List

Consolidated lessons learned from the successive transitions.

- ATBT Log

Actions to be taken in order to improve the end-to-end transition process.

e.g.: update checklist; adapt template; enhance the support tools; etc.

4.16.6.8 Execute Final Transition

Some implementation activities are scheduled after the launch week-end, usually on the first few days of the following week. These tasks are part of the global transition process and therefore are managed the same way as the transition week-end activities.

During the "post-implementation" period, production problems are managed by the production support, with the help of development. For major projecst, development and testing provide additional support during the next few weeks following the implementation. In addition, a test environment is kept one or two weeks after the implementation in order to reproduce some production issues, to investigate and to test production fixes before applying them in production.

Deliverable: Transition Closedown Report

Transition performance assessment:

- Comparison between planned and actual.
- Analysis of issues encountered during and after the transition.
- Proposed solutions and actions to be taken.

4.16.7 Multiple systems configuration Activities and Work Products

4.16.7.1 Collect Project Information and Update Transition Documents

Collect and consolidate any project information related to the transition

- Main environment changes:

Infrastructure/HW changes are usually known at the beginning of the project; in general, it concerns hardware, security or infrastructure changes directly linked to the project. In some projects which includes new applications or a completely new infrastructure, they are defined at the early stages of the project (Concept/Specification) and all consolidated during the testing phases.

- Impacted applications:

Usually, the complete list of the applications impacted by the projects are known at the early stages of the project initialization process, but it may occur that other application changes are identified later during the testing phase.

- Changes in the project context:

Changes might require an update to the high-level Transition Team Plan and control the transition cut-over position. For each existing application impacted in a project, there is at least one document which describes the technical checklist of the implementation activities. If major changes occur in the application, a user guide is also produced for production support and additionally the business users.

- Transition Strategy

- The key elements defining the strategy are:
  - The cut-over point of each application:

This is the exact point where each application will be stopped in order to apply the migration to the new level of code. It can be different from one application to another, but must remain consistent and guarantee the overall integrity of the systems. This cut-over point is usually defined during the WE, either on Friday evening before the NTP BOX1, or over the Friday night between NTP boxes, or even on Saturday morning after all batch processing is completed.

  - The verification method of the transition:

In order to insure technical and functional verification, some production activities or specific business inputs or other processing will be planned at an out of business time during the week-end. All this must be correctly validated and agreed with business users and eventually external counterparties.

  - The fall back possibilities and plan at several steps of the transition:

At several steps of the transition, severe problems may occur. In major cases those will be fixed and the transition will continue. In the case of a show-stopper problem, all fall back possibilities must be anticipated, evaluated and planned.

  - The customer inputs and reports deliveries:

The implementation of a project will interrupt customer's input and reports delivery for several hours. Some report generation may also be suspended in order to re-enable them after the migration, using the new level of code, either for verification purposes or simply for business requirements. The close and re-opening of the input and reporting channels will be part of the strategy and the plan.

- Transition Macro Timeline

Set of project information collected by the transition team during development in order to define macro tasks of the time line.

Timing and dependencies not defined.

4.16.7.2 Develop First Draft of the Timeline

The first draft of the Time Line represents the sequence and duration of activities at a high level, with a first notion of duration. This enables a global view of the transition so that a first level of optimization can be made.

The Transition Plans contains the Transition Timeline and the Detailed Plan documents. These documents are the key documents which will be used during the transition week-end and contain specific information for this purpose.

- Draft Transition Plans

First version of the Transition Plans based on development and testing inputs from PT and STT cutover executions.

- Transition Timeline

Time Sequence of transition and production activities organized to be compliant with other parallel activities and sequence constraints.

Each transition activity is described as follows:

- ID
- Label
- Start date and time
- End date and time
- Application

The official version is available on TII Web.

- Transition Plan

It consists of:

- Global Transition Schedule (detailed description of transition and production activities)
  - Start Task ID
    - Last Task ID
    - Predecessor Task ID List
    - Activity Label
    - Ownership
    - Estimated Available Time/Start Time/End Time
    - Status
    - Actual Start Time/End Time
    - Activity Description - Additional notes
- Detailed Transition Schedule per system/application/team (detailed description of transition activities)
- For each transition activity
    - Description of transition tasks
    - System/application impacted or team
    - Task ID
    - Predecessor Task ID List
    - Activity Label
    - Ownership
    - Start Time
    - End time
    - Duration
    - Activity Description - Additional notes

It is the key reference for all participants in the transition weekend. The official version is available on TII Web.

4.16.7.3 Refine Transition Plans

Based on more accurate information coming from BAT and TAT, the subsequent versions of the Transition Plans are built which includes all Timeline activities in addition to more detailed information.

- Updated Transition Plans
- Intermediate versions of the Transition Plans based on TAT (Technical Acceptance Testing) and BAT (Business Acceptance Testing performance) information inputs.

4.16.7.4 Execute Transition Rehearsal

The items addressed during transition rehearsal are: risks/timing and readiness of the Transition Execution staff.

- Transition rehearsal takes place to minimize the risks associated with the introduction of new or modified components into the production environment.
- In addition, the time window available to apply the modifications is limited in order to reduce to the minimum the interruption of the normal production operations.
- The Transition Plan is assessed and adjusted through several test iterations against the following aspects:
  - Timing of each activity
  - Dependencies on external events
  - Transition tasks cross dependencies
  - Feasibility in terms of parallelism of activities
  - End-to-End timing of the Transition
  - Resourcing of the activities

4.16.7.5 Complete and Updated Transition Documents

The documentation required for the Transition Execution and Rollout is finalized, validated and adjusted during the different test iterations taking place in the Transition Rehearsal.

For the launch week-end, resources are requested in order to implement the project. Each involved party is contacted and has to provide the needed resources for the week-end. Resources are allocated based on two different ways.

- Completed Transition Plans

Based on the last detailed information coming from each Transition cycle, last version of the Transition Plans is built.

- Validated Delivery List

Validation of the code to be delivered to production based on last deliveries received from Development teams.

- Back Out Plan

Activities to be performed to restore the service if the implementation fails.

Structure is the same as Transition Plan if it has been tested specifically during STT or, if not, only a summary of activities to be performed.

- Transition Support List

List of resources assigned for transition execution and support.

  - Team.
  - System/application.
  - Resources (Unit, Userid, Function, Phone, Mobile, On Site/ On Call 1, On Call 2, Location, Assigned period).

4.16.7.6 Execute Transition

This covers all activities occurring on the production environment in order to bring it to the correct level defined by the project.

Testing teams support the readiness of IT Operations and the transition execution, to ensure an optimal handover to IT Operations and Business Teams.

The execution team registers the start and end time of each transition activity. The transition team validates the entries and monitors the progress.

Any delay in a task may have impacts to be assessed in order to keep safe the overall Transition.

Plantool (Win32 Application) has been deployed in production in order to either permit people to send the statuses of the tasks or to provide a real-time of the timeline to all participants.

In case of critical delays, the status and issues are reported to the on-site management team in order to take appropriate decisions.

- Activity Status

Information that a transition activity in the transition plan is still to be done, ongoing or completed.

Actual end time is recorded in the Transition Plan.

- Incident Report

Incident encountered when performing a transition activity.

It is assessed first by the team responsible for this activity. Transition team and testing team are informed of the issue.

If the issue is solved, the activity status is updated in the transition plan. The transition records it for further analysis. If not, escalation procedure is initiated.

- Lessons Learned and Requests for Improvement

Inputs from all teams involved in the transition about the issues encountered during and after the transitions, suggestions for improvements and lessons learned from IT Management.

4.16.7.7 Implement Work Commitments

Implementation of work commitments planned to be released in the post-implementation, as described in the SAC.

4.16.7.8 Fix Post-Implementation Issues

Implementation of defects fixes, for defects that have been met in post-implementation tests.

4.16.7.9 Prepare Post-Implementation Transition Plans

Prepare Transition Plans for the post-implementation, if required.

4.16.7.10 Post-Implementation Transition

Some implementation activities are scheduled after the launch week-end, usually on the first few days of the following week. The tasks are part of the global transition process and therefore are managed the same way as the transition week-end activities.

During the "post-implementation" period, production problems are managed by the production support, with the help of the development. For major projects/, development and testing provide additional support during the following one or two weeks after the implementation. In addition, a test environment is kept one or two weeks after the implementation in order to reproduce some production issues, to investigate and to test production fixes before applying them into production.

4.16.7.11 Collect Lessons Learned and Track ATBTs

To aggregate and filter the different lists of lessons learnt coming from any Transition Participants.

Once validated by the implementation management, a list of ATBTs is created and distributed to the relevant teams. The Transition team is responsible for tracking the ATBTs statuses.

- Lessons Learned & ATBTs

Consolidated lessons learned from the successive transitions and actions to be taken in order to improve the end-to-end transition process.

Deliverable: Transition Closedown Report

Transition performance assessment:

- Comparison between planned and actual.
- Analysis of issues encountered during and after the transition.
- Proposed solutions and actions to be taken.

4.16.8 Project Management Activities

4.16.8.1 Information Security Risk Register Update

In case the information security Risk Assessment has revealed residual risk the information risk register has to be updated by the impacted Information owner. It is the project managers' responsibility to ensure risk register update during the transition.

4.16.8.2 Prepare SAC and Identify Work Commitments (ICSD specific)

In the Service Acceptance Certificate, work commitments are identified.

Work Commitments are functional or technical requirements that were planned in the scope of the release and in the IT Contract Offer, but that have not been developed or have been tested and were unsuccessful.

A deadline needs to be given in the SAC for these work commitments. It can be either in the post-implementation, or in another release.

- Service Acceptance Certificate (SAC)

The certificate provides an explicit written confirmation of the successful completion of all necessary steps to enable the production launch, as well as the written order to launch production on the foreseen date.

This work product will be prepared by the project manager, who will then double-check that all relevant documents have been properly approved and no blocking open issue remains. It will be provided by the project manager, and final approval will be given by the product owner, the project owner, the managing director IT delivery and the managing director application development of the relevant application development sub-division.

4.16.8.3 Organize Implementation Management Progress Status Meetings

- Report Progress Status Meeting

During the transition week end, the project manager has regular coordination status meetings with stakeholders, these meetings, organized every 2 or 3 hours, perform the following:

- Present latest refresh of status visually on a screen.
- Present the summary of issues & delays and their impacts & dependencies.
- Decide the action points for the next period.

Decision / Action

Decision to execute contingency action or activate the roll out plan.

- Report Status and Escalate Issues Meeting

This meeting, organized with IT Management, aims to:

- Present last refresh of status visually on a screen.
- Present the summary of issues & delays and their impacts & dependencies.
- Decide the action points for the next period.

Decision / Action

Decision to execute contingency action or activate the roll out plan.

4.16.8.4 Escalated Issue - Progress Reporting

4.16.8.5 Organize Lessons Learned Meeting

- Lessons Learned Meeting

This meeting is organized after the implementation and aims at analysing how the implementation went, in order to collect lessons learned and improve the process.

Deliverable: Project Closedown Report

The main objective of the closedown report is to review the end-to-end execution of a release, with a view to improving in the future.

The report will review the results of a release, compared with what it set out to do. It will not attempt to compare between releases.

The closedown report will avoid any form of audit on procedures or staff members.

Each recommendation will show how to reduce costs, increase quality, and reduce risk or combinations of these items.

The document structure will be based on the SUMMIT Modules, plus a section for Project Organization and Administration (joint teams, tools, issue tracker, workflow tools, collaboration and use of service providers, etc.).

Any topics that remain open from the release's scope have to be documented in the Closedown Report. If open items remain in a workflow tool (e.g. JIRA), these items have to be documented properly regarding

- reason for not having completed the item in the release
- indication of the relevant management approval (e.g. Steering Committee minutes)
- description of the further treatment of this item (e.g. hand-over to following release)

4.16.9 Quality Keynotes

- Transition preparation activities generally start at the beginning of the project, defining the implementation date, impacted systems and eventual new or decommissioned components.
- The execution of the transition rehearsal is not reasonable before the new release has been proven to be stable during the module STT. Furthermore, if a simulation during MMR is planned, transition procedures could be trained and thus a transition rehearsal could possibly be executed before or during the simulation.

5 Glossary

Acceptance Testing: Any test performed on a system version whose purpose is to verify that the version meets the user-defined objectives for that portion of the system.

AD: Application Development.

Application: A facility, which provides automated support to a set of functions, which form part or all of a business system.

Application Repository: The application repository is a central inventory where all applications must be registered.

Application Life Cycle: Usually refers to that portion of the systems life cycle that deals with an individual project. Includes Analysis, Design, Development, Installation, and Maintenance.

Architecture: The physical configuration of application residing on or distributed across a network of hardware and its associated operating system software. It may be regarded as the blueprint for the executable system(s). Architecture may be depicted graphically showing the layers of hardware, system software, middleware, and application software by physical location.

Batch: That general category of computer system that (1) interacts little, if at all, with the operator or user, (2) processes a stream of identical (or very similar) data, and (3) processes its inputs at a scheduled time and at its own speed, regardless of when the data first becomes available.

Batch Processing: An automated approach in which data items to be processed are grouped and then either coded U or sequenced prior to processing.

Benchmarking: A technique for comparing an organization's internal performance, products, and services to the external standards of excellence by quantitative and qualitative measures. This information is then used to attain and sustain best in class performance.

BIS: Building Institutional Skills - initiative within Deutsche Börse Group to consolidate, document and establish best practices.

Black Box Testing: A testing procedure that validates that a particular function is being done correctly by simply validating that, given a particular input, the output is correct. Tests performed without regard to the actual program structure.

Bug: A defect, error, or mistake; a glitch or anomaly. First used by hardware engineers to indicate an unspecified problem. Said to have originated with an early hardware engineer who found an actual insect to have caused a hardware failure. Both the term and the story behind it are generally attributed to Grace Hopper.

Change Management: A systematic process of aligning the organization's people and culture with changes in business strategy, organization structure, systems, and processes resulting in ownership of and commitment to change, sustained and measurable improvement, and improved capability to manage future change.

Code Review: The process of reading through programs (code) and discovering ways that the code does not meet the requirements. Usually refers to a process that is less formal than a structured walkthrough.

Critical Path: That patch through a precedence network that connects all the activities that have no slack time. This path defines the shortest possible time that the overall activity will take, and indicates clearly those activities which, if delayed, will delay the completion of the overall activity.

Data: The raw material required by the process in order to accomplish its task, the basic pieces of an event. That material that describes real-world things or events and which serves as a basis for discussion and/or decisions. In the information systems arena, this material is represented symbolically (as numbers, letters and/or special characters) so it can be processed more readily (and its meaning understood more clearly). A fact, observation, or detail.

Data Base Schema: Any of the many diagrams that illustrate how records are logically organized in a file or data base which may be as a hierarchy, a network, or as a set of tables.

Data Dictionary: A compilation of definitions for all data stores, data elements, data flows, and processes of a system.

Data Flow: A data item that is composed of more than one data element; a parcel of data that moves between processes as well as to and from terminators on a data flow diagram.

Data Flow Diagram: A graphic tool used to show the flow of data through an information system and the functions performed on and with that data. A number of graphical approaches exist. The most widely-used, developed, and popularized by Sarson and Gane, uses four constructs to represent the interrelationships between data and processing logic. The model defines the system in terms of sources and receivers of the system's data, data transformations, data being held, and data in motion.

Data Model: A graphic representation of the data structure of an information system; examples are entity-relationship model and the data structure diagram. A diagram used to illustrate data and data relationships, it is the primary diagramming tool of data analysis.

Data Set: A group of stored data elements; which should be attributes of the same object.

DBAG: Deutsche Börse AG

DBG: Deutsche Börse Group

DBS: Deutsche Börse Systems

Debug: To locate and remove errors from a computer system (either software or hardware).

Defect: Anything that may cause an information system to operate incorrectly. The term emphasizes that the responsibility for such things lies with the author/creator of the system.

Deliverable: see First Level Work Product definition. This is a synonym for First Level Work Product.

Detailed Plan: In addition to the master plan a detailed plan is worked out for each consecutive phase. For example, the "SRA Phase Plan" describes the required resources (budget, manpower) for the SRA phase. It should include a detailed phase schedule.

Development Phase: During this phase the system is actually constructed (coded and tested). The output of this phase is a system that is running acceptably stand-alone and for which approval to begin implementation (installation) has been obtained.

Internal

Enhancements: Changes to a system to fulfil new or unforeseen requirements that involves either new programs or new file structures, or both.

Enterprise Data Model: The data model component of a Business Information Model.

Error: A mistake; a situation or condition that is not supposed to occur. The occurrence of such a condition is usually interpreted to mean that an information system has malfunctioned.

Hardware: The computer and its peripherals.

Infrastructure: The "support" mechanisms necessary to build, extend, and maintain an architecture. Mechanisms include people (resources), standards, procedures, and guidelines. Often such standards and guidelines will constrain the design of an architecture by limiting the sets of supportable hardware/software platforms afforded to a given business unit or an enterprise as a whole.

IT Architecture: A framework for the future IT environment.

Logical Data Model: The name given to the entire body of data definition, including graphics, developed for the new information system just prior to the file/database design effort.

Methodology: The basic approach providing the basis for a strong logical and a consistent work structure for any assignment. A way of doing something; specifically, a set of procedures, deliverables, roles, tools, and techniques that define, organize, implement, and manage a systems development effort appropriate to the business and the information system being built. A methodology is a pattern for developing and using information systems that can be followed in whole or in part, as necessary, to achieve results.

Metrics: key numbers and statements, which enable the assessment of quality and performance in application development processes. Statement and documentation of a basis-set of numbers (effort, duration, line of codes, test cases, test errors, etc.)

Project: A planned undertaking of well-defined scope, outputs and ordering of tasks.

Project/development activity manager: Any role (including a development activity manager) that is currently responsible for determining what subordinates will be doing in the next time period, reviewing progress reports from these subordinates, taking corrective action as necessary to solve problems reported, scheduling and conducting status meetings, preparing status reports, and ensuring that change control procedures are being followed.

Project Plan: it provides the basis for effective project control during the first phase and therefore has to document project factors in a manner sufficient to establish realistic expectations and commitment from the upper levels of management.

Project/development activity Status Meeting: A meeting of project participants called by the project/development activity manager to either disseminate information to the group as a whole or assess project progress as a whole. Problems affecting the group as a whole may also be discussed.

Prototyping: The process of constructing or simulating an information system, in whole or in part, specifically for the purposes of either uncovering requirements or validating potential solutions to them.

Regression Testing: Rerunning test cases that have been previously executed correctly in order to ensure that modifications to the system have not introduced new errors. Rerunning test cases during top-down, integration testing to ensure that at each level of integration errors have not been introduced into the tested system unintentionally. This involves performing tests that extend beyond the effort needed to simply test the integration of the current module.

Relation: A file in a relational data base. A two-dimensional table in which the left-to-right order of columns were changed, or the sequence of rows were changed, no information would be lost. Any data structure proceeding through the normalization technique.

RFP: Request For Proposal. An invitation from a buyer to a potential vendor for the vendor to submit a bid for a product or work to be supplied.

Service Level: That level of information support provided to an external entity as measured by system accessibility, response time, data freshness, etc. One of the characteristics used in strategic design to refine the design units.

Sign-Off for Understanding: Signatories confirm only that the document is complete and accurate with the right level of detail and quality and that they understand the content sufficiently to proceed with further steps. They are not expected to pass judgement on the content.

Solution: A solution is an implementation of people, processes, information and technologies in a distinct system to support a set of business or technical capabilities that solve one or more business problems

Solution Architect/IT Archictect: Project role, responsible for designing a solution architecture, which is in line with the group standards and guidelines and therefore fits into the enterprise architecture.

Issue Tracker: Tool to support the software change management process and other processes, such as Jira or SAP

Software Engineering: The application of a well-defined set of life cycle tools and techniques to do the general activities of analysing, designing, and developing information systems. This implies that the tools and techniques are used in a consistent way; that each tool or technique is to be used in a rigorous fashion; that the tools and techniques provide the capability to break the effort into portions and deal with one at a time; and that we have the capability of being able to deal with system characteristics independently and at various levels of detail.

Specification: A clear, complete, and accurate statement of requirements, as in "requirements specification." Also, the procedure to be followed to determine if the requirements are met.

Stress Testing: Tests designed to confront programs with abnormal situations, tests that execute a system in a manner that demands resources in abnormal quantity, frequency, or volume.

Terms of Reference (ToR): Terms of Reference document the general rules along which the decision bodies (Technology / Business Committees, ...) within Deutsche Börse Group operate and take their decisions. It lists the documents they expect to base their decisions on, and the sequence of when and by whom these documents must be provided.

Test Case: Any situation or occurrence which will need to be tested to ensure that the information system is functioning correctly. Test cases should result in the highest likelihood of finding the

most errors with the minimum amount of time and effort. A set of test data executed in one run to test at least one requirement.

Test Plan: The overall plan for the testing effort. The plan will detail who will be involved, how much testing will be done, what criteria will be for a successful test, and how long the testing will take. At a detail level the test plan covers the area to be tested, the types of tests that will be run, the test scripts and the test environments, and execution plans for the effort.

User: Any person who accesses a computer information system, from a clerk to the president.

User Acceptance Testing: Those tests that are run in addition to acceptance testing to identify the ways that the system does not adequately meet the client's requirements. These tests normally involve the adequacy of the man/machine interface.

User Documentation: A written record detailing current processing procedures for a system; should be complete and clear so users can depend on it for the bulk of their initial training and for answers to many of their later questions.

User Representative: Person with detailed knowledge of a part of the project application area who describes the detailed procedures of the current or required system, assists in developing test specifications, data, and outcomes, and compares actual results of acceptance testing to predicted outcomes.

User View: Any data view consisting of data required by an external entity or process to answer an inquiry or make a decision; normally equivalent to a simple data flow. Shown as data flows on data flow diagrams, user views may be either input views from external entity to process or output views from process to external entity.

Utility: Any type of purchased or pre-written software used to perform a system function which, while necessary to accomplish the business function, was not specifically one of the functions requested by the System Procurement Group.

Value-Added Analysis: A tool for identifying which activities in a process are important from the perspective of the process customer, and which activities are not important. Value-added analysis asks a series of questions that help us determine whether an individual activity adds value for customers (value-added), or whether it does not add value (non-value-added). Non-value-added activities generate cost, but do not create anything of value for process customers.

Work Products: any tangible, verifiable outcome, result or item that may be produced to complete phases, modules, tasks or activities. They represent the smallest constituents to determine the entire procedure and chronology of activities. Work Products undergo a sign-off procedure.

When mentioning Work Products, it is recommended to mention the process they belong to in order to avoid confusion. For example, PDC (Product Development Committee) Work Products refer to Work Products resulting from the PDC phases. SUMMIT Work Products are not restricted to the final deliverables of the project, but encompass everything that the project has to create or change, however physical or otherwise these might be. Work Products include the intermediate products created by the project on the way to the final product or outcome (for example: Project Charter, System Prospectus as first level work products and Project Scope, System Requirements, Business Requirements as second level work products), management products such as project plans, schedules and status reports, and intangibles such as agreements.

Internal

Work Product Matrix: A formalised, consistent and comparable project framework including a distinct work product ownership and a mandatory sign-off procedure for each work product.

Internal

6 Abbreviations

Abbreviation	Long-form
AIP	Architecture and Infrastructure Planning
BAT	Business Acceptance Test
BPEL	Business Process Execution Language
BRA	Business Requirements Analysis
DTP	Day Time Processing
E2E	End-to-End Phase
INR	Internal Readiness
IS	Information Security
LEEA	Legal Entity Enterprise Architect
LUW	Logical Unit of Work
MMR	Market and Member Readiness
NTP	Night Time Processing
PAI	Product Acquisition and Installation
PER	Product Evaluation and Recommendation
PPI	Project Planning and Initiation
PSD	Package Solution Definition

Abbreviation	Long-form
RTP	Real Time Period
RTS	Real Time System
SDLC	System Development Lifecycle
SDS	System Delivery Specification; not referring to legal context/contractual document “Service Definition Statement”
SLA	Service Level Agreement
SRA	System Requirements Analysis
STT	System Test
TAT	Technical Acceptance Test
TISI	Technical Infrastructure and Software Inventory Phase
TPD	Technical Procedure Development
TRA	Transition
TSD	Technical System Design

7 Appendix A: Reference Information

All information provided in this section applies to distributed as well as project, but for readability, the term “project” in this section refers as well to development activities undertaken outside of a project context.

7.1 Committees Structure

Projects must apply to the Committees in place. Please refer to the current Committee structure at Company and Group level.

7.2 Distributed Development Projects

Distributed Development Project identifies those activities necessary to perform application development projects which are across sites or companies.

In particular, a framework should be developed how to cover peak workload demands at one location by staff from the other location.

No evidence has been found of definite models how to perform distributed development; however, there are various best practices which have developed over time.

Most of the related recommendations do not explicitly depend on the way the work is split between locations. The recommendations focus on minimizing the potential for misunderstandings by clearly defining roles, responsibilities, accountabilities and processes from the very beginning. In addition, methodologies, processes and tools should be aligned upfront, and team building should be supported by some co-located work. Specific recommendations on key roles and processes/methodologies are also analysed.

Existing practices how to split work between locations are summarized. Two models are recommended for further use within Deutsche Börse IT, namely Distribution by Project Phase and Distribution by Component. For both models, a brief characteristic of a potential project is presented.

In addition, recommendations on how to perform future Distributed Development projects within Deutsche Börse Group IT are provided. Two cases are described. For projects which are naturally distributed because the related applications are spread among different locations, the distribution onto the locations arises more or less naturally. Specific focus has to be put onto interface definitions. For projects which are intentionally distributed, a mix between Distribution by Component and by Phase is recommended. To a reasonable degree, management tasks in distributed projects overlap with the standard tasks to be performed in any development project. These activities only need to be followed more strictly. Specific management actions to be performed in addition to the normal ones are summarized as part of this thread.

Corporate Business and Project Selection are also described. No show stopper has been found for starting distributed development in the short term. In the medium term, methodologies, tools and processes must be aligned company-wide to become more efficient. It is recommended to start with projects with medium to low complexity and time-to-market needs to account for the additional complexity.

Conclusion: the recipe for successful distributed development can be summarized as follows:

- Apply clear management by accountability and aligned process management.
- Projects with lower criticality should be taken as first projects to prepare the organization.
- A mixture of Distribution by Phase and by Component is recommended.

7.2.1 Do's and Don'ts

Most of the recommendations on how to perform Distributed Development do not depend on the way the work is distributed between locations. Therefore, they are presented separately in this section. Distributed projects lack the unplanned communication, the common environment and the common cultural background of co-located projects. Thus, there is more potential for

misunderstandings, issues sometimes need more time to be solved, and it is difficult to create a common team spirit. Consequently, measures focus mainly on minimizing the potential for misunderstandings by clearly defining roles, responsibilities, accountabilities and processes from the very beginning. In addition, methodologies, processes and tools should be aligned upfront, and team building should be supported by some co-located work.

In conclusion, many of the measures seem simple. And indeed, achieving success in Distributed Development depends mainly on being aware of the nature of Distributed Development, namely the potential for misunderstandings, and on applying simple but appropriate measures to fill the communication gap arising from distribution.

7.2.1.1 Work Locations

Recommendation 1

Do not distribute Requirements and Specification phase. Relocate remote experts during these phases to the core location. Activities which need intense communication should always be co-located (also applies to R&D activities). People participating in these phases should always work in one location during these phases. In intentionally distributed projects Business Acceptance Test should be co-located as well.

Recommendation 2

The project manager, the remote delivery manager and the architects office should work in a roaming mode to ensure visibility in all project locations. Other team members can remain at their locations (exception: requirement & specification phase).

A roaming project manager is recommended, mainly to achieve a status view across all locations, to ensure synchronization across locations, to avoid that the remote site starts to drift, to be visible and to show management commitment to the remote locations, and to facilitate communication. For the same reasons, roaming work mode is also recommended for the remote delivery manager, i.e. the responsible for the part of the project which is performed at the remote location.

Architects have a central role in solution definition and issue resolution. Therefore, they should maintain a balance between co-located work in the core location and availability in the remote location to ensure consistency in solutions as well as consultancy to developers.

Other team members can normally stay at their location. However, the remote team should be put together in a common office.

7.2.1.2 Processes and Tools

Recommendation 3

Upfront define project execution process (org chart, timelines, work products, roles and responsibilities, processes) and publish in a Cross-site Development Approach.

Location-specific culture, i.e. the way things are being done, is normally part of the environment in which everybody acts. Because of the implicit assumption that the other side acts similarly, cultural gaps will create misunderstandings. Any clarification about the way how things are planned to be done thus decreases the potential for misunderstandings.

Therefore, a Cross-site Development Approach document should be one of the first deliverables of the project to define the execution process. Contents could be the development process to be used,

the work products and deliverables, tools to be used, basic processes such as status reporting, review and sign-off, meetings, information distribution, issue resolution, and escalation, and a project org-chart containing key project members with roles, responsibilities and decision authorities and number of team members (including positions where currently nobody is assigned to). Experts of both locations should be involved in this work product. The work product should be communicated to the project team.

Recommendation 4

Adhere to formal processes for reviewing, incorporating comments, sign-off, etc. Adhere to deadlines, with reminder before and rejection after the deadline.

By adhering to defined processes, each side acts as expected from the other side and thus minimizes the impact of cultural differences in Distributed Development. Non-adherence to the process should normally not be accepted.

Recommendation 5

Use a common methodology within the project. If this is not possible, focus on the most critical deliverables and map them upfront.

Recommendation 6

Use the same tools on both sides. Align issue tracking, management, communication, and development tools upfront. Choose the tools which are used at the main location of the project if possible. Focus on harmonization of tools and processes within the project, but do not try to convert the company with the project.

Recommendation 7

Do not experiment with processes and tools during the project. Process changes cause disturbance. Validate the processes but do not change them if it can be avoided.

7.2.1.3 Roles and Responsibilities

Recommendation 8*

Avoid double heads. Do not duplicate key roles. Achieve single responsibilities. Assign a single owner to each work product and all other deliverables.

Double-headed management always requires additional effort for alignment and creates ambiguities about responsibilities, accountabilities, contact persons and decision authorities. Every management role should be occupied by just one person. Similarly, each work product should have a single owner.

For Interface Specifications, the ownership and the responsibility should lie on the receiving side. A contact person at the providing side should also be defined upfront.

There are occasions where double heads are effective, e.g. in order to prepare the in-sourcing of applications currently developed by an external service provider.

* One of the authors (SH) proposes the following wording for this instruction: "Ensure organization chart reflects clear accountability. ‘Double heads’ should only be used when complementary skills can be brought together and when it is made clear that both parties are jointly responsible for

success. This model of responsibility can be used to reflect the bringing together of two areas/sites in close co-operation".

Recommendation 9

If possible, only 100% assignments should be used in Distributed Development. People prioritize their local work higher than distributed work. In addition, when remaining in their local environment, people tend to local patriotism, i.e. to preferring the local way of doing things. To achieve a common project view and commitment, the remote team should completely work in the project for the duration of the project, i.e. employees should work on a 100% assignment and, moreover, should completely be transferred to the project for the duration of the project.

In particular, in addition to the line manager, the project manager should have to confirm vacations or other off-project time like trainings. The remote team should be relocated from their normal working environment and put together in a new office for the duration of the project. If 100% assignments are not possible, the assignment percentage should be maximized. A team member in a remote team should not work on more than one other project.

Recommendation 10

Project roles should be defined upfront by assignment agreements between project manager, employee and related Line Head.

In order to minimize ambiguities and misunderstandings, project roles and related activities and responsibilities should be defined upfront, agreed between the project/development manager, employee and the related Line Head and be fixed in an assignment agreement to have a basis for solving possible role-related misunderstandings.

7.2.1.4 Information Management and Communication

Recommendation 11

Assign one contact person per functional area - preferably a functional architect - so that team members know where to address their questions.

Clearly communicate these responsibilities to the project members. Architects can be appropriate contact persons but for specific areas other persons might be more appropriate.

Recommendation 12

Embed functional and/or technical experts from the main location into the remote team (20% ratio recommended).

Embedding experts is the most efficient way to provide know-how to the remote team. Thus, this recommendation is preferably applicable to projects where the remote location develops software for applications which are normally located at the main location. Experts (preferably the contact persons mentioned above) should be embedded into the remote team for the duration of the respective work. If this is not possible, they should be embedded for at least a few weeks to distribute know-how. After a sufficient relationship has been built, they could be relocated to their main location and then be contacted via phone.

Recommendation 13

Internal

Issue resolution should be performed by using a common issue log tool, with pre-defined experts - preferably the architects - that are responsible to solve the issues and communicate the solutions. Issue resolution meetings should be performed face-to-face rather than via video conference.

Recommendation 14

When Build and Test is spread across location, a configuration team should be set up as centralized authority to plan, clarify and perform the hand-over of code from build to test teams and to transfer all other necessary information. The centralized team should be located where the major part of the build is being performed.

Recommendation 15

Project management should inform the team regularly about the project status on both sides. Regular status updates help achieving a common project view and a team spirit.

Recommendation 16

At least 50% of the meetings with involvement of people from both locations should be co-located. Distributed meetings should be performed as video conference if possible. Meetings should be performed with agenda defined upfront, minutes and action items distributed after the meeting. In a distributed project, the additional level of communication helps pinpointing different understandings and clarifies meeting outcome. The remote team lead should participate in the different Committee meetings. In addition to the standard project meetings, regular meetings specifically addressing cross-location coordination and discussion needs can compensate the lack of informal communication.

Recommendation 17

For issue resolution, synchronous communication (e.g. phone) should be preferred, mail discussion should be avoided. Response time across location should be shorter than within location.

Generally, choose the right communication media. Issues can be discussed much faster by phone than in lengthy mail discussion. Mail can be used to verify outcomes. To prevent an out-of-sight / out-of-mind culture, a rule should be established that response time to the remote location should be shorter than within the same location. In addition, key people should ensure general availability for calls, e.g. by re-directing calls to their mobiles when they are not available via their office phone.

7.2.1.5 Project Planning

Recommendation 18

Include distribution specific activities (ramp-up, knowledge transfer, cross-location project setup, cost allocation and reporting, distributed infrastructure testing, etc.) into the project planning.

Ramp-up of the remote team is expected to need around one month. A one to two week period for initial knowledge sharing activities should also be foreseen, cross-location cost allocation and reporting requires the setup of mirror projects. Their setup should allow for a consolidated view of costs for teams from different locations working on the same tasks (as is the case for embedded experts). Further tasks to be considered are described in section "IT Project Manager's Tasks in Distributed Projects".

Recommendation 19

Adjust budget and time planning to account for the cross-site communication interface overhead.

Distribution specific costs are relocation costs and travel costs for roaming team members. In a top-down view, from offshoring industry experience cross-site communication interface overhead is expected to increase costs by 10-15% for the distributed parts of the project. Include time contingencies into the plan.

Recommendation 20

Plan for more effort in the Specification phase and enhance the phase containment. A very detailed specification and strong phase containment are widely applied industry best practices in the offshoring context.

7.2.1.6 Infrastructure

Recommendation 21

Centralize project information, including project status information. Preferably, use a shared project drive. Inform the project team where which information is stored (there may be different practices at different locations).

Recommendation 22

Centralize the code repository. Give full access to both sides. Avoid multi-site version control.

Recommendation 23

Ensure full access from remote side early (firewalls!). Consider the need to access project drives and other sources of information, the code repository, specific tools like an issue log tool, testing environments, etc.

Recommendation 24

Check bandwidth requirements for network connections upfront. Normal connection bandwidth may be too small for specific needs, e.g. for GUI testing.

7.2.1.7 Team Building

Recommendation 25

As early as possible, perform a co-located kick-off with the initial team. Alignment in early phases is vital to achieve a common project vision. If the projects start without an aligned vision, even after a few days each location develops an own vision and drifts away. Re-aligning after this vision has stabilized is difficult.

In case there is no team member from the remote location in the initial team, include at least the key people from remote team into the kick-off.

Recommendation 26

Perform a few days of co-located work with the initial team in the beginning to build effective working relationships.

The best way to achieve team building and to get acquainted to the contacts across locations is working together in a co-located way in the beginning of the project, including some key persons from the remote location.

Team building can be supported by some informal evening activities where people can also exchange their private background. Team building should be performed in smaller, work-centric teams targeted at the people that need to communicate across locations. The main effect is achieved in the first two occasions. Further opportunities have only little effect

7.2.1.8 Project Evaluation

Recommendation 27

Perform regular project reality checks in order to identify shortcomings, specifically in the project areas where work is split between locations.

In regular intervals (e.g. one months after a new project phase has started), someone independent from the project should check whether there are problems in communication between locations and suggest mitigating measures. Results should be communicated to the project management as well as the project sponsor. IT internal BIS Project Audit team could perform these reality checks.

Recommendation 28

Perform ex post evaluations of Distributed Development Project about their lessons learnt. Successful Distributed Development is essentially a best practice Business. Thus, at the end of a project, lessons learnt should be collected, the recommendation in this document should be revised and the document should be updated accordingly. This activity could be assigned to the Quality Management section.

7.2.2 Models How to Distribute Work between Locations

The project team has not found evidence of definite models how to distribute work between locations; however, there are several best practices which have developed over time. These practices follow two main concepts:

- Distribute development such that the cut between locations will minimize the need for cross-location issue resolution. Models that fit to this category are Distribution by Phase, Distribution by Component and Distribution by Requirement.
- Shift work to the remote location where a reduced speed of issue resolution does not affect the work directly. This approach is mainly used in the Distribution by Priority model and partly for Distribution by Requirement which is generally applied in an environment without time-to-market pressure.

Considering the specific situation of Deutsche Börse Group IT, the feasibility of the four main models, Distribution by Phase, Component, Requirement and Priority, can be summarized as follows:

- Recommended: Distribution by Phase, Distribution by Component.
- Not recommended: Distribution by Requirement, Distribution by Priority.

Recommendation 29

Use a combination of Distribution by Component and Distribution by Phase in order to shift work to the remote location in future projects. Explicitly, perform some phases in the application development process for some components (or applications) in the remote location. The decision about which phases and which components depends on the project, the availability of know-how and the resource workload and has to be specifically defined for any new project.

The four main models are briefly explained in the subsections.

7.2.2.1 Distribution by Phase

This model is one of the most common models to distribute work between locations as it is essentially the Indian Onsite-Offshore model for development projects.

The idea is to perform different project phases at different locations. Typical phases that are shifted between locations are Design, Build and Test; sometimes also the Specification is performed at the remote location.

Requirements and Acceptance Test are normally performed in the main site.

The distribution between locations can be applied in different ways. Examples are:

- Design and Build at the main location, Test at the remote location.
- Design at the main location, Build and Test at the remote location.
- Design, Build and Test at the remote location.

Generally, the better the know-how about the related application at the remote location, the more phases can be shifted to this location.

Example: Distribute Build and Test onto different locations.

The model could be applied within Deutsche Börse Group IT in the following situation:

- The workload is low in the remote location, while there is a large project scheduled in the main location.
- The remote location has only generic know how about the applications in the main location.

A work distribution by phase where the product test is performed in the remote location is now an appropriate approach to allow shifting part of the work to the remote location while building up additional application know-how.

After the Specification is finished, the test team at the remote location starts the test preparation (defining the test approach, the test cycles, cases and conditions) while design and build is performed at the main location in parallel.

- The test approach needs to be strongly aligned between project management and the remote test manager. This work should be performed in a co-located or roaming mode.
- Embedded experts from the main location ensure sufficient know-how in the test preparation at the remote location. Vice versa, some key experts from the remote location should participate in the requirement and specification phase.
- While the test is performed at the remote location, a bug fixing team is working in parallel at the main location.

- Acceptance test is then completely performed at the main location.

Recommendation 30

In the short term, shifting the product test to the remote location should be used to build up functional know-how on the remote site to prepare the re-location of additional project phases. After sufficient know-how has been built at the remote location, further project phases can be considered to shift workload between locations.

Industry experience with offshoring shows that a stable project scope, strong phase containment and very detailed description of deliverables positively influence projects which are distributed by phase.

7.2.2.2 Distribution by Component

This model is a widespread model how to distribute work between locations. It is primarily applied in large companies, but also common in the context of outsourcing or offshoring application development.

The idea is to develop different parts of the software in different locations. The model is applied for two main reasons: either the remote location has specific know-how not available at the main location (e.g. hardware-related programming performed by a development unit located together with the hardware development) or non-core parts of the software are outsourced for cost-saving reasons.

Instead of Distribution by Component, projects can also be distributed by:

- Application in case of projects where many applications are involved.
- Functional or technical blocks in case the software is not composed of components.

For Distribution by Component the emphasis should be put on detailed interface specifications and on the identification of cross-component issues in order to minimize gaps (see recommendations 1, 2, 12, 13).

Example: Shift development work for some applications to the remote location. The model could be applied within Deutsche Börse Group IT in the following situation:

- A new system needs to be developed or a major release is planned.
- Major changes need to be applied in various different applications.

A work distribution where the development work for some of the related applications is performed in the remote location is now an appropriate approach to use resources from the remote location instead of filling staffing gaps with externals.

A project would be set up as follows:

Requirements and Specifications are performed in the main location. In the Design, Build and part of the Test phase, the development work for some of the related applications is performed in the remote location. Assembly, Integration and System test are performed jointly again.

- The distribution of the application development work onto the locations is defined as early as possible.
- The major part of work in the requirement and specification phase is performed by local experts.

However, key experts from the remote location are already involved in order to build know-how and working relations.

- Classroom sessions and application walk-through exercises build know-how build in the remote team. Additional time and budget is planned in the project plan for these activities.
- When the design phase starts, remote experts that participated in the requirements and specification phases move back to their location. Additionally, experts from the core locations are embedded into the remote team and work together with the remote team during design, build and test.
- For assembly, integration, system and security tests, embedded experts and key experts from the remote team move to the main location in order to perform these tasks co-located.

Recommendation 31

A Distribution by Component is recommended as a model to shift work to the remote location for large scale projects. Specific know-how acquisition phases should support building the necessary know-how in the remote location.

Presumable, the applied split will be rather a split by application than by component.*

Similarly, to Distribution by Phase, a stable project scope, strong phase containment and very detailed description of deliverables positively influence projects which are distributed by Component.

* e.g. Except for the front-end / back-end split, there is no obvious cut to be applied to Xetra, Eurex, CCP or the Risk Engine.

7.2.2.3 Distribution by Requirement

This model is primarily used in Open Source software development projects where a globally spread pool of independent developers works on the same software. The setup of the GNOME project, one of the largest open source projects with around 500 registered developers, may serve as an example:

- Program Management is the responsibility of the GNOME Foundation, which coordinates the decision process about future direction of development.
- The software itself is split into 19 modules. Each module is assigned to a module maintainer who is responsible for providing a vision for the further development of the module, for attracting developers and for defining which requirements are implemented in which order.
- The module maintainer assigns the work to the developers, preferably with a division of tasks into parts with minimum interaction so that developers can work most independently. Every developer more or less independently implements new requirements into the software by coding, testing and then providing a software patch to a centralized configuration management.

As a model for Distributed Development, Distribution by Requirement would thus mean that different locations take care of independent functional requirements and implement them through the entire application from Analysis of Requirements to Build & Test. Assembly, Integration and

System Test are again centralized. All locations need to have the same know how about the application. Sophisticated configuration management is needed to ensure consistency of the code.

Recommendation 32

Although proven to be successful in the Open Source environment, Distribution by Requirement is NOT RECOMMENDED for use in Distributed Projects in the future within Deutsche Börse Group IT.

Contraindications are as follows:

The Open Source environment differs in two central topics. There is no time-to-market pressure; and there are no committed service levels. There is no proof that the model could also work in a time-to-market environment or with pre-defined SLAs. In addition, a high level of functional and technical integration is required - both sides need to have a complete in-depth view of the application at the remote location in order to implement new functionality throughout the entire application. Finally, configuration management is highly complex, especially to achieve consistency when both sites work on the same component in a situation where tasks cannot be divided along distinct functional lines without overlapping technical components.

7.2.2.4 Distribution by Priority

This model is the standard model for application maintenance in the context of onsite-offshore application development. Characteristics of the Distribution by Priority model are as follows:

- Teams at both locations have the same know-how of the application and should be capable of doing all necessary work.
- Errors/Issues are analysed onsite.
- Critical, urgent fixes are performed onsite, other work is performed offshore.

Offshore companies typically work with a duplicated project management structure onsite and offshore. A clear process of communication and decision is defined upfront, and both teams follow it very strictly. However, all activities that need fast response time are performed onsite.

In the context of Distributed Development Projects, one would be shifting less prioritized work to the remote location, e.g. work which is not so urgent and complex. Examples are programming of reports, data archiving tasks, data consistency checkers or performance-related tasks. In this way, the need for expertise and knowledge sharing is reduced at the remote location, and there is slightly more time to resolve upcoming issues without directly influencing the progress of the entire project.

Recommendation 33

Although proven to be successful in the onsite/offshore context, Distribution by Priority is NOT RECOMMENDED for use in Distributed Projects in the future within Deutsche Börse Group IT.

Contraindications are as follows:

- First, it is difficult to find sufficient low priority development work to justify setting up a remote site.
- Second, work classified as 'low priority' at the start of the project can become 'high priority' at later stages in the life cycle, in particular as there is always the risk that the remote team with the "low priority" is being overlooked from the main location.

- Finally, team motivation and commitment are difficult to achieve when a team is designated to perform only "low priority" work (in the onsite/offshore context, this is levelled out by the possibility to work "at home") and thus the remote team will drift away from the main project team.

7.2.3 Recommendations for Example Projects

7.2.3.1 Distribution of Project Work

Distribution defined by the location of the relevant applications.

Situation: some projects have to be performed in a distributed way due to their nature. Typically, they cover one main application which gets data from and/or delivers data to several other up and/or downstream systems where some of the applications are located at the remote location. Examples for such projects are Collateral Management, OTC CCP or Reference Data Factory.

The distribution model to be applied to these projects is the natural distribution of work according to the location of the individual applications. Thus, the areas where the cut between locations is applied are the interfaces, and consequently, these are the main areas to specifically focus on when performing Distributed Development.

The most important recommendations contained in the recommendation section "Do's and Don'ts in Distributed Development Projects" are the following ones:

- Achieve single responsibilities. Assign a single owner to each work product or any other deliverable. In particular, for each interface define one responsible owner (preferably one the receiving side) and a contact person on the providing side (see recommendation 8).
- Co-locate Requirements and Specification phase, including interface definitions and impact analysis. Embed remote experts into the main team for these phases (see recommendation 1).
- A general overview of a project manager's tasks in projects where the distribution model is defined by the distribution of the related applications is provided in section "IT project manager's tasks in Distributed Projects".

Intentionally distributed projects

Intentionally distributed projects are projects which are not naturally distributed but where a distribution of work is applied in order to take advantage of capacity in a remote site.

Thus, a situation where there is a large project in the main location and low workload in the remote location is the case where an intentional distribution is applied in order to shift workload to the remote location. Instead of hiring service providers to fill the resource shortfall in the main location, capacity available in the remote location is used. In this scenario, a planning horizon of 2 to 3 months is expected.

The distribution model applied in this situation is a mix between a Distribution by Component (or Application) and a Distribution by Phase as elaborated in sections "Distribution by Phase" and "Distribution by Component". Initially, a distribution of Build and Test onto different locations seems the most appropriate distribution.

No other phase distribution model should be applied. Other models such as splitting analysis or integration test require too much collaborative effort. These are the most communication intense parts of the software life cycle and are therefore less suited to distribution.

The most important generic topics for project management in distributed projects are:

- Achieve a clear definition of roles, responsibilities and processes upfront. Prefer single responsibilities.
- Do not experiment with tools and processes during the project. Validate processes but do not change them if changes can be avoided.
- Focus on team building and trust.
- Teach the team managers to be ready for Distributed Development.
- As project manager, be visible and open for everybody.

7.2.3.2 IT Project Manager's Tasks

This section describes the steps and deliverables required for the IT Project Manager to set up and run a Distributed Development Project. The intention is not to provide a general management guide but to highlight the specifics of distributed development.

The chart is set out as follows:

- The first column contains the standard SUMMIT project phases.
- The next two columns describe the project/development activitiy manager's tasks and default deliverables for all distributed projects regardless of naturally or intentionally distributed.
- The last two columns describe specific actions to be taken and related deliverables when the project is intentionally distributed to take advantage of resource availability in the remote site.

SUMMIT Phase	Actions for PM All Distributed Models	Deliverables	Additional Actions for PM Distributed Projects	Additional Deliverables - Intentionally Distributed Projects
BRA	Project manager retains E2E architect from architecture team.		Project manager and E2E architect agree work distribution model across sites.	Distribution Model - presentation showing how phases and/or components will be distributed across locations.
SRA/AIP	Project manager and E2E architect build a high-level view of resources required: -Impacted areas.	Updated Project Budget - to include cost of embedding experts from remote site into main site	Identify experts from remote site to work on SRA at main site in knowledge transfer mode.	Updated Project Budget - to include cost of embedding experts from main site into remote site from

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SUMMIT Phase	Actions for PM All Distributed Models	Deliverables	Additional Actions for PM Distributed Projects	Additional Deliverables - Intentionally Distributed Projects
	-Number of people for each site/application area. -Level of expertise. -Number of team leaders. -Number of experts for co-located SRA phase. -Key dates.	throughout SRA phase.		SDS phase onwards.
	Resource Arrangements: -For Luxembourg and for CSD Applications Department resource requirement project manager to contact line managers responsible for application areas ('ICSD Settlement and Custody Applications', 'ICSD Connectivity Applications' or 'CSD Applications'). -Where necessary Assignment Agreements are drawn up to clarify arrangements between project and line managers.	Resource Plan.	Identify experts from main site to embed in remote site.	Resource Plan - to include list of experts willing to relocate to remote site.
	Points of contact, team leader, roles and responsibilities established in all sites.	Organization chart.	PM and team leads develop and implement knowledge transfer plan for the remote site.	K.T. Plan.
	Itemise and agree:			

Internal

SUMMIT Phase	Actions for PM All Distributed Models	Deliverables	Additional Actions for PM Distributed Projects	Additional Deliverables - Intentionally Distributed Projects
	-Specific development environments required. -Communication technology. -Office requirements: project teams to be co-located within each site. -Tools and processes.			
	Project manager and team leads work out: -Development approach. -Cost estimates per application area. -Cross site/system impacts. -Cross site/system development plan.	-Cross Site Development Approach. -Cross Site Impact Matrix -E2E plan.		
	Project manager re-validates initial re-sourcing approach with: -Luxembourg line managers. -Head of consulting in FFM.	Updated resource plan.		
	E2E architect together with local experts, from all sites, design high level architecture.		Reiterate in more detail, within the development approach, how the work will be distributed across sites.	Updated Distribution Model - presentation showing how phases and/or components will be distributed across locations.

SUMMIT Phase	Actions for PM All Distributed Models	Deliverables	Additional Actions for PM Distributed Projects	Additional Deliverables - Intentionally Distributed Projects
	Cross site communication infrastructure in place.			
SDS	Face-to-face team building exercise.			
	Local experts move back to remote site to participate in SDS.	Updated Organization chart.	Experts from main site relocate to remote site to facilitate knowledge sharing - they remain in the remote site for the duration of the distribution.	Updated Organization chart.
	Project manager and team leads develop a cross site E2E testing approach.	E2E Test Approach.		
	Project manager, architect, and key development team leads go into roaming mode.			
TSD	Cross site development infrastructure tested.			
TPD	Senior development experts go into roaming mode to support integration testing.			
BAT			If development has been split by component PM may decide to co-locate the fix team in main site for duration of SAT/BAT and post implementation support.	Updated Organization chart.

7.2.4 Corporate Readiness

Recommendations to achieve Corporate Readiness are split into pre-requisites, medium-term and long-term recommendations.

Pre-Requisite activities Deutsche Börse Group should secure before proceeding with distributed development:

- There are no corporate pre-requisites.
- However, currently, there is limited experience with the other side's applications, processes and culture. Thus, as preparatory measure, it is recommended that some expected key persons for future distributed projects participate in projects of the other side in order to build related know-how.

Medium Term activities Deutsche Börse Group should progress in the medium term to ensure distributed development becomes increasingly effective.

- Currently, there is no cross-location overview about available resources with skills and project history. In particular, there are different assignment processes. A common process would be preferable.
- Align Tools:
  - Align Management Tools - Issue, defect, change request, decision processing - in progress with Quality Management team.
  - Align Development Tools - design, review, test tools.
- Align Processes:
  - Align Development Standards
  - Align Test Standards
  - Align Development Methodology.
  - Align Project Processes - change process, issue resolution process.
- Provide dedicated or mobile communication facilities (i.e. video conference).
- Optimize process for managing project budget across Deutsche Börse Group locations.
- Ensure flexible facilities management for relocating people between sites.
- Develop process for temporary staff relocation remaining in sync with human resource limitations.
- Increase relocation opportunities across Deutsche Börse Group sites:
  - Reduce cross site cultural barriers.
  - Increase cross site application knowledge.

Longer Term activities Deutsche Börse Group should progress to optimize distributed development.

- Align product development process.
- Align process for determining resource availability - update current resource requisition process to enable checks across Deutsche Börse Group sites.

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7.2.5 Project Selection

Concerning the projects to be chosen for distribution:

- It is preferable to distribute projects with low-medium time-to-market needs and complexity in order to account for the additional complexity.
- The minimum duration of a project should be 6 months. For short term projects, inevitable ramp-up costs (estimation: 2 weeks for know-how build) and costs for temporary re-location (rough estimation: approx. +30% for re-located work compared to work in the home location) in early project phases may offset the savings.
- A minimum number of people should be involved particularly in the remote site, at least around 10 people. It is recommended that no less than 20-30% of the project team is located in the remote site.
- Project work shifted to the remote site should have sufficient relevance and priority to achieve team commitment.
- The project should allow for a clear breakdown of work across sites. An approved SRA document should be the basis for this decision.

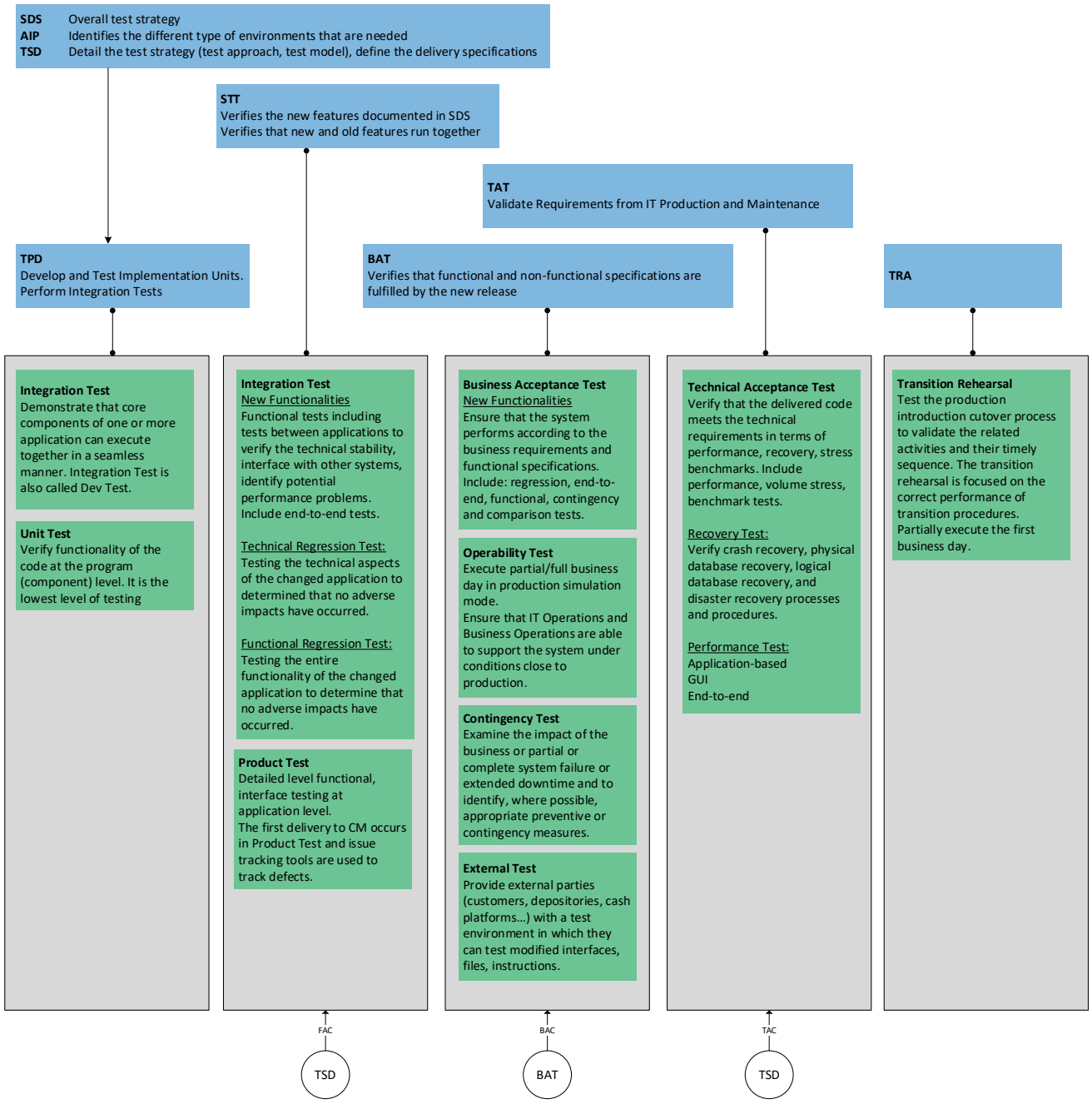
In order to start the first distributed projects:

- Board involvement is needed in order to trigger the distribution of work. Normally, information about low workload in the remote location, i.e. about the reason to distribute the work, is not available to a project manager.
- Additional preparation time of 2-3 months is expected for the first distributed projects to spread the idea, achieve buy-in of all related actors and to get experience with the new processes for setting up distributed projects.

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8 Appendix B: Testing

Test phases



Internal